

GLOBAL CREDIT TRADER

An Update on AI-Related Debt Issuance

- In late July, we outlined our top-down framework for AI-related debt issuance from the hyperscalers, using the CapEx estimates from our colleagues in GIR equity research as a key input. At the time, our estimate for 35% of 2027 CapEx to be financed with debt mapped to roughly \$400 billion of gross debt issuance, globally, *directly* issued from the group.
- Several weeks later—and with the benefit of the most recent round of earnings results—we refresh the gross hyperscaler issuance implied by our forecasts. We now expect \$420 billion of gross debt issuance in 2027, which reflects some modest upward revisions in CapEx forecasts.
- This forecast would represent more than a 60% increase vs. our full-year 2026 estimate of \$250 billion of direct global hyperscaler IG debt issuance. So far this year, \$229 billion has been issued across a range of currencies. We continue to expect 65% to 75% of the 2027 hyperscaler gross debt issuance to price in the USD IG bond market, subject to market conditions and investor receptivity in light of issuer concentration considerations.
- So far this year, we have tracked \$578 billion of global, gross AI-related debt issuance, which includes a broad universe of firms. Hyperscalers continue to represent ‘just’ 40% of overall AI-related debt issuance, underscoring the wide range of AI-related exposure in the global credit markets.
- The performance bifurcation between AI-related spreads and the rest of the market persists. We believe the relative resilience of the broader USD IG market reflects two key factors. First is the combination of a supportive macro, including solid economic activity, constructive credit fundamentals, and demand from yield-based investors. The second factor is the diversification value provided by bonds in larger sectors more directly removed from the AI theme, including Banks, Energy, and Healthcare, among others.
- The share of AI-related notional across the IG and HY corporate credit indices has been increasing. The largest exposure is in the USD IG market, at 12% of total notional, followed by USD HY at 6%. In the EUR market, the figures are more contained at 3% for IG and 1% for HY. We see meaningful scope for this to grow as the multi-year issuance cycle extends.

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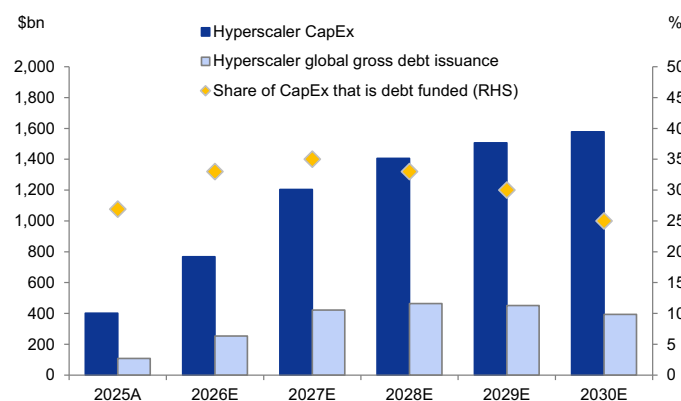
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In [late July](#), we outlined our top-down framework for AI-related debt issuance from the hyperscalers, using the CapEx estimates from our colleagues in GIR equity research as a key input. At the time, our estimate for 35% of 2027 CapEx to be financed with debt mapped to roughly \$400 billion of gross debt issuance, globally, *directly* issued from the group. (Note: this excludes project finance issuance for data center and chip financing, which we forecast separately, expecting \$300 billion in 2027).

Several weeks later—and with the benefit of the most recent round of earnings results—we refresh the gross hyperscaler issuance implied by our forecasts. As [Exhibit 1](#) illustrates, we now expect \$420 billion of gross debt issuance, which reflects some modest upward revisions in CapEx forecasts. Our colleagues in GIR equity research have recently [highlighted](#) the ongoing supply-demand imbalance related to the AI build out and expect the CapEx cycle through the end of 2027 to remain elevated.

Exhibit 1: We expect 35% of CapEx in 2027 will be debt financed

GIR Equity Research CapEx estimates and our estimates of hyperscaler IG gross debt issuance (\$bn), as a share of CapEx (RHS)



Note: As of September 15, 2026. Excludes data center financings issued by special purpose vehicles. Hyperscalers include Amazon, Alphabet, Meta, Microsoft, and Oracle.

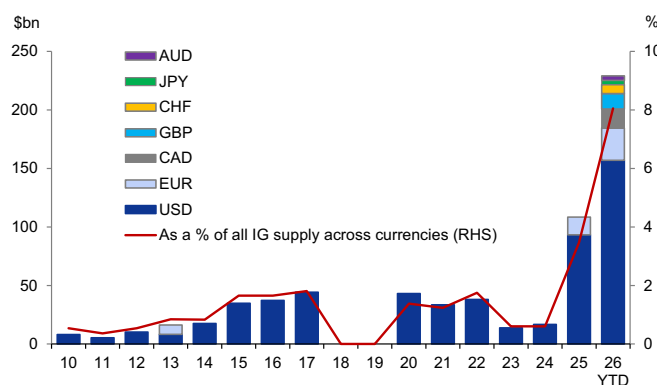
Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

This forecast would represent more than a 60% increase vs. our full-year 2026 estimate of \$250 billion of direct global hyperscaler IG debt issuance. So far this year, \$229 billion has been issued, across the range of currencies outlined in [Exhibit 2](#).

We continue to [expect](#) 65% to 75% of the 2027 hyperscaler gross debt issuance to price in the USD IG bond market, subject to market conditions and investor receptivity in light of issuer concentration considerations. As [Exhibit 3](#) illustrates, there is still meaningful capacity for the group to grow their debt levels before reaching the size of the largest issuers.

Exhibit 2: We expect \$420 billion of global IG debt issuance directly from the hyperscalers in 2027

Hyperscaler debt issuance across currencies (in USD-equivalent, \$bn), and the share of IG supply across those regions

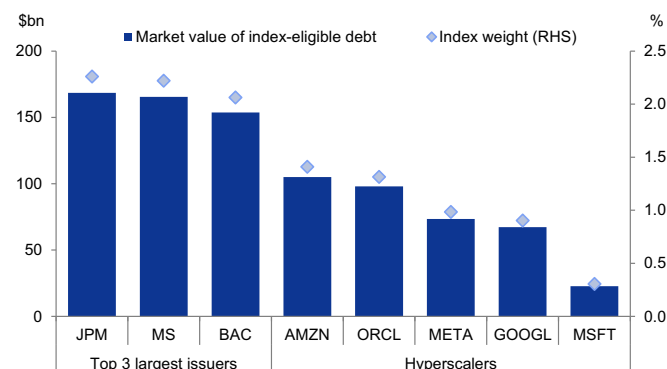


Note: As of September 14, 2026. Captures debt issued by the following companies: Alphabet, Amazon, Meta, Microsoft, and Oracle.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

Exhibit 3: We ultimately expect the hyperscaler debt capital structures to grow to the size of today's largest issuers

For the three largest issuers in the Bloomberg USD IG Corporate Index and the hyperscalers: market value of index-eligible debt outstanding (\$bn), and index weight (%), and index weight (%), and index weight (%)



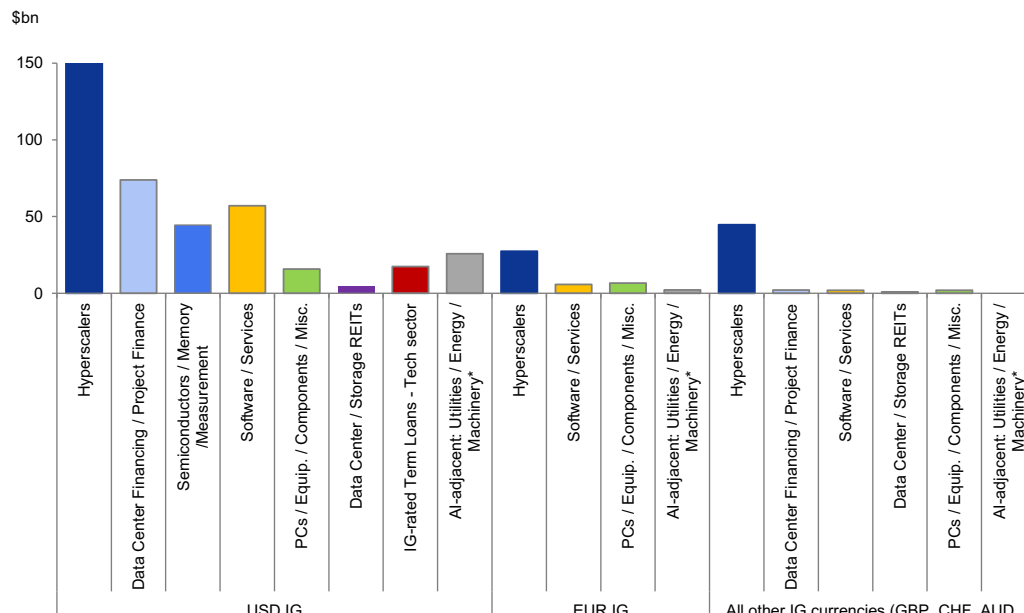
Note: As of September 13, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

So far this year, we have tracked \$578 billion of global, gross AI-related debt issuance, which includes a broad universe of firms across the hyperscalers, project finance issuers, and a wide range of other AI-related entities. The components of the IG AI-related supply are listed in [Exhibit 4](#), and the leveraged finance components are outlined in [Exhibit 5](#). Hyperscalers continue to represent 'just' 40% of overall AI-related debt issuance, underscoring the wide range of AI-related exposure in the global credit markets.

Exhibit 4: We estimate \$490 billion of AI-related supply across the global IG markets so far this year

Gross AI-related issuance (\$bn) in the global IG markets

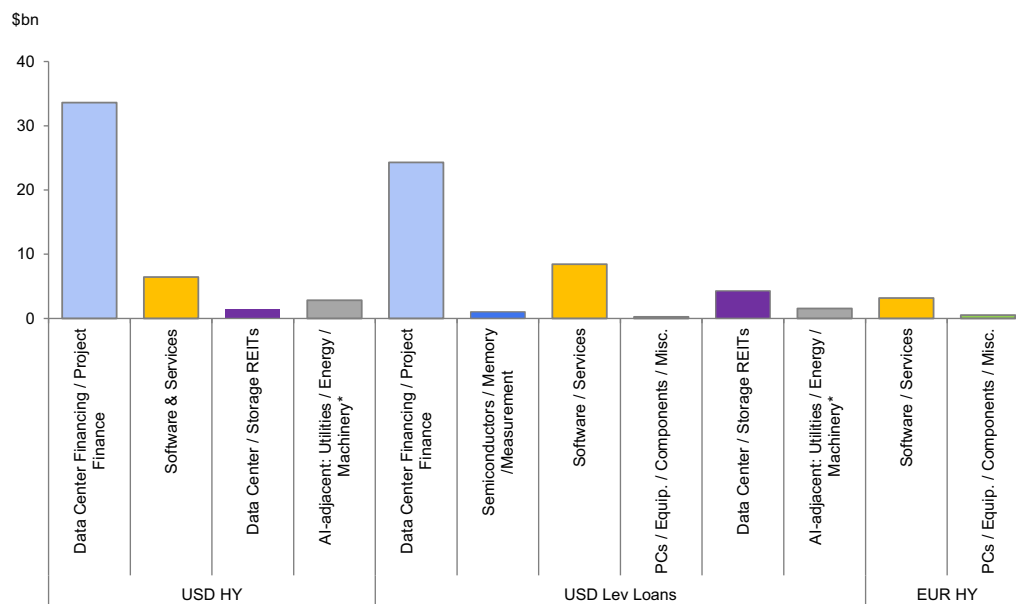


Note: As of September 14, 2026. Excludes undrawn credit facilities and revolvers. For AI-adjacent sectors (Energy & Utilities, Oil & Gas, Machinery, Chemicals, Metal & Steel), we assume only 30% of gross issuance is AI-related. For non-USD markets, we use USD-equivalent figures. Excludes pending financings.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

Exhibit 5: We estimate \$88 billion of AI-related supply across the global leveraged finance markets so far this year

Gross AI-related issuance (\$bn) in the global leveraged finance markets



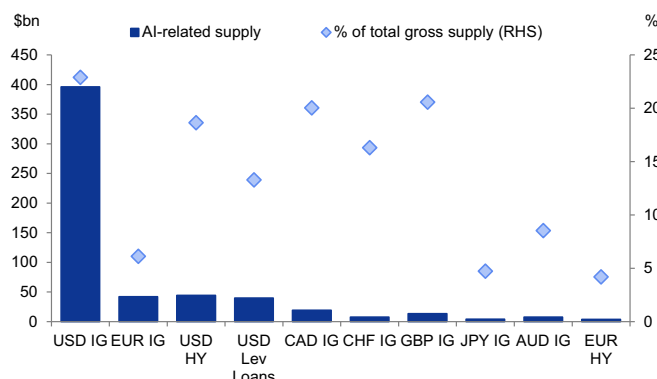
Note: As of September 14, 2026. Excludes undrawn credit facilities and revolvers. For AI-adjacent sectors (Energy & Utilities, Oil & Gas, Machinery, Chemicals, Metal & Steel), we assume only 30% of gross issuance is AI-related. For non-USD markets, we use USD-equivalent figures. Excludes pending financings.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

A diverse mix of global syndicated credit markets have participated in the AI-related financing, as shown in [Exhibit 6](#). We continue to expect the share of AI-related issuance from the EUR market will increase, considering its year-to-date share has lagged a range of other global IG markets. Moreover, as [Exhibit 7](#) illustrates, the other non-USD global IG corporate credit markets pale in comparison to the size of the EUR IG market, limiting the amount of ‘heavy lifting’ they might do. Away from the traditional credit markets, we also see a larger role for [convertible bond issuance](#) in financing the AI buildout.

Exhibit 6: We expect the share of AI-related issuance in the EUR market to increase

Amount of AI-related gross supply (\$bn) in each market for YTD 2026, and the share it represents of total gross supply (RHS)

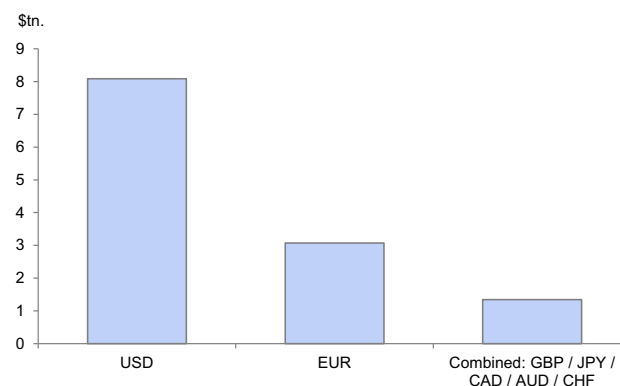


Note: As of September 14, 2026. For non-USD markets, we use USD-equivalent figures. Excludes undrawn credit facilities and revolvers. For AI-adjacent sectors (Energy & Utilities, Oil & Gas, Construction, Machinery, Chemicals, Metal & Steel), we assume only 30% of gross issuance is AI-related.

Source: Dealogic (ION Analytics), PitchBook LCD, Goldman Sachs Global Investment Research

Exhibit 7: We expect the EUR IG and HY markets will absorb more of the AI-related financing, going forward, given the limited size in other non-USD regional IG corporate credit markets

Amount outstanding (\$tn) for the currency-specific Bloomberg IG Corporate indices



Note: Excludes index-ineligible debt in all markets. Based on currency of issue, not country of issuer. As of September 16, 2026. In USD equivalents.

Source: Bloomberg, Goldman Sachs Global Investment Research

The performance [bifurcation](#) between AI-related spreads and the rest of the market persists, as shown in [Exhibit 8](#) and [Exhibit 9](#). We believe the relative resilience of the broader USD IG market reflects two key factors. First is the combination of a supportive macro backdrop, including solid economic activity, generally constructive credit [fundamentals](#), and an attractive all-in yield opportunity that continues to generate demand from [yield-based](#) investors. The second factor is the diversification value provided by bonds in larger sectors more directly removed from the AI theme, including Banks, Energy, and Healthcare, among others.

While spreads for AI-related names, especially hyperscalers, have found some relief locally, we remain cautious on adding meaningful exposure to the segment. Near term, our supply estimates imply that much of hyperscaler funding this year has been completed, but the nature of the multi-year build-out keeps risks skewed to the upside, as sanguine sentiment could instead catalyze issuance. As the build-out has grown in scope, we [continue](#) to think that hardware names will remain less insulated from credit pressure as compute shipments, and related financing, grows.

Issuance for the data center and compute ‘project finance’ segment remains even more difficult to assess, but we continue to think that it marches forward. We maintain that investors should be selective when investing in this segment, [underwriting](#) project-related risks in the face of a technical that is unlikely to abate in the near term.

Exhibit 8: The broader USD IG credit market has remained resilient, even as AI-related spreads have widened

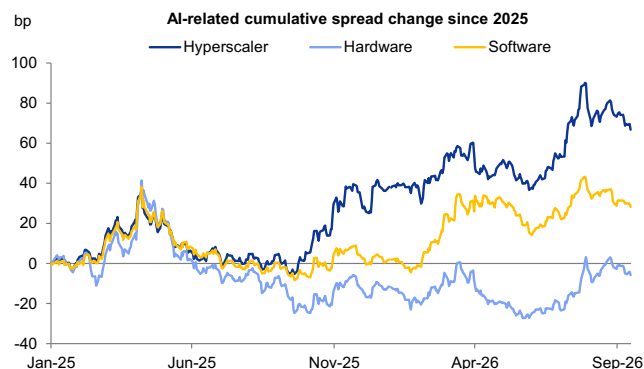
Weighted-average spreads (market value times duration) for AI and non-AI bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 9: There is a significant amount of spread performance bifurcation within the AI ecosystem

Weighted-average spreads (market value times duration) for the AI-related ecosystem

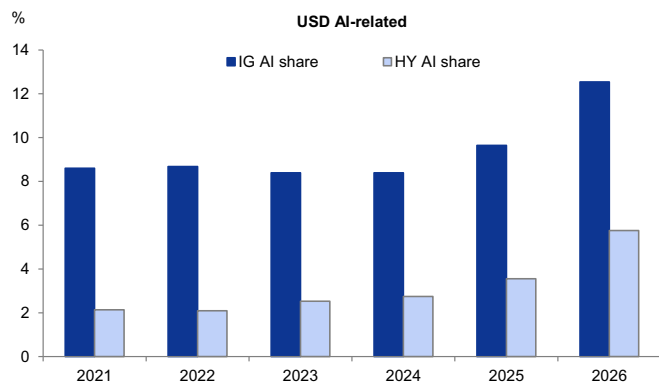


Source: iBoxx, Goldman Sachs Global Investment Research

As [Exhibit 10](#) and [Exhibit 11](#) highlight, the share of AI-related notional across the IG and HY corporate credit indices has been increasing in the USD and EUR markets. The largest exposure is in the USD IG market, at 12% of total notional, followed by USD HY at 6%. In the EUR market, the figures are more contained at 3% for IG and 1% for HY. We see meaningful scope for this to grow as the multi-year issuance cycle extends. For context, the largest sector in the USD IG Corporate Index—Banking—is 22% of the total market value. In the EUR IG Corporate Index, Banks are even larger at 31%.

Exhibit 10: We estimate AI-related bonds represent 12% of the Bloomberg USD IG Corporate Index

AI-related notional as a percentage of the Bloomberg USD corporate credit indices

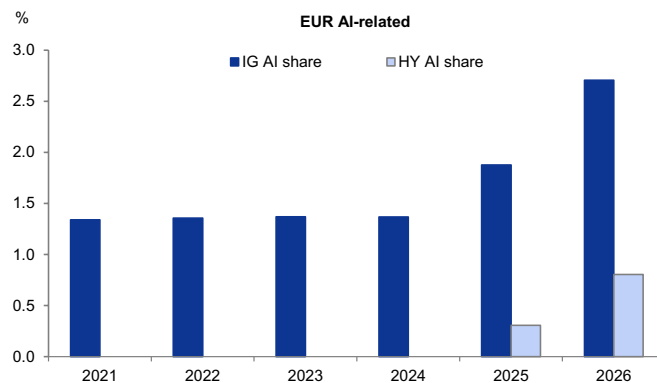


Note: Captures only index-eligible debt.

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 11: The share of AI-related notional in the EUR market is more contained but is expected to grow

AI-related notional as a percentage of the Bloomberg EUR corporate credit indices



Note: Captures only index-eligible debt.

Source: Bloomberg, Goldman Sachs Global Investment Research

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 12: USD and EUR spread forecasts

Market	Updated as of Sep. 16, 2026				
	Current	2026Q3	2026Q4	2027Q1	2027Q2
USD spreads (bp)					
USD IG	76	85	82	80	78
USD HY	269	305	295	290	280
EUR spreads (bp)					
EUR IG	83	91	88	86	84
EUR HY	292	320	315	310	305

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 14: Issuance forecasts

Market	Updated as of Sep. 16, 2026	
	2026 YTD (bn)	2026 Full-year forecast (bn)
USD market		
Gross issuance		
IG	\$1,759	\$2,300
HY	\$241	\$400
Leveraged loans	\$302	--
Net issuance		
IG	\$748	\$1,000
HY	\$65	\$160
EUR market		
Gross issuance		
IG	€ 638	€ 850
HY	€ 80	€ 125
Leveraged loans	€ 59	--
Net issuance		
IG	€ 183	€ 265
HY	€ 11	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 13: USD and EUR excess returns forecast

Market	Updated as of Sep. 16, 2026		
	Current duration	2026 YTD (%)	2026 Full-year forecast (%)
USD excess returns			
USD IG	6.7	0.7%	0.5%
USD HY	3.4	2.0%	1.3%
EUR excess returns			
EUR IG	4.5	0.6%	0.6%
EUR HY	3.2	1.7%	1.4%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 15: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of Sep. 16, 2026	
	2026 YTD	2026 Full-year forecast
USD market		
Rising stars (bn)	\$78	\$75
Fallen angels (bn)	\$45	\$60
HY defaults (12-month trailing)	3.4%	4.0%
Leveraged loan defaults (12-month trailing)	4.9%	4.75%
EUR market		
Rising stars (bn)	€ 23	€ 20
Fallen angels (bn)	€ 5	€ 30
HY defaults (12-month trailing)	4.7%	5.0%
Leveraged loan defaults (12-month trailing)	3.5%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 16: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	We have a slight preference for USD credit vs. its EUR peer, owing to a somewhat more challenging growth-inflation-monetary policy mix in the Euro Area. That said, the lines are blurring between the regional classifications, given the acceleration in cross-border issuance.
	Overweight EUR IG CLO tranches vs. USD IG CLO tranches	EUR IG tranches offer carry pickup relative to USD IG CLO tranches. While the policy and macroeconomic picture is less supportive in Europe, subordination should protect tranches up in the capital structure from a continued idiosyncratic default picture.
USD market	Neutral IG vs. HY	While we previously had a slight preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
	Overweight Bs, Neutral BBs, Underweight CCCs	We expect BBs will continue to encounter a challenging supply technical due to AI-related supply. As a result, we favor the B rated cohort. The CCC universe is highly idiosyncratic and requires careful credit selection, in our view.
	Overweight HY bonds vs. loans	The substantial software sector exposure in the loan market leaves us favoring the HY market, at least until we receive more clarity on the refinancing of the 2028 loan maturity walls. The lower-rated distribution of loans and our economists' base case for no rate hikes in 2026 are also contributing factors.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight Hybrids vs. BBBs	We view the additional spread compensation of hybrids vs. BBBs as moderately attractive in the context of the tight spread environment. We also expect the hybrid spread premium to compress vs. BBBs, as the hybrid market grows further.
	Neutral IG vs. HY	While we previously had a preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
EUR market	Overweight BBBs vs. higher-rated cohorts	We recently shifted our view to favor BBBs over both AAs and As in the EUR market, owing to an improved macroeconomic backdrop which now favors a broad-based, down-in-quality stance within IG.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB is increasing borrowing costs for floating rate issuers, given the central bank's focus on inflationary pressures.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.

Source: Goldman Sachs Global Investment Research

Earnings calendar

Exhibit 17: USD IG earnings report dates

Upcoming earnings report dates for the Bloomberg USD IG Index (LUACTRUU); minimum \$1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
9/22/2026			9/23/2026			9/24/2026			9/25/2026		
AZO	AutoZone Inc	7.5	GIS	General Mills Inc	7.5	SNX	TD SYNEX Corp	2.8			
			PAYX	Paychex Inc	4.2	COST	Costco Wholesale Corp	2.8			
			CTAS	Cintas Corp	1.2	DRI	Darden Restaurants Inc	1.6			
9/28/2026			9/29/2026			9/30/2026			10/1/2026		
JEF	Jefferies Financial Group Inc	8.5	CNXC	Concentrix Corp	2.0	CAG	Conagra Brands Inc	6.5	ACN	Accenture PLC	9.7
						JBL	Jabil Inc	2.9	NKE	NIKE Inc	6.0
									MKC	McCormick & Co Inc/MD	2.8

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 18: USD HY earnings report dates

Upcoming earnings report dates for the Bloomberg USD HY index (LF98TRUU); minimum \$500 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
9/22/2026			9/23/2026			9/24/2026			9/25/2026		
KBH	KB Home	1.0									
THO	Thor Industries Inc	0.5									
9/28/2026			9/29/2026			9/30/2026			10/1/2026		
MTN	Vail Resorts Inc	1.1	AIR	AAR Corp	0.7						

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 19: EUR IG earnings report dates

Upcoming earnings report dates for the Bloomberg EUR IG Index (LECP TREU); minimum €1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
9/23/2026			9/24/2026			9/25/2026			9/28/2026		
GIS	General Mills Inc	2.5	HMBSS	H & M Hennes & Mauritz AB	1.5				JEF	Jefferies Financial Group Inc	1.4
9/29/2026			9/30/2026			10/1/2026			10/2/2026		
CCL	Carnival Corp Ltd	2.1									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 20: EUR HY earnings report dates

Upcoming earnings report dates for the Bloomberg Pan-Euro HY index (LP01 TREU); minimum €100 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
9/23/2026			9/24/2026			9/25/2026			9/28/2026		
GIS	General Mills Inc	1.7									
9/29/2026			9/30/2026			10/1/2026			10/2/2026		

Source: Bloomberg, Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 21: Performance of USD and EUR markets

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	77	5.8	-1.14	-2.37	-1.54	-1	0.97
AA	57	5.6	-1.06	-2.80	-2.31	8	0.98
A	66	5.6	-1.17	-2.29	-1.43	1	0.89
BBB	96	6.0	-1.07	-2.08	-0.86	-5	0.87
HY	270	7.7	-0.94	-0.23	1.72	4	1.21
BB	155	6.7	-0.92	-0.31	1.52	-14	0.77
B	278	8.1	-0.86	0.20	2.65	-23	0.72
CCC	1076	15.9	-1.28	-1.44	-1.30	191	2.96
USD Leveraged Loans							
Leveraged loan index	419	8.1	0.36	2.10	3.44	24	0.2
BB	248	6.2	0.25	1.64	4.22	-8	-0.2
B	406	7.9	0.54	2.67	3.77	14	0.1
CCC	2137	26.7	-1.18	-1.65	-5.02	573	6.1
USD Securitized products							
Agency MBS	36	5.6	-1.63	-2.97	-1.56	14	1.0
ABS (Autos)	55	5.1	-0.25	0.30	1.77	-11	0.8
ABS (Credit cards)	29	4.9	-0.46	-0.07	1.13	-4	1.0
ABS (Student loans)	93	5.0	0.22	1.20	3.49	120	0.1
USD Macro indices							
CDX IG	52	--	-0.06	0.06	0.53	2	--
CDX HY	314	--	-0.45	0.09	1.68	-2	--
EUR Corporate bonds							
IG	98	4.3	-1.11	-2.34	-0.99	8	0.9
AA	72	4.2	-1.07	-2.72	-1.46	10	0.8
A	90	4.3	-1.09	-2.36	-1.03	6	0.8
BBB	112	4.5	-1.14	-2.23	-0.86	10	0.9
HY	274	6.4	-1.07	-1.09	0.79	4	0.8
AT1	268	6.0	-1.21	-1.24	2.11	-18	0.7
EUR Leveraged loans							
Leveraged loan index	450	7.2	0.34	1.18	3.07	0	0.7
BB	268	5.3	0.19	1.03	3.83	-84	-0.3
B	418	6.9	0.37	1.35	3.12	4	0.7
EUR Macro indices							
iTraxx Main	54	--	-0.10	0.00	0.5	3	--
iTraxx Xover	258	--	-0.41	-0.09	3.2	14	--
iTraxx SenFin	56	--	-0.08	0.03	0.8	2	--
iTraxx SubFin	90	--	-0.08	0.17	1.6	-3	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96. EUR IG we use the iBoxx QW5A index with history back to Jan-99. EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 22: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	177.0	76		5.8	7.2	BAA1/BAA2	-1	-2	-5	-14	-0.1	-1.2	-2.1	-0.8
Chemicals	1%	81.3	80		5.9	7.4	BAA1/BAA2	-2	-2	-5	-18	-0.1	-1.2	-2.0	-0.3
Metals & Mining	1%	70.1	68		5.7	7.5	A3/BAA1	-1	-2	-3	-9	-0.2	-1.3	-2.4	-1.3
Paper	0%	25.6	81		5.8	5.8	BAA2/BAA3	0	-3	-8	-15	-0.3	-1.1	-1.6	-0.7
Capital Goods	6%	420.0	58		5.6	6.5	A3/BAA1	-3	-2	2	-8	-0.1	-1.2	-2.3	-1.3
Aerospace & Defense	2%	146.4	65		5.7	7.9	BAA1/BAA2	-2	-2	2	-9	-0.1	-1.2	-2.8	-1.6
Building Materials	1%	45.9	73		5.7	6.5	BAA1/BAA2	-1	-1	0	-4	-0.2	-1.3	-2.3	-1.4
Diversified Manufacturing	2%	127.0	56		5.5	6.2	A3/BAA1	-4	-5	0	-8	-0.1	-1.2	-2.3	-1.3
Construction Machinery	1%	56.3	37		5.3	4.6	A1/A2	-4	0	8	-10	-0.2	-0.9	-1.4	-0.6
Packaging	0%	13.9	68		5.6	4.3	BAA2/BAA3	-1	2	1	-14	-0.3	-1.2	-1.5	-0.5
Environmental	0%	30.5	45		5.4	5.7	A3/A3	-2	0	5	-8	-0.3	-1.2	-2.0	-1.4
Communications	8%	569.4	106		6.2	8.2	A3/BAA1	-3	-3	7	12	0.1	-1.1	-3.3	-3.2
Cable & Satellite	2%	127.2	159		6.7	8.5	BAA1/BAA2	-1	0	8	21	0.1	-1.4	-3.7	-3.7
Media & Entertainment	2%	128.9	94		6.0	8.2	A1/A2	-4	-6	10	11	0.1	-0.8	-3.5	-4.7
Wireless	2%	128.1	81		5.8	6.9	BAA1/BAA2	-3	-3	2	0	0.0	-1.0	-2.5	-1.7
Wirelines	2%	185.3	96		6.1	8.8	BAA1/BAA2	-3	-4	4	8	0.2	-1.1	-3.3	-2.8
Consumer Cyclical	7%	542.9	72		5.7	6.2	A2/A3	-1	2	8	1	-0.1	-1.1	-2.2	-1.5
Automotive	2%	155.3	84		5.7	4.2	BAA1/BAA2	2	6	5	-5	-0.2	-1.0	-1.1	0.0
Leisure	0%	5.8	102		6.0	6.1	BAA2/BAA2	-2	5	13	1	-0.4	-1.8	-3.0	-2.3
Gaming	0%	23.9	104		5.9	4.6	BAA2/BAA3	0	-2	0	-6	-0.2	-1.0	-1.5	-0.4
Home Construction	0%	9.4	79		5.7	4.9	BAA1/BAA2	-1	-1	1	-5	-0.2	-1.2	-1.7	-0.8
Lodging	0%	20.4	71		5.6	4.7	BAA2/BAA3	-1	0	9	-4	-0.3	-1.3	-1.8	-1.0
Retailers	4%	269.2	62		5.7	7.5	A1/A2	-2	0	16	17	0.0	-1.2	-3.0	-2.5
Restaurants	0%	31.2	59		5.7	7.5	BAA1/BAA2	-2	-2	4	-7	-0.1	-1.2	-2.6	-1.6
Consumer Cyc Services	0%	27.6	73		5.7	6.0	A3/BAA1	-2	1	1	2	-0.2	-1.3	-2.0	-1.5
Consumer Non-Cyclical	15%	1096.2	64		5.7	7.6	A2/A3	-3	-1	3	-5	-0.1	-1.2	-2.7	-1.8
Consumer Products	1%	50.7	42		5.4	6.0	A1/A2	-13	-10	-6	-17	-0.2	-1.2	-2.0	-1.2
Food & Beverage	3%	223.8	68		5.7	7.4	A3/BAA1	-3	-1	2	-2	-0.1	-1.3	-2.6	-1.9
Healthcare	4%	316.7	70		5.8	7.6	A3/BAA1	-2	-1	1	-8	-0.1	-1.2	-2.6	-1.6
Pharmaceuticals	6%	411.3	57		5.7	8.1	A2/A3	-3	-1	6	-3	0.0	-1.2	-2.9	-2.0
Supermarkets	0%	12.3	86		6.0	9.7	BAA1/BAA2	-4	-3	2	0	0.2	-1.2	-3.0	-2.3
Tobacco	1%	81.4	71		5.7	6.5	A3/BAA1	-3	-1	4	-8	-0.1	-1.2	-2.3	-1.3
Energy	7%	506.3	73		5.8	7.5	A3/BAA1	-3	-3	-6	-18	-0.1	-1.1	-2.1	-0.5
Independent Energy	1%	85.9	73		5.8	7.4	BAA1/BAA2	-2	-3	-9	-23	-0.1	-1.1	-1.9	0.3
Integrated Energy	1%	97.4	51		5.6	8.1	AA3/A1	-4	-3	-7	-14	0.0	-1.2	-2.3	-1.4
Oil Field Services	0%	23.8	77		5.8	7.4	A3/BAA1	0	-1	-5	-20	-0.2	-1.3	-2.3	-0.7
Refining	0%	28.9	74		5.8	7.0	BAA1/BAA2	-2	-3	-13	-30	-0.1	-1.1	-1.5	0.6
Midstream	4%	270.2	81		5.9	7.4	BAA1/BAA2	-3	-3	-5	-16	-0.1	-1.1	-2.1	-0.5
Technology	11%	800.0	84		5.9	7.0	A2/A3	-3	-4	6	10	0.0	-0.9	-2.7	-2.2
Transportation	2%	139.0	66		5.8	8.8	A3/BAA1	-1	-1	2	-9	0.0	-1.3	-3.1	-1.7
Airlines	0%	16.4	78		5.6	3.7	A2/A3	-1	0	3	-14	-0.3	-1.0	-1.1	0.2
Railroads	1%	84.9	63		5.8	10.6	A3/BAA1	-2	-1	0	-6	0.1	-1.4	-3.7	-2.5
Transportation Services	1%	37.6	67		5.7	7.0	A3/BAA1	-1	1	5	-10	-0.1	-1.2	-2.7	-0.9
Other Industrial	1%	40.2	63		5.8	9.7	AA3/A1	-2	-2	6	0	-0.1	-1.4	-3.4	-2.2
Utility	10%	756.6	81		5.9	8.0	A3/BAA1	-2	-1	3	-4	0.0	-1.2	-2.8	-1.7
Electric	9%	681.9	81		5.9	8.0	A3/BAA1	-2	-1	3	-4	0.0	-1.2	-2.8	-1.7
Natural Gas	1%	59.0	86		5.9	7.5	A3/BAA1	-2	-3	-1	-11	-0.1	-1.2	-2.6	-1.3
Other Utility	0%	15.7	69		5.8	8.2	BAA1/BAA2	-2	0	4	-1	-0.1	-1.3	-2.7	-2.0
Financial Institutions	32%	2394.5	78		5.7	5.3	A2/A3	-2	-1	3	0	-0.2	-1.1	-1.8	-1.1
Banking	22%	1625.8	73		5.6	4.8	A2/A3	-2	0	5	2	-0.2	-1.1	-1.7	-1.0
Brokerage Asset Managers Exchanges	2%	123.5	78		5.7	6.0	A3/BAA1	-1	1	0	5	-0.2	-1.2	-2.0	-1.5
Finance Companies	1%	111.1	134		6.2	3.5	BAA2/BAA3	-1	-1	-8	-5	-0.2	-0.8	-0.4	0.6
Insurance	5%	335.8	84		5.9	8.0	A3/BAA1	-2	-2	2	-5	0.0	-1.3	-2.9	-1.9
REITS	3%	191.3	72		5.6	5.2	A3/BAA1	-1	0	6	-6	-0.3	-1.2	-1.9	-1.0
Other Financial	0%	6.9	105		6.0	5.3	BAA1/BAA2	-1	4	6	4	-0.3	-1.3	-1.5	-0.1
US Corporate IG		7442.0	77		5.8	6.7	A3/BAA1	-2	-2	3	-1	-0.1	-1.1	-2.4	-1.5

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 23: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	93.6	251		7.6	3.4	BA3/B1	3	2	-4	-14	-0.3	-0.7	0.1	2.8
Chemicals	3%	47.2	235		7.4	3.6	BA3/B1	5	5	-3	-27	-0.3	-0.8	0.1	4.1
Metals & Mining	3%	39.7	183		7.0	3.4	BA3/B1	1	-3	-10	-12	-0.2	-0.5	0.5	2.2
Paper	0%	6.7	763		12.6	2.5	B1/B2	4	7	9	52	-0.3	-0.9	-1.4	-3.0
Capital Goods	11%	161.7	217		7.3	3.4	BA3/B1	1	5	4	2	-0.3	-1.2	-0.7	0.1
Aerospace & Defense	2%	33.1	106		6.3	3.4	BA3/B1	12	19	3	6	-0.6	-1.3	-0.7	1.0
Building Materials	3%	39.1	369		8.8	3.9	BA3/B1	3	8	14	29	-0.3	-1.4	-1.6	-1.5
Diversified Manufacturing	2%	29.4	162		6.7	3.1	BA3/B1	1	13	6	54	-0.2	-0.9	0.0	1.3
Construction Machinery	2%	23.0	98		6.0	3.2	BA3/B1	-3	2	-27	-26	-0.2	-0.9	-0.1	2.0
Packaging	2%	28.2	313		8.2	3.2	B1/B2	0	3	5	-25	-0.1	-1.1	-0.3	-1.2
Environmental	1%	8.9	150		6.5	3.2	BA3/B1	0	2	9	1	-0.2	-0.9	-0.9	0.8
Communications	14%	202.9	458		9.6	4.0	BA3/B1	-1	2	7	32	-0.3	-1.2	-1.2	0.3
Cable & Satellite	5%	73.4	629		11.2	3.3	B1/B2	-6	-4	9	57	-0.2	-1.1	-1.5	-1.6
Media & Entertainment	5%	76.0	323		8.3	4.4	BA3/B1	1	5	-2	-4	0.0	-1.1	0.2	2.1
Wireless	1%	18.0	316		8.2	4.2	BA3/B1	2	7	8	26	-0.3	-1.3	-1.1	-0.4
Wirelines	2%	35.5	463		9.7	4.2	B1/B2	5	6	18	48	-0.9	-1.6	-3.6	0.4
Consumer Cyclical	18%	268.4	245		7.4	3.3	BA3/B1	6	9	4	-3	-0.3	-1.0	-0.2	2.0
Automotive	3%	39.0	296		8.0	3.3	BA3/B1	10	18	21	7	-0.5	-1.3	-0.5	1.5
Leisure	1%	20.6	235		7.2	3.2	B2/B3	-2	8	9	31	-0.5	-1.6	-1.4	0.6
Gaming	3%	45.4	268		7.7	3.0	BA3/B1	3	5	6	10	-0.4	-1.1	-0.4	1.6
Home Construction	1%	16.8	179		6.8	3.1	BA3/B1	5	10	5	-11	-0.4	-1.2	-0.1	2.6
Lodging	1%	16.2	116		6.2	4.2	BA2/BA3	5	12	4	-12	-0.4	-1.5	-1.0	0.6
Retailers	5%	76.4	242		7.4	3.6	BA3/B1	18	17	1	-26	-0.2	-0.9	0.1	2.9
Restaurants	1%	11.2	157		6.5	3.8	BA3/B1	6	12	3	-3	-0.5	-1.4	-0.7	1.2
Consumer Cyc Services	3%	42.9	281		7.9	2.7	B1/B2	0	0	-9	1	0.0	-0.4	1.0	2.9
Consumer Non-Cyclical	11%	155.0	232		7.4	3.3	B1/B2	-3	-4	-8	-6	-0.3	-0.9	0.2	2.4
Consumer Products	1%	10.6	227		7.3	3.7	BA3/B1	-26	-23	-29	-30	-0.4	-1.6	-0.7	-0.4
Food & Beverage	2%	29.6	194		7.0	4.1	BA3/B1	6	14	12	14	-0.5	-1.3	-0.8	1.0
Healthcare	5%	78.8	236		7.4	3.1	BA3/B1	2	2	4	13	-0.3	-0.8	0.1	2.7
Pharmaceuticals	2%	26.2	288		8.0	2.9	B2/B3	2	-8	-26	-34	-0.1	0.0	2.9	7.1
Supermarkets	1%	9.4	181		6.8	4.1	BA2/BA3	3	8	16	162	-0.4	-1.6	-1.2	-0.6
Tobacco	0%	0.3	186		7.1	3.1	BA3/BA3	0	-4	-4	9	-0.2	-0.7	-0.3	0.3
Energy	11%	167.1	135		6.5	3.4	BA2/BA3	4	1	-23	-50	-0.3	-0.7	0.8	5.0
Independent Energy	3%	46.2	129		6.5	3.1	BA3/B1	4	0	-34	-48	-0.2	-0.4	1.8	5.8
Oil Field Services	2%	28.1	151		6.8	3.3	BA3/B1	2	-5	-23	-50	-0.3	-0.5	1.0	5.0
Refining	0%	5.3	136		6.5	2.2	BA3/B1	1	-10	-44	-63	-0.2	-0.2	2.9	8.9
Midstream	6%	87.6	133		6.4	3.6	BA2/BA3	5	4	-12	-51	-0.4	-0.9	0.1	4.4
Technology	10%	144.5	352		8.6	3.3	BA3/B1	1	0	5	12	0.3	-0.3	0.0	1.2
Transportation	2%	33.0	448		9.5	2.9	BA3/B1	1	10	11	24	-0.5	-1.9	-1.7	-1.9
Airlines	1%	13.4	436		9.2	2.7	BA3/B1	5	19	34	58	-0.8	-3.0	-3.3	-1.9
Railroads	0%	1.8	932		14.5	3.4	B1/B2	-13	6	4	-3	-1.0	-1.6	-1.5	-8.3
Transportation Services	1%	17.8	409		9.2	3.0	BA3/B1	2	5	-3	16	-0.3	-1.1	-0.5	-1.1
Other Industrial	1%	15.2	225		6.0	3.3	BA3/B1	2	4	-5	-13	-0.3	-1.1	0.4	1.5
Utility	3%	45.1	172		6.8	4.2	BA2/BA3	3	2	7	17	-0.4	-1.1	-1.1	1.5
Financial Institutions	13%	186.9	241		7.4	3.2	BA3/B1	2	6	-1	8	-0.3	-1.0	0.0	1.1
Banking	0%	2.8	151		6.5	4.3	BA1/BA2	1	-9	-16	-30	-0.2	-0.8	-0.4	0.5
Brokerage Asset Managers Exchanges	1%	13.1	201		7.1	3.1	B1/B2	1	1	12	6	-0.2	-0.7	1.5	3.5
Finance Companies	4%	66.0	247		7.5	3.3	BA2/BA3	4	6	0	19	-0.4	-1.0	0.0	0.7
Insurance	4%	58.7	240		7.4	3.3	BA3/B1	3	7	-1	25	-0.4	-1.3	-0.4	-0.3
REITS	2%	28.3	227		7.3	2.8	B1/B2	1	8	-6	-17	-0.1	-1.1	-0.4	2.2
Other Financial	1%	18.0	293		7.9	3.0	BA3/B1	-1	-1	-9	-7	0.0	-0.5	0.8	3.4
US Corp High Yield		1473.5	270		7.7	3.4	BA3/B1	2	3	0	1	-0.2	-0.9	-0.2	1.7

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 24: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.5	€ 388.1	92.7		4.3	4.5	A	5	7	5	8	-0.58	-1.11	-2.19	-1.01
Automobiles & Parts	5.2	€ 161.7	115.8		4.5	3.7	A	7	10	7	14	-0.58	-1.04	-1.70	-0.76
Beverages	2.9	€ 89.1	86.0		4.3	5.7	A	4	5	4	5	-0.61	-1.27	-2.85	-1.34
Food Producers	1.9	€ 59.9	73.4		4.1	4.9	A	4	6	3	3	-0.59	-1.16	-2.48	-1.22
Household Goods	0.2	€ 6.1	57.9		4.1	5.7	AA	4	8	7	7	-0.60	-1.23	-2.92	-1.55
Personal Goods	1.7	€ 52.1	69.3		4.0	4.2	A	4	4	4	6	-0.54	-1.00	-2.08	-1.06
Tobacco	0.6	€ 19.1	110.8		4.5	4.7	BBB	4	7	5	-1	-0.59	-1.18	-2.32	-0.69
Consumer Services	4.6	€ 144.0	93.9		4.4	5.1	A	4	6	7	12	-0.58	-1.15	-2.64	-1.34
Food & Drug Retailers	0.6	€ 19.9	92.6		4.3	4.4	BBB	4	6	9	7	-0.59	-1.10	-2.26	-0.93
General Retailers	0.6	€ 19.4	102.3		4.6	6.2	AA	2	6	10	23	-0.47	-1.12	-3.42	-2.05
Media	1.1	€ 33.9	95.6		4.3	4.2	BBB	5	7	8	12	-0.59	-1.07	-2.21	-0.94
Travel & Leisure	2.3	€ 70.9	90.9		4.4	5.4	A	4	5	5	10	-0.60	-1.21	-2.74	-1.46
Energy	3.1	€ 96.8	100.5		4.4	4.8	A	4	8	4	3	-0.59	-1.17	-2.28	-0.68
Oil & Gas Producers	2.8	€ 86.3	100.7		4.4	4.7	A	4	8	4	2	-0.58	-1.15	-2.24	-0.65
Oil Equipment / Services & Distribution	0.3	€ 8.1	100.3		4.5	5.3	A	6	5	8	24	-0.69	-1.30	-2.79	-1.04
Renewable Energy	0.1	€ 2.4	91.8		4.2	4.9	BBB	2	3	-7	-10	-0.57	-1.07	-1.96	-0.41
Industrials	9.5	€ 295.3	86.9		4.2	4.4	A	5	7	6	6	-0.60	-1.13	-2.23	-1.01
Aerospace & Defense	0.5	€ 15.5	76.8		4.1	4.3	A	6	9	5	12	-0.63	-1.14	-2.16	-1.28
Construction & Materials	1.5	€ 47.1	91.8		4.3	4.2	BBB	5	7	9	8	-0.60	-1.12	-2.24	-0.99
Electronic & Electrical Equipment	0.9	€ 28.0	77.8		4.1	4.8	A	5	6	6	8	-0.65	-1.26	-2.63	-1.44
General Industrials	1.2	€ 36.6	73.4		4.2	5.2	A	5	5	3	-1	-0.67	-1.23	-2.65	-1.27
Industrial Engineering	1.1	€ 34.5	80.5		4.1	3.3	A	4	7	3	3	-0.47	-0.89	-1.47	-0.32
Industrial Transportation	3.3	€ 103.7	91.7		4.3	4.6	BBB	5	7	7	7	-0.63	-1.18	-2.36	-1.17
Support Services	1.0	€ 29.9	102.9		4.3	4.0	BBB	4	6	8	8	-0.53	-1.02	-1.78	-0.43
Health Care	5.6	€ 174.4	82.8		4.3	5.6	A	4	5	5	3	-0.59	-1.18	-2.82	-1.30
Health Care Equipment & Services	2.4	€ 74.5	89.9		4.4	5.5	A	4	6	7	6	-0.59	-1.18	-2.80	-1.36
Pharmaceuticals & Biotechnology	3.2	€ 100.0	77.6		4.3	5.6	A	4	4	4	1	-0.59	-1.18	-2.83	-1.26
Basic Materials	2.6	€ 82.1	91.1		4.3	4.7	A	6	10	9	5	-0.67	-1.28	-2.42	-0.95
Chemicals	2.0	€ 62.7	90.1		4.3	4.9	A	6	11	8	6	-0.69	-1.34	-2.56	-1.10
Forestry & Paper	0.2	€ 5.0	97.7		4.3	3.9	BBB	6	8	20	11	-0.58	-1.06	-1.41	0.15
Industrial Metals	0.1	€ 3.6	89.7		4.2	4.8	BBB	6	5	4	-3	-0.70	-1.31	-2.59	-0.94
Mining	0.3	€ 10.8	95.8		4.3	4.0	BBB	5	8	9	-1	-0.58	-1.05	-2.09	-0.60
Technology	2.8	€ 86.9	102.3		4.5	5.4	A	2	4	7	16	-0.47	-1.03	-2.93	-1.90
Software & Computer Services	2.4	€ 74.4	105.4		4.5	5.7	A	1	4	7	16	-0.46	-1.04	-3.11	-2.07
Technology Hardware & Equipment	0.4	€ 12.4	72.1		4.0	3.5	A	5	7	10	12	-0.53	-0.95	-1.84	-0.89
Telecommunications	5.5	€ 172.2	101.8		4.4	5.4	BBB	3	6	10	10	-0.60	-1.24	-2.92	-1.48
Fixed Line Telecommunications	4.5	€ 139.5	101.4		4.4	5.5	BBB	3	7	11	11	-0.60	-1.26	-2.96	-1.53
Mobile Telecommunications	1.1	€ 32.7	103.7		4.5	5.1	BBB	4	5	7	8	-0.59	-1.15	-2.77	-1.27
Financials	42.2	€ 1,312.2	103.0		4.3	4.0	A	6	6	10	8	-0.61	-1.02	-2.08	-0.74
Banks Sen	26.5	€ 824.1	90.2		4.2	3.8	A	5	4	8	6	-0.55	-0.91	-1.90	-0.65
Banks Sub	4.0	€ 125.3	132.5		4.6	3.6	BBB	7	6	8	6	-0.57	-0.92	-1.68	-0.22
Financial Services Sen	2.8	€ 86.9	86.7		4.2	4.4	A	4	5	5	3	-0.55	-1.03	-2.17	-0.72
Financial Services Sub	0.0	€ 1.0	105.3		4.2	1.8	A	-2	3	-7	-33	-0.14	-0.34	-0.21	1.53
Insurance Sen	1.4	€ 42.5	83.4		4.2	4.1	A	4	3	7	8	-0.54	-0.98	-2.17	-1.05
Insurance Sub	2.6	€ 81.7	146.3		4.8	5.0	A	7	7	14	13	-0.75	-1.24	-2.80	-1.15
Real Estate Sen	4.8	€ 148.7	128.4		4.6	4.7	BBB	11	15	19	20	-0.92	-1.56	-2.87	-1.45
Real Estate Sub	0.1	€ 2.0	214.8		5.4	4.2	BBB	16	49	30	0	-1.05	-2.54	-3.02	0.00
Utilities	11.6	€ 361.0	102.1		4.4	5.2	BBB	5	8	7	8	-0.66	-1.30	-2.72	-1.23
Electricity	6.9	€ 215.7	104.3		4.5	5.2	BBB	5	8	7	8	-0.65	-1.31	-2.72	-1.18
Gas / Water & Multiutilities	4.7	€ 145.3	98.8		4.4	5.1	BBB	5	7	7	8	-0.66	-1.30	-2.73	-1.30
Index	100.0	€ 3,113.1	97.7		4.4	4.5	A	5	6	8	8	-0.60	-1.11	-2.33	-0.99

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 25: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	27.2	€ 122.5	285		6.1	2.9	BB	9	9	-5	10	-0.52	-0.93	-0.58	1.39
Consumer Goods	14.9	€ 66.8	260		5.9	2.9	BB	9	15	-10	24	-0.48	-0.92	-0.41	1.44
Consumer Services	12.4	€ 55.7	316		6.4	2.9	BB	7	-3	0	-3	-0.57	-0.94	-0.79	1.33
Energy	10.1	€ 45.4	243		5.7	2.9	BB	5	7	-13	-15	-0.44	-0.75	-0.28	1.02
Energy	2.3	€ 10.5	213		5.3	2.5	BB	5	4	-31	-88	-0.34	-0.50	0.30	3.72
Utilities	7.8	€ 34.9	250		5.8	3.1	BB	5	8	-9	3	-0.48	-0.82	-0.45	0.25
Industrials	30.6	€ 137.8	311		6.4	2.9	BB	5	20	-16	-33	-0.40	-0.82	-0.21	2.42
Basic Materials	6.5	€ 29.3	383		7.1	2.9	B	3	2	-35	-181	-0.41	-0.52	0.25	6.83
Construction & Materials	1.4	€ 6.3	561		8.9	3.0	BB	14	36	32	-47	-0.79	-1.45	-1.14	-0.09
Health Care	7.5	€ 33.8	305		6.3	2.9	B	-6	36	-7	38	-0.01	-0.55	-0.12	1.98
Industrials	15.2	€ 68.5	262		5.9	2.9	BB	10	16	-15	-24	-0.55	-1.03	-0.37	1.34
TMT	20.1	€ 90.2	354		6.8	3.5	BB	4	12	24	68	-0.35	-0.93	-1.41	-0.20
Technology	4.7	€ 21.2	444		7.7	3.5	B	4	6	13	66	-0.16	-0.41	0.37	0.07
Telecommunications	15.4	€ 69.1	327		6.6	3.5	BB	5	16	26	54	-0.40	-1.08	-1.94	-0.43
Financials	12.0	€ 53.8	390		7.1	3.1	BB	22	44	26	62	-0.88	-1.56	-1.13	-0.40
Core Financials	4.8	€ 21.6	202		5.3	3.1	BB	14	14	8	23	-0.57	-0.83	-0.98	0.23
Financial Services	1.6	€ 7.4	832		11.6	2.3	B	4	7	-135	231	-0.05	-0.60	1.95	-2.42
Real Estate	5.5	€ 24.9	451		7.8	3.3	BB	36	75	68	73	-1.39	-2.47	-2.12	-0.34
Index	100.0	€ 449.7	317		6.4	3.1	BB	7	16	2	15	-0.48	-0.95	-0.67	1.14

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

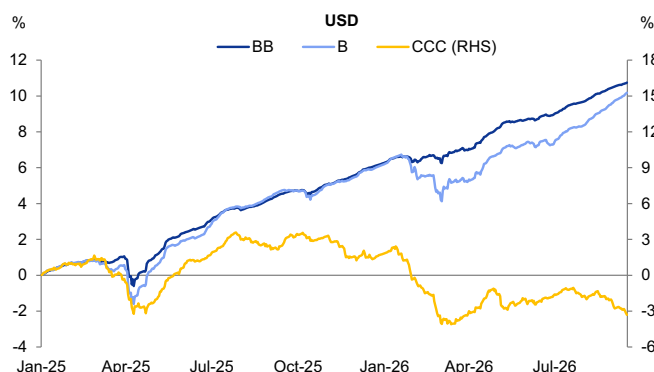
Exhibit 26: USD ETF performance

Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$27.99B	-1159	-1.1%	268	14.04	105.01	0.04%	-1.7	-2.6	-0.7	iBoxx USD Liquid Investment Grade Index
	VCSH	\$44.21B	-325	13.2%	570	1.30	77.77	0.01%	0.5	-0.5	-0.6	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$67.79B	-541	19.7%	852	2.00	79.91	0.00%	-1.5	-2.1	-1.0	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$10.93B	137	21.4%	215	--	50.98	0.05%	3.0	0.9	0.2	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$23.44B	21	11.2%	455	0.34	51.57	0.01%	0.5	-0.5	-0.6	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.74B	259	13.8%	366	0.49	51.41	0.03%	-1.4	-2.1	-1.1	Bloomberg Barclays U.S. Interim
	SPSB	\$10.83B	-430	30.9%	365	1.50	29.70	0.04%	1.2	0.0	-0.4	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$11.27B	6	6.7%	346	1.72	32.68	0.01%	-0.4	-1.2	-0.8	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCIT	\$8.77B	-92	23.4%	124	14.36	71.48	0.08%	-2.2	-3.6	-0.5	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$14.88B	-253	-23.1%	190	42.28	78.70	-0.04%	1.6	-0.1	-0.9	iBoxx USD Liquid High Yield Index
	JNK	\$6.71B	-76	-9.2%	71	8.72	94.70	-0.05%	1.8	-0.1	-0.9	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.7B	-7	-7.7%	191	1.51	24.63	0.01%	1.9	0.1	-0.6	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.75B	25	-0.3%	186	0.30	41.83	0.01%	2.2	0.3	-0.6	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.78B	-5	20.8%	19	0.05	91.98	-0.06%	1.7	0.1	-0.6	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.61B	-12	-5.4%	40	0.03	40.23	-0.04%	0.5	0.1	-0.9	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.18B	10	4.2%	112	0.95	28.56	-0.13%	1.6	-0.6	-1.0	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$6.72B	-41	-0.8%	325	16.66	20.66	-0.27%	2.5	2.5	0.4	S&P/LSTA U.S. Leveraged Loan 100 Index
Loans	SRLN	\$5.6B	0	-13.2%	138	6.10	40.66	-0.14%	3.3	2.7	0.3	Markit iBoxx USD Liquid Leverage Loans Total Return Index
	FTSL	\$2.44B	7	5.9%	54	0.30	44.97	-0.01%	2.2	1.5	0.1	S&P/LSTA US Leveraged Loan 100 Market Value Index
	EMB	\$14.01B	-234	-12.2%	151	9.21	93.60	-0.02%	0.6	-1.7	-0.8	J.P. Morgan EMBI Global Core Index
EM	EMLC	\$5.07B	0	23.9%	201	1.01	25.27	-0.58%	2.0	0.4	-0.8	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
CLOs	JAAA	\$30.94B	78	27.2%	611	7.08	50.65	0.02%	3.3	1.1	0.3	
	PAAA	\$13.09B	334	110.8%	255	--	51.45	0.02%	3.5	1.1	0.3	
	JBBB	\$1.52B	0	27.8%	32	--	47.45	-0.29%	3.5	1.2	0.2	
	CLOI	\$1.65B	106	24.4%	31	12.05	52.91	0.06%	3.5	1.2	0.3	
	CLOA	\$2.28B	0	65.5%	44	--	51.91	0.01%	3.5	1.2	0.3	

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

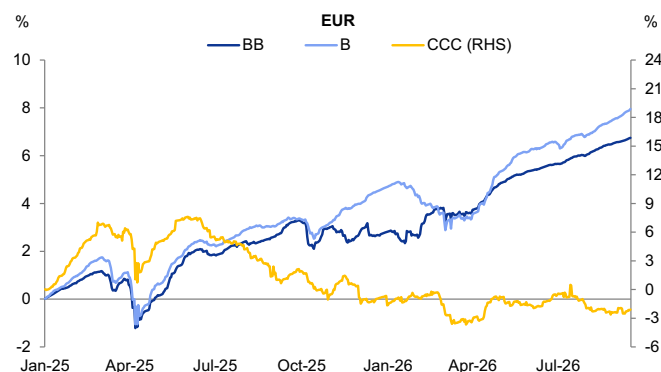
Performance dashboard: Leveraged Loans

Exhibit 27: Cumulative total return by rating for USD Leveraged Loans



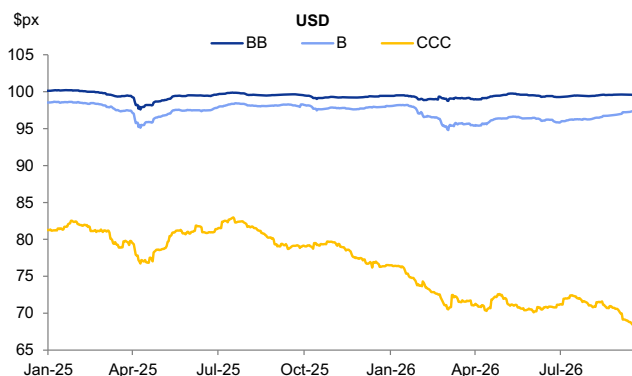
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 28: Cumulative total return by rating for EUR Leveraged Loans



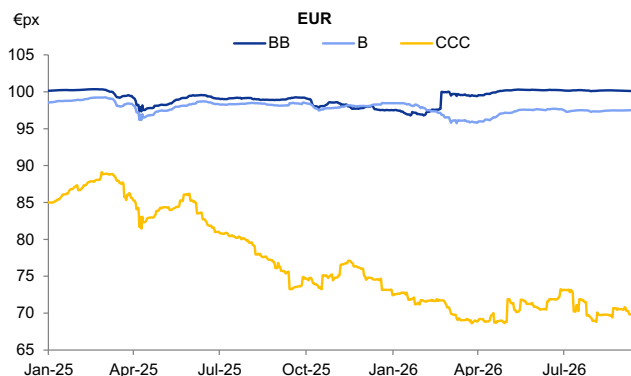
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 29: Bid price for USD Leveraged Loans by rating



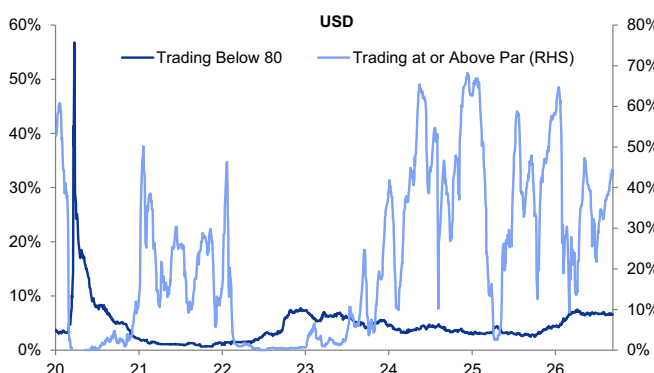
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 30: Bid price for EUR Leveraged Loans by rating



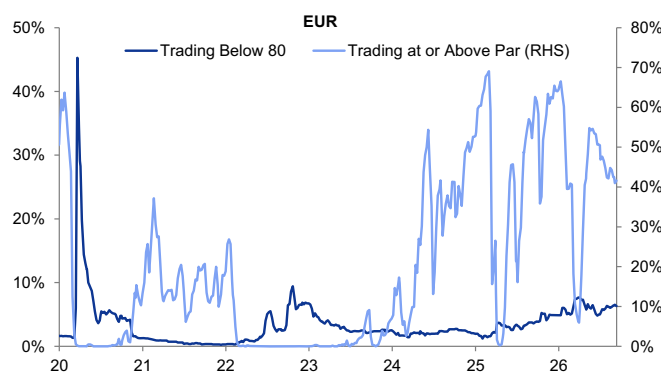
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 31: Price breakdown for USD Leveraged Loans



Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 32: Price breakdown for EUR Leveraged Loans

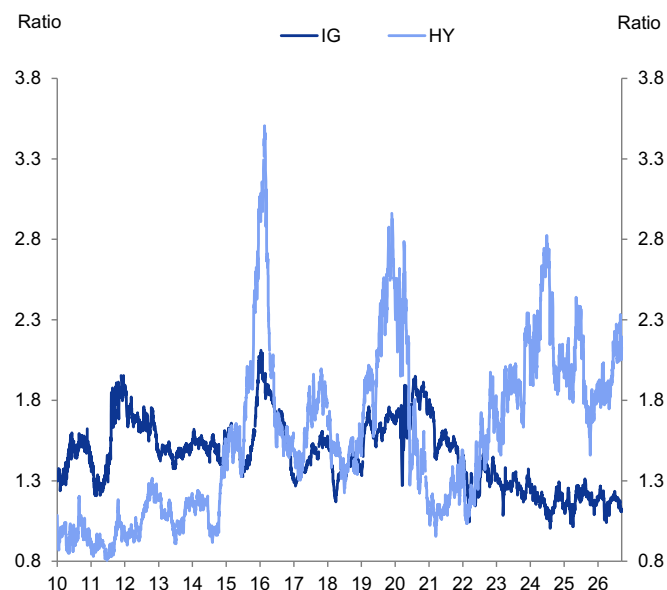


Source: PitchBook, Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

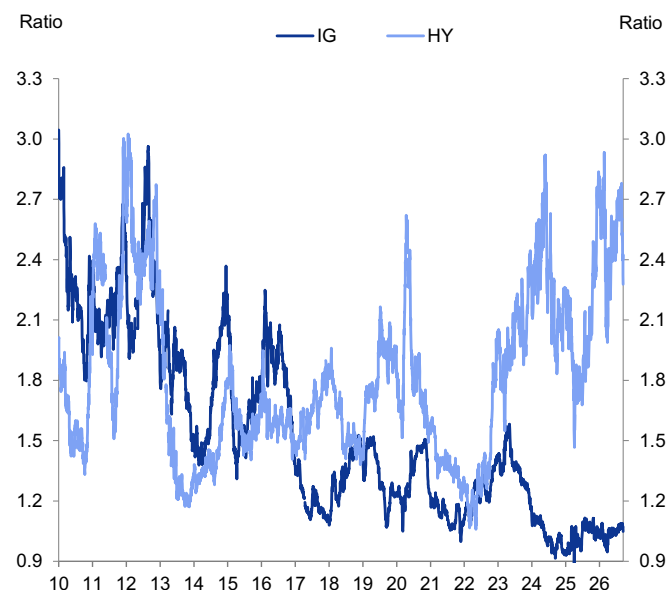
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 33: USD spread dispersion



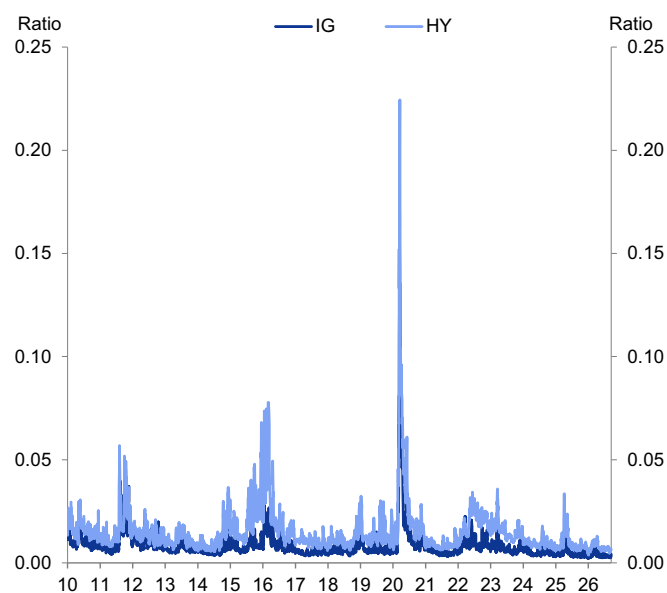
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 34: EUR spread dispersion



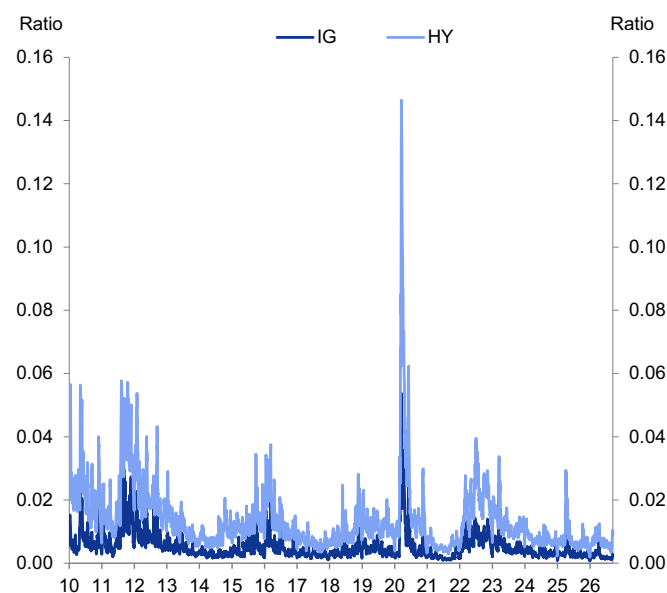
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 35: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 36: EUR return dispersion

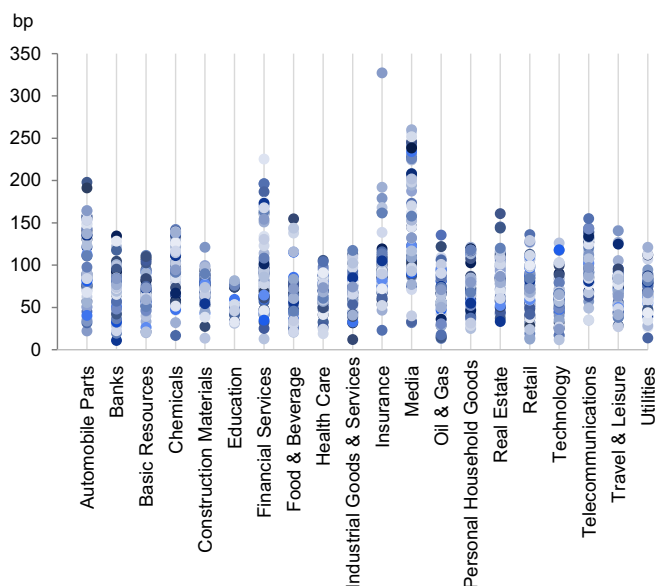


Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Sector dispersion

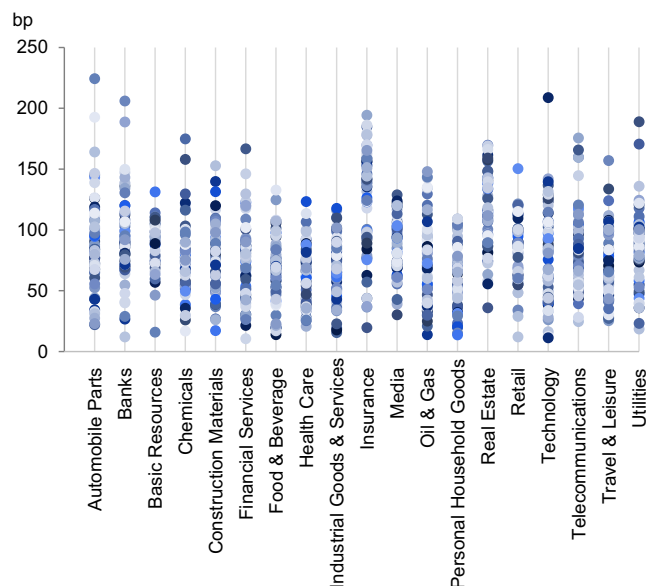
Issuer-level spreads are calculated using a market-value weighted average across the capital structure.

Exhibit 37: Distribution of issuer-level spreads by sector in the USD IG market



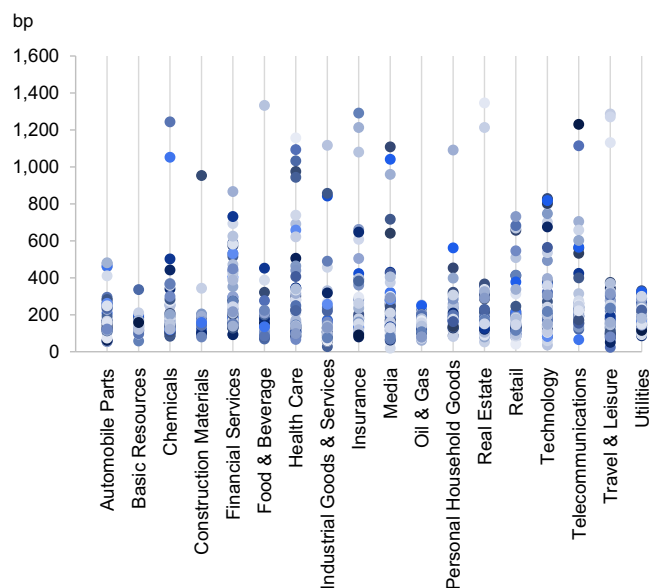
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 38: Distribution of issuer-level bond spreads by sector in the EUR IG market



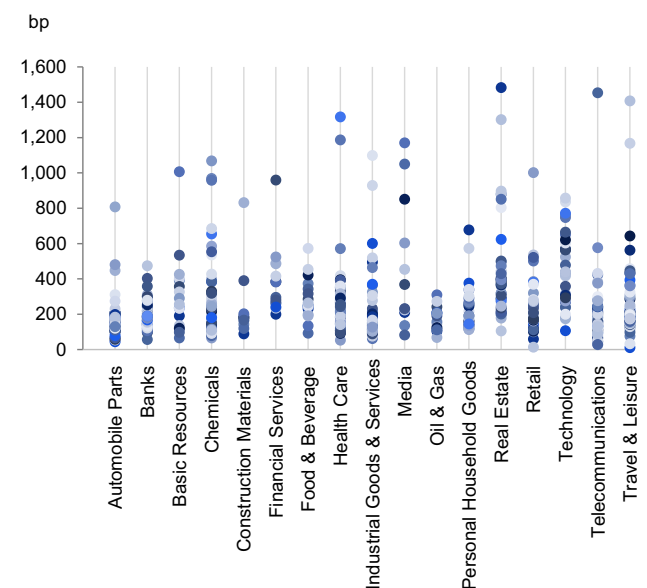
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 39: Distribution of issuer-level bond spreads by sector in the USD HY market



Source: Goldman Sachs Global Investment Research

Exhibit 40: Distribution of issuer-level bond spreads by sector in the EUR HY market



Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 41: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Oracle Corp	Software	ORCL 4.1 03/25/61	1,500	Baa2/BBB-/BBB	4.1%	2.4	60.05
Campbells Company/The	Food	CPB 5 1/4 10/13/54	350	Baa2/BBB-/BBB-	3.2%	2.5	78.82
Meta Platforms Inc	Internet	META 6.3 05/15/56	6,000	Aa3/AA-/--	2.9%	2.6	90.90
Jpmorgan Chase & Co	Banks	JPM 4.95 06/01/45	1,750	A2/A-/A	2.7%	2.4	86.76
Broadcom Inc	Semiconductors	AVGO 3 1/2 02/15/41	2,985	A3/A-/A-	2.3%	1.7	74.03
Jpmorgan Chase & Co	Banks	JPM 6.4 05/15/38	2,500	A1/A/AA-	2.1%	2.2	105.14
Microsoft Corp	Software	MSFT 4.1 02/06/37	851	Aaa/AAA/--	2.0%	1.8	89.91
Texas Instruments Inc	Semiconductors	TXN 4.85 02/08/34	600	Aa3/A+/-	2.0%	1.9	96.84
Prudential Financial Inc	Insurance	PRU 6 09/01/52	1,200	Baa1/BBB+/BBB	1.9%	1.9	99.24
National Grid Plc	Electric	NGGLN 5.418 01/11/34	750	Baa2/BBB/BBB+	1.9%	1.9	98.19
Dell Int Llc / Emc Corp	Computers	DELL 5.4 04/15/34	1,000	Baa2/BBB+/BBB+	1.8%	1.8	97.83
Anheuser-Busch Inbev Wor	Beverages	ABIBB 5.45 01/23/39	2,000	A2/A-/--	1.8%	1.7	96.86
Dominion Energy Inc	Electric	D 6 5/8 05/15/55	1,250	Baa3/BBB-/BBB-	1.7%	1.7	99.95
Meta Platforms Inc	Internet	META 5 3/4 05/15/63	1,750	Aa3/AA-/--	1.7%	1.4	81.58
Arcelormittal Sa	Iron/Steel	MTNA 5 3/8 05/19/36	1,000	Baa2/BBB/-	1.7%	1.6	95.70
At&T Inc	Telecommunications	T 3 1/2 09/15/53	7,496	Baa2/BBB/BBB+	1.7%	1.0	60.11
Royal Bank Of Canada	Banks	RY 4.305 11/03/31	1,000	A1/A/AA-	1.5%	1.5	96.03
Oracle Corp	Software	ORCL 5 3/8 09/27/54	1,750	Baa2/BBB-/BBB	1.5%	1.1	72.47
Charter Comm Opt Llc/Cap	Media	CHTR 3 1/2 06/01/41	1,005	Ba1/BBB-/BBB-	1.5%	1.0	65.59
Amazon.Com Inc	Internet	AMZN 2 1/2 06/03/50	2,500	A1/AA/AA-	1.5%	0.8	53.19
Bottom 20 performers							
Verizon Communications	Telecommunications	VZ 6 3/4 02/15/57	1,750	Baa2/BBB-/BBB	-2.7%	-2.8	101.25
Salesforce Inc	Software	CRM 6.4 03/15/46	1,500	A2/A+/-	-2.3%	-2.3	99.75
Hp Enterprise Co	Computers	HPE 6.35 10/15/45	1,492	Baa2/BBB/BBB+	-2.1%	-2.1	100.22
Csx Corp	Transportation	CSX 5 1/2 04/15/41	550	A3/BBB+/A-	-2.0%	-2.0	98.06
Athene Holding Ltd	Insurance	ATH 6.15 08/15/36	1,000	Baa1/A-/BBB+	-2.0%	-2.0	99.31
Athene Holding Ltd	Insurance	ATH 6 5/8 05/19/55	1,000	Baa1/A-/BBB+	-2.0%	-1.9	94.55
Merck & Co Inc	Pharmaceuticals	MRK 5.2 05/22/36	1,500	Aa3/A+/-	-1.9%	-1.9	98.82
Dell Int Llc / Emc Corp	Computers	DELL 8.35 07/15/46	646	Baa2/BBB+/BBB+	-1.8%	-2.2	121.81
Lincoln National Corp	Insurance	LNC 7 06/15/40	500	Baa2/BBB+/BBB+	-1.7%	-1.8	106.22
Blackstone Private Cre	Investment Companies	BCRED 6 01/29/32	1,000	Baa2/BBB/-	-1.6%	-1.6	98.24
Vale Overseas Limited	Iron/Steel	VALEBZ 3 3/4 07/08/30	1,500	--/-/-	-1.5%	-1.5	95.68
Eli Lilly & Co	Pharmaceuticals	LLY 4.55 10/15/32	1,000	Aa3/AA-/--	-1.5%	-1.5	97.58
Conagra Brands Inc	Food	CAG 4.85 11/01/28	1,300	Baa3/BBB-/BBB-	-1.4%	-1.4	99.92
Santander Holdings Usa	Banks	SANUSA 5.22 06/05/32	750	Baa2/BBB+/A-	-1.4%	-1.4	98.59
Astrazeneca Finance Llc	Pharmaceuticals	AZN 4.9 02/26/31	1,000	A1/A+/-	-1.4%	-1.4	100.34
Lockheed Martin Corp	Aerospace/Defense	LMT 3.8 03/01/45	1,000	A2/A-/A	-1.4%	-1.1	75.64
Ibm Corp	Computers	IBM 7 1/8 12/01/2096	316	A3/A-/A-	-1.4%	-1.5	108.43
General Motors Co	Auto Manufacturers	GM 6.8 10/01/27	1,000	Baa2/BBB/BBB	-1.4%	-1.4	101.82
Hsbc Holdings Plc	Banks	HSBC 5.546 03/04/30	1,500	A3/A-/A+	-1.3%	-1.3	100.52
Cheniere Energy Partners	Pipelines	CQP 4 1/2 10/01/29	1,500	Baa2/BBB+/BBB	-1.3%	-1.3	99.06

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 42: Top and bottom 20 performers in the USD HY market

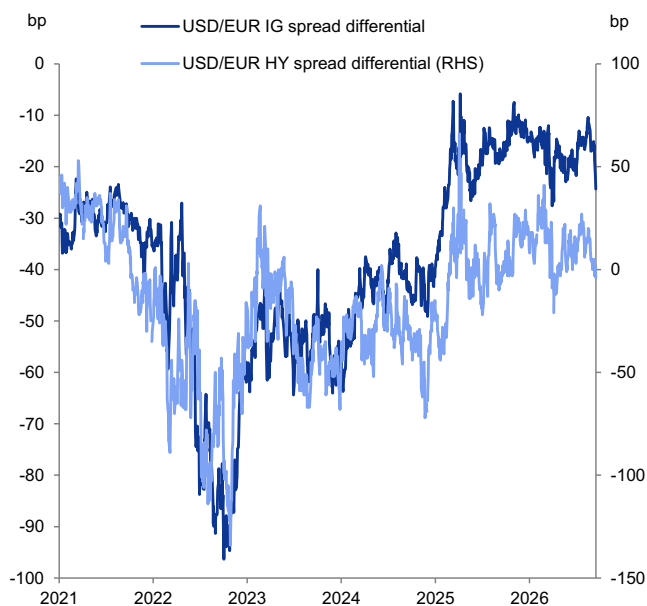
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Angi Group Llc	Internet	ANGI 3 7/8 08/15/28	400	B2/B/--	9.6%	7.5	78.25
Xerox Corporation	Office/Business Equip	XRX 6 3/4 12/15/39	350	Ca/CCC/--	6.8%	2.4	35.23
Whirlpool Corp	Home Furnishings	WHR 4 1/2 06/01/46	500	B2/B+/BB-	2.5%	1.3	54.00
Csc Holdings Llc	Media	CSCHLD 6 1/2 02/01/29	1,750	Caa2/CCC/--	2.3%	1.1	48.25
Baldwin Insur Grp/Fin	Insurance	BWIN 7 1/8 05/15/31	600	B2/B/--	2.2%	2.3	101.75
Sv Rno Property Owner 1	Telecommunications	TRACTC 5 7/8 03/01/31	3,800	Ba1/BB+/BB	2.1%	1.9	90.88
Csc Holdings Llc	Media	CSCHLD 4 1/2 11/15/31	1,500	Caa2/CCC/--	2.0%	1.0	47.53
Getty Images Inc	Internet	ABEGET 11 1/4 02/21/30	540	Caa2/CCC+/-	2.0%	1.5	74.00
Michaels Cos Inc/The	Retail	MIK 11 03/15/34	750	Caa1/CCC/--	1.7%	1.6	96.00
Pr Rno Property Owner 1	Telecommunications	TRACTD 6 1/2 05/01/31	4,585	Ba1/BB/BB	1.6%	1.5	91.88
Csc Holdings Llc	Media	CSCHLD 11 3/4 01/31/29	2,050	Caa2/CCC/--	1.6%	0.8	49.48
Csc Holdings Llc	Media	CSCHLD 4 1/8 12/01/30	1,100	Caa2/CCC/--	1.5%	0.7	48.03
Icahn Enterprises/Fin	Investment Companies	IEP 10 11/15/29	1,000	B1/B+/-	1.4%	1.4	96.89
Icahn Enterprises/Fin	Investment Companies	IEP 9 06/15/30	749	B1/B+/-	1.4%	1.3	90.50
Clydesdale Acquisition	Packaging&Containers	NOVHOL 6 3/4 04/15/32	1,415	B2/B+/BB+	1.3%	1.2	95.39
Altice France Sa	Telecommunications	SFRFP 6 7/8 07/15/32	1,524	Caa1/CCC+/-	1.2%	1.2	96.01
Alta Equipment Group	Commercial Services	ALTG 9 06/01/29	500	Caa1/B+/-	1.2%	1.1	95.13
Icahn Enterprises/Fin	Investment Companies	IEP 9 3/4 01/15/29	696	B1/B+/-	1.1%	1.1	94.88
Staples Inc	Retail	SPLS 10 3/4 09/01/29	2,375	B3/B+/-	1.0%	1.0	96.50
Brand Industrial Service	Engineering&Construction	BRANDI 10 3/8 08/01/30	1,435	Caa1/CCC+/-	1.0%	0.8	84.76
Bottom 20 performers							
Embarq Llc	Telecommunications	EMBARQ 7.995 06/01/36	1,215	Ca/CCC/--	-17.9%	-3.0	16.50
Dye & Durham Limited	Software	DNDN 8 5/8 04/15/29	509	Caa1/CCC+/-	-9.0%	-5.3	58.29
Cogent Comms Grp / Fin	Internet	CCOI 6 1/2 07/01/32	600	Ba2/B+/-	-5.7%	-4.3	74.75
Avianca Midco 2 Plc	Airlines	AVIAGP 9 5/8 02/14/30	1,000	B1/-/B+	-4.5%	-4.0	89.03
Merlin Entertainments Gr	Entertainment	MERLLN 7 3/8 02/15/31	500	Caa2/B+/-	-4.4%	-3.3	74.25
Connect Holding li Llc	Telecommunications	BSPEED 10 1/2 04/03/31	2,225	B2/CCC+/-	-4.1%	-3.6	88.25
Csc Holdings Llc	Media	CSCHLD 5 3/8 02/01/28	1,000	Caa2/CCC/--	-3.9%	-2.0	53.04
Coronado Finance Pty Ltd	Coal	CRNAU 9 1/4 10/01/29	400	Caa2/B+/-	-3.8%	-3.3	86.85
Whirlpool Corp	Home Furnishings	WHR 5 3/4 03/01/34	300	B2/B+/BB-	-3.5%	-2.6	75.74
Tronox Inc	Chemicals	TROX 4 5/8 03/15/29	1,075	Caa2/CCC+/-	-3.2%	-2.0	64.04
Motion Finco Sarl	Entertainment	MERLLN 8 3/8 02/15/32	410	Caa2/B+/-	-3.1%	-2.3	73.00
Lcpr Sr Secured Fin Dac	Media	LILAPR 5 1/8 07/15/29	820	Caa3/CCC+/CCC-	-3.1%	-1.8	57.00
Virgin Media Finance Plc	Media	VMED 5 07/15/30	925	B3/B-/B-	-2.8%	-1.8	65.04
Vmed O2 Uk Finan	Telecommunications	VMED 4 3/4 07/15/31	1,400	Ba3/B+/BB	-2.8%	-2.3	81.75
Iheartcommunications Inc	Media	IHRT 7 3/4 08/15/30	661	Caa1/CCC+/-	-2.8%	-2.3	81.75
Vmed O2 Uk Finan	Telecommunications	VMED 4 1/4 01/31/31	1,350	Ba3/B+/BB	-2.5%	-2.1	81.57
Asurion Llc/Asurion Co	Insurance	ASUCOR 8 3/8 02/01/34	3,300	B3/B+/-	-2.5%	-2.2	88.87
Acrisure Llc / Fin Inc	Insurance	ACRISU 8 1/4 02/01/29	925	Caa2/CCC+/-	-2.5%	-2.1	85.50
Fmc Corp	Chemicals	FMC 6 3/8 05/18/53	500	Ba2/BB+/BB+	-2.4%	-1.9	78.89
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 5/8 04/30/33	450	B2/B+/BB-	-2.1%	-1.8	83.30

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

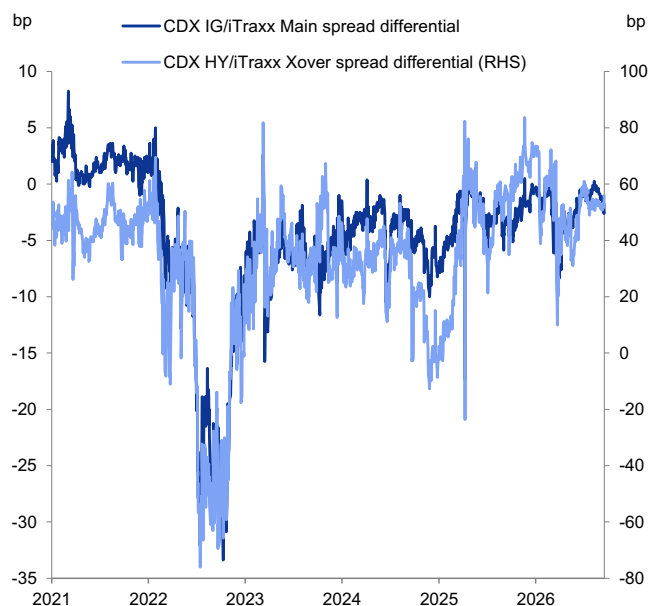
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 43: USD/EUR IG and HY spread differential



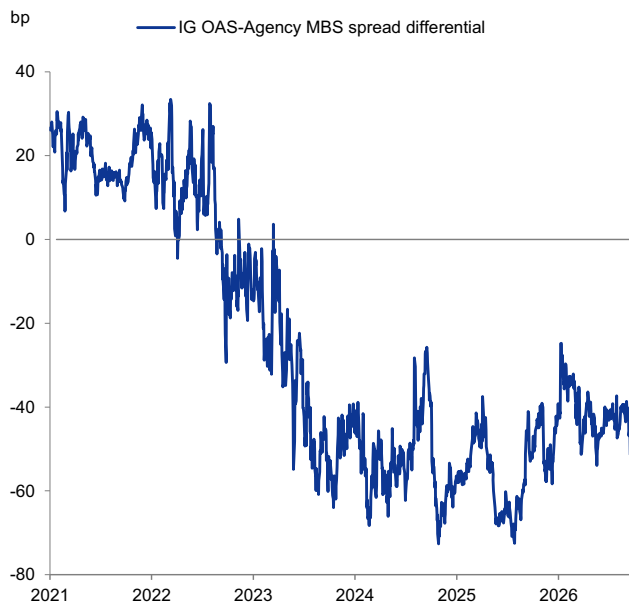
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 44: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



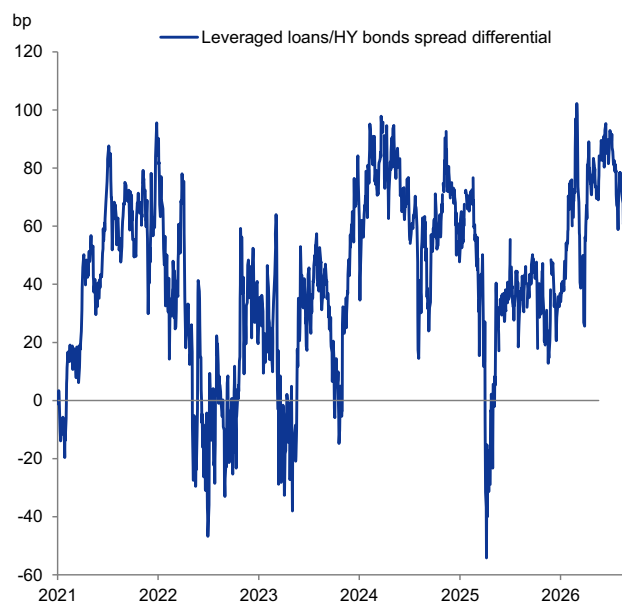
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 45: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

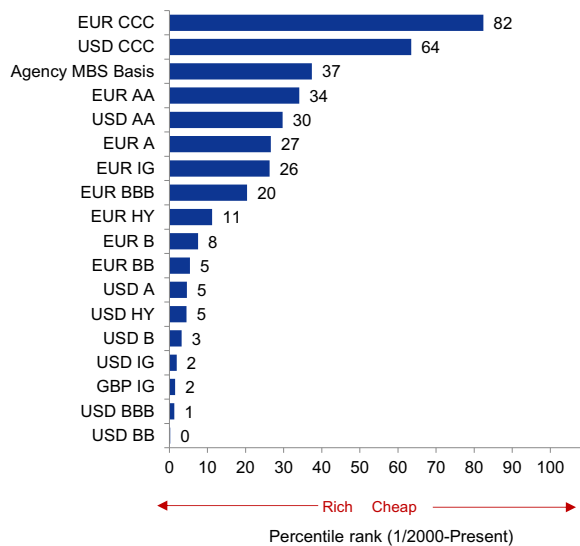
Exhibit 46: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

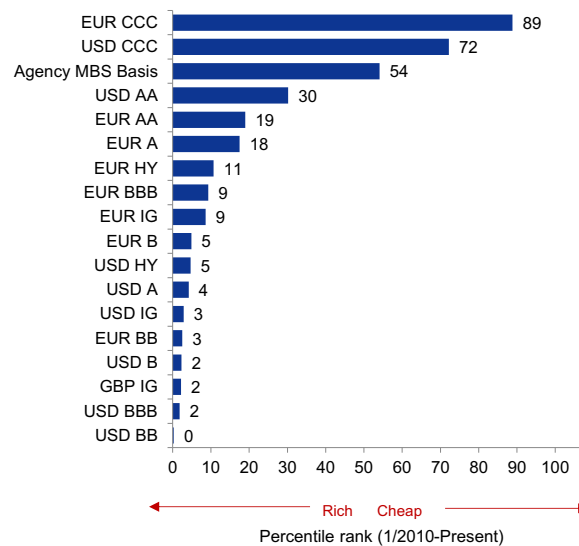
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 47: Percentile ranks for spreads across markets (2000-current)



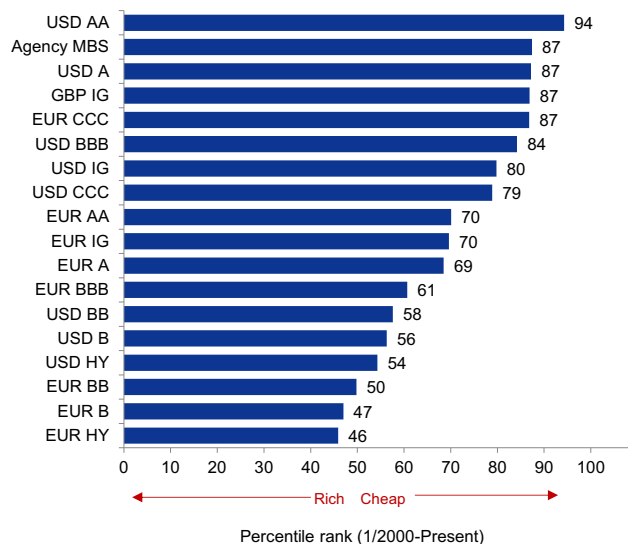
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 48: Percentile ranks for spreads across markets (2010-current)



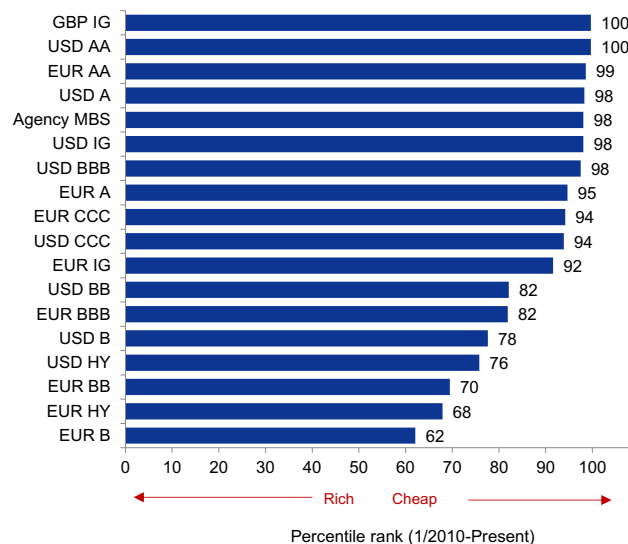
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 49: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

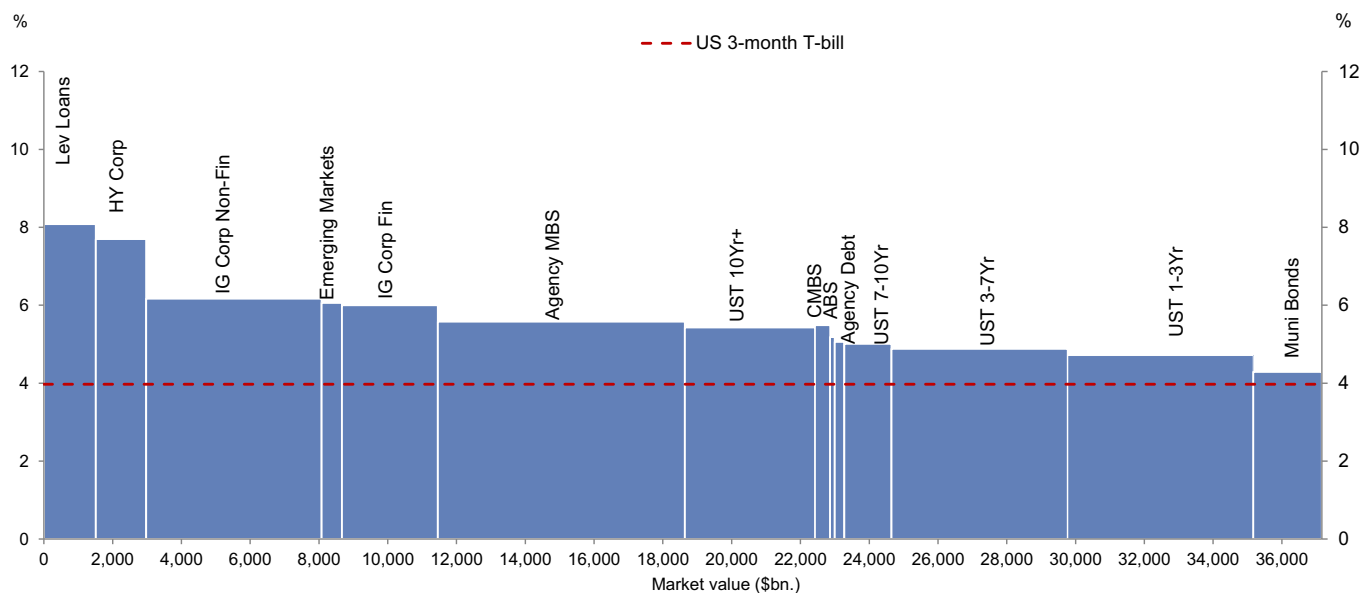
Exhibit 50: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

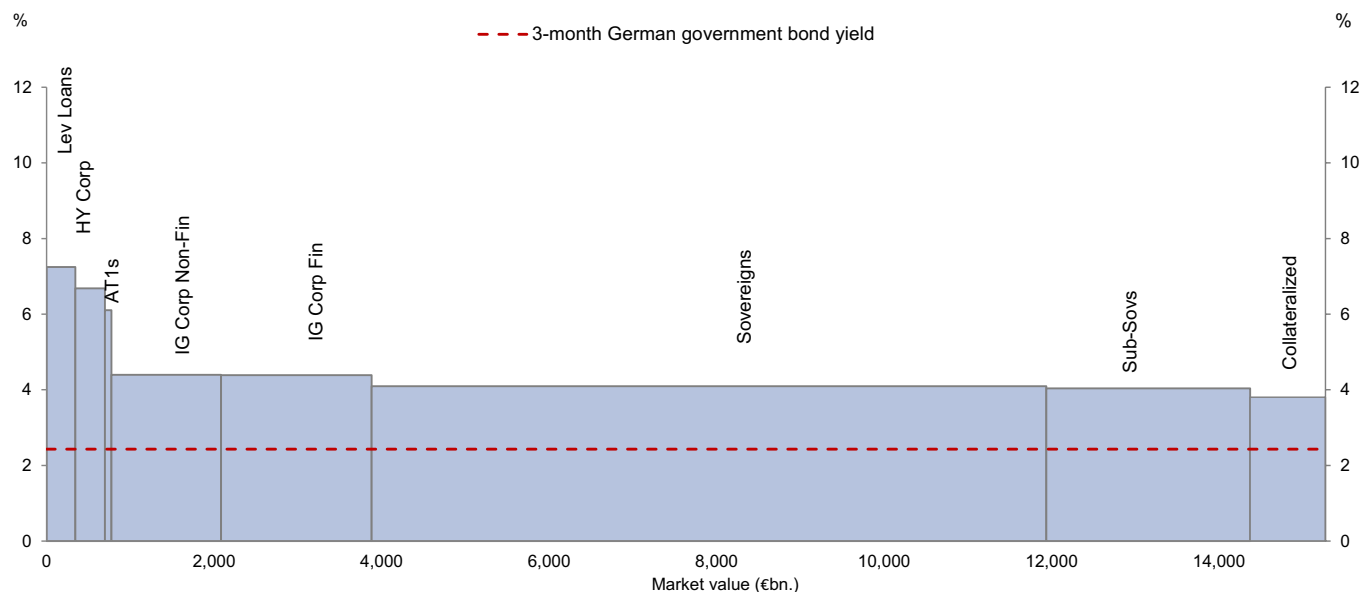
Exhibit 51: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 52: Average yield vs. total market value outstanding for the EUR fixed income markets

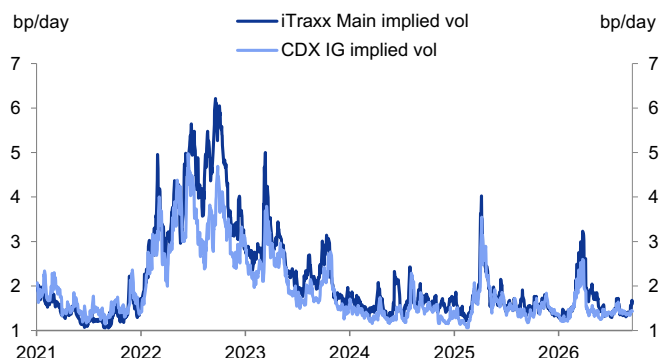


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

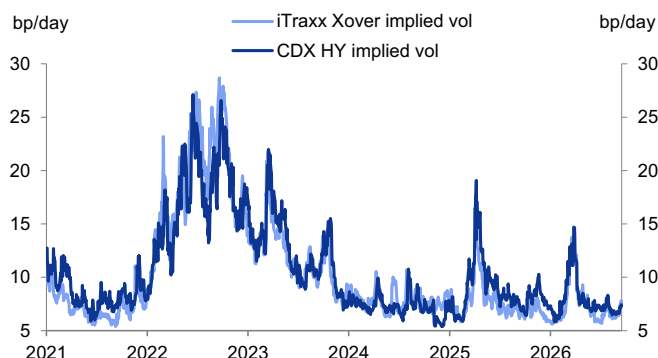
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 53: CDX IG and iTraxx Main vol 3m ATM implied volatility



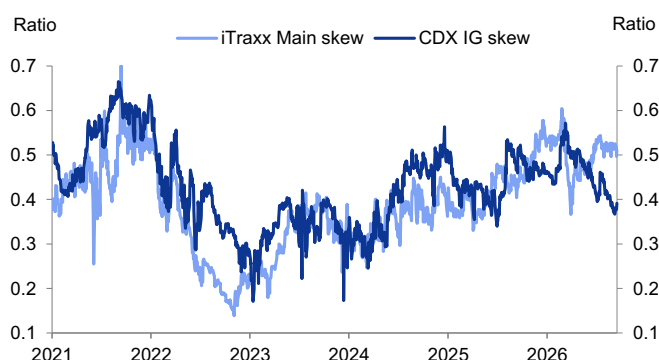
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 54: CDX HY and iTraxx Xover vol 3m ATM implied volatility



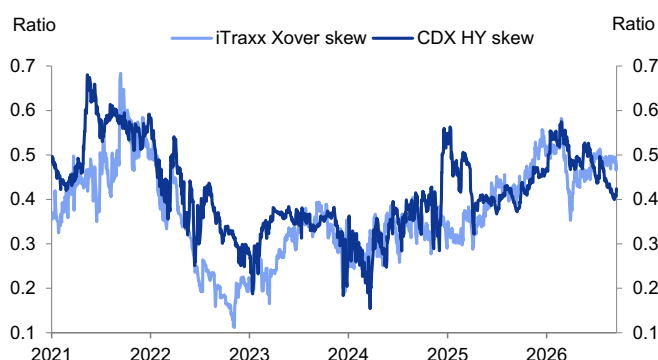
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 55: CDX IG and iTraxx Main skew



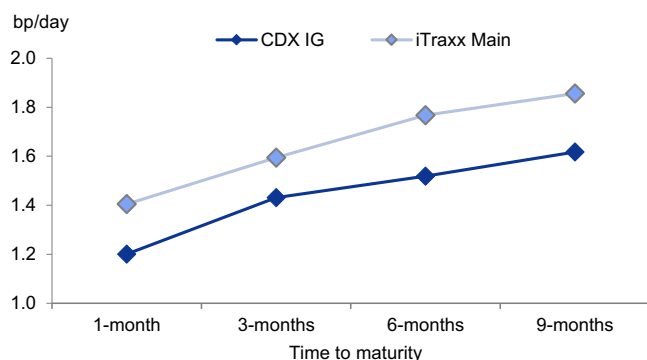
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 56: CDX HY and iTraxx Xover skew



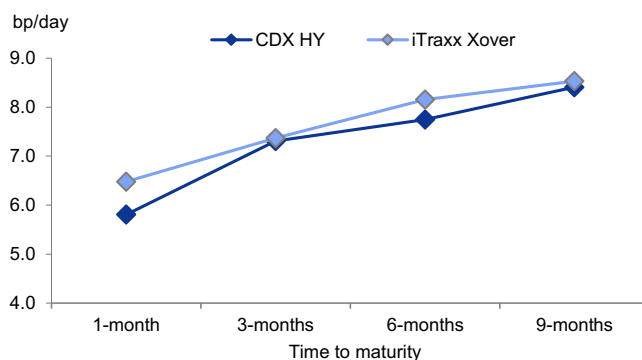
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 57: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 58: CDX HY and iTraxx Xover ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

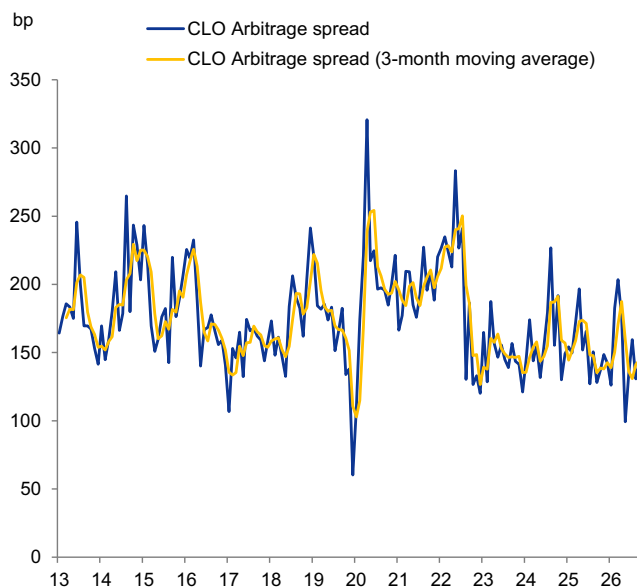
Exhibit 59: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	107	14%	8%	5%
AA	142	0%	0%	0%
A	165	3%	2%	1%
BBB	270	14%	8%	7%
BB	762	65%	54%	64%

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 61: CLO new issue arbitrage spread

New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



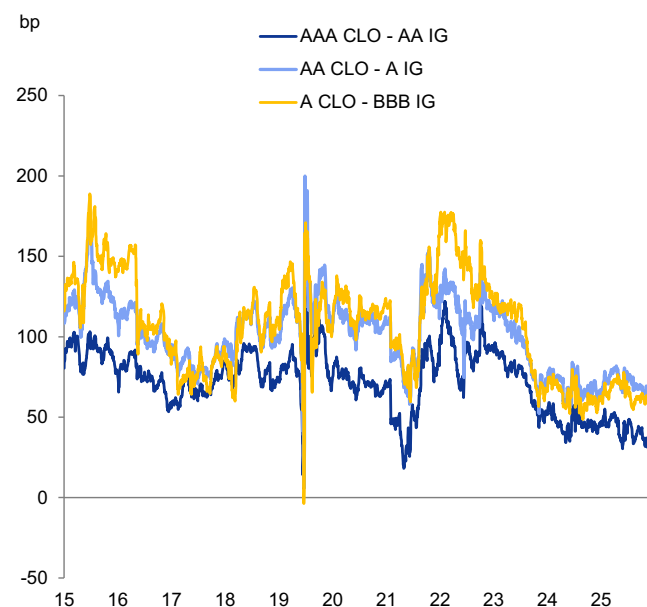
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 60: CLO equity vs. CLO BB yield

Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 62: CLO vs. USD IG spread differential

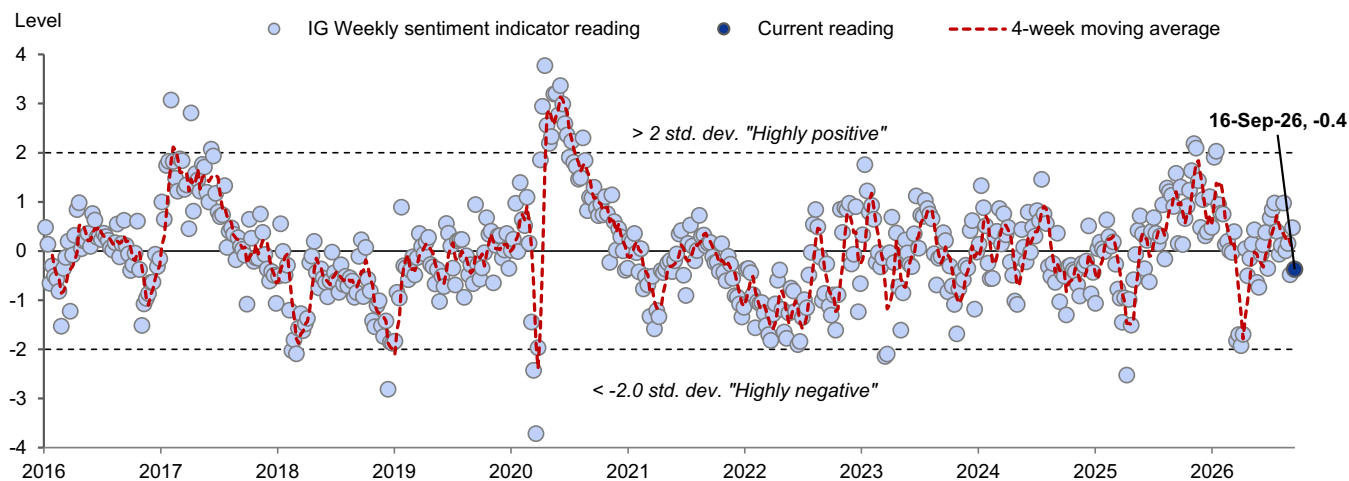
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

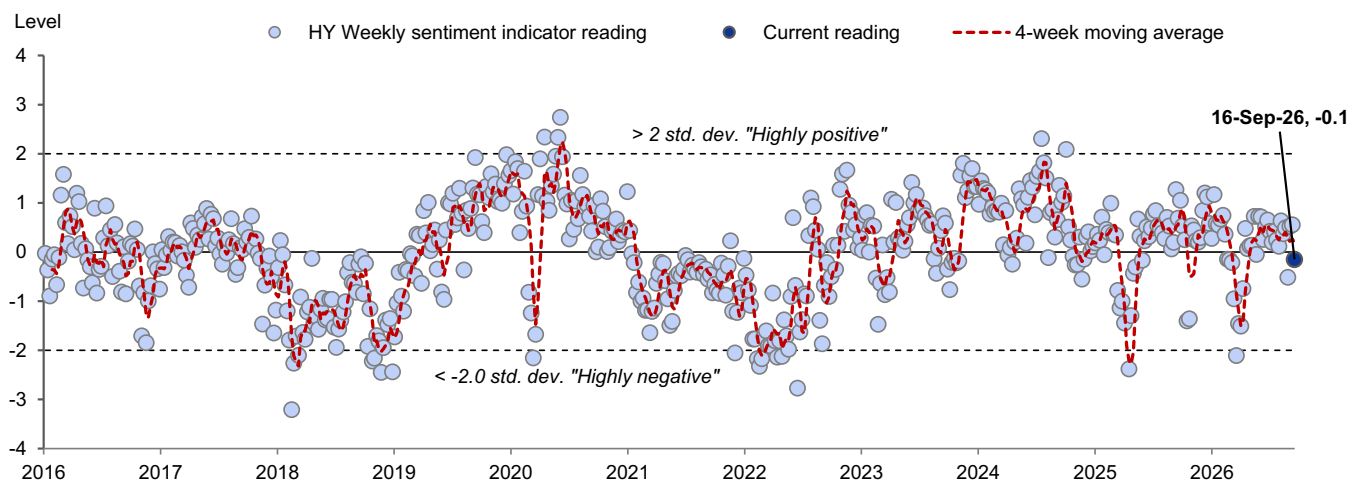
USD IG and HY Sentiment indicators

Exhibit 63: IG Weekly sentiment indicator



Source: See table below.

Exhibit 64: HY Weekly sentiment indicator



Source: See table below.

Exhibit 65: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 66: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
AMD	Advanced Micro Devices Inc	7.1	300.5	4.2	A	1.4	5.1
NVDA	NVIDIA Corp	32.5	916.9	2.8	AA	1.5	6.9
GOOGL	Alphabet Inc	73.8	2069.4	2.8	AA	1.1	9.3
PWR	Quanta Services Inc	6.2	134.0	2.2	BBB	2.3	5.5
CPB	The Campbell's Company	5.7	118.4	2.1	BBB-	4.7	6.0
CAG	Conagra Brands Inc	5.8	107.6	1.9	BBB-	5.7	5.5
AON	Aon North America Inc	11.6	209.1	1.8	BBB+	4.6	8.0
META	Meta Platforms Inc	81.3	1444.3	1.8	AA-	1.4	9.5
TEVA	Teva Pharmaceutical Finance Netherlands III BV	6.7	117.8	1.7	BBB-	8.2	6.2
NOW	ServiceNow Inc	5.5	93.9	1.7	A	1.9	5.9
MLM	Martin Marietta Materials Inc	10.4	174.5	1.7	BBB+	2.4	7.9
AMZN	Amazon.com Inc	117.0	1942.1	1.7	AA-	2.5	8.7
ORCL	Oracle Corp	117.2	1909.0	1.6	BBB	4.5	8.4
AVGO	Broadcom Inc	54.1	868.7	1.6	A-	1.7	6.2
ACN	Accenture Capital Inc	9.7	155.0	1.6	AA-	1.1	4.5
SUZANO	Suzano Austria GmbH	7.8	119.9	1.5	BBB-	6.1	5.0
INTU	Intuit Inc	5.5	83.1	1.5	A-	2.4	6.9
MAR	Marriott International Inc	15.1	227.3	1.5	BBB	2.9	5.2
HPE	Hewlett Packard Enterprise Co	14.9	210.2	1.4	BBB	2.8	5.3
AMXLM	America Movil SAB de CV	8.6	115.6	1.3	A-	11.1	7.7

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

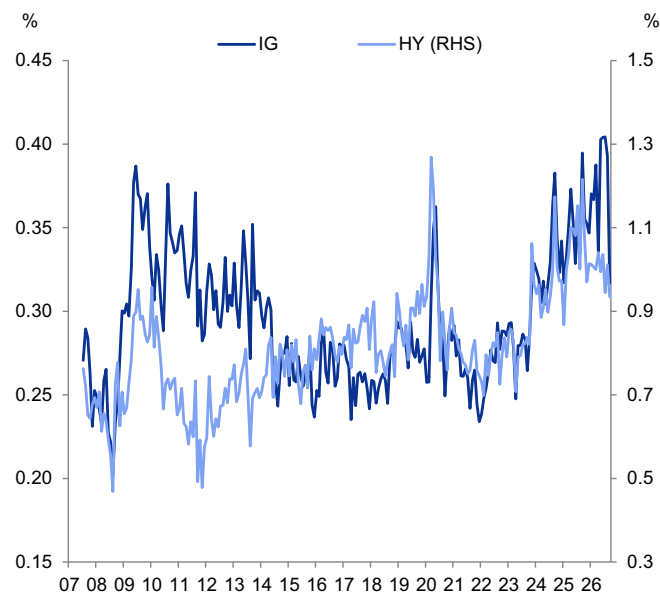
Exhibit 67: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	83.3	4.2	B	2.0	3.8
MIK	Michaels Cos Inc/The	2.8	75.7	2.8	B-	0.5	5.2
FUN	Six Flags Entertainment Corp/Canada's Wonderland Co/Millen	3.1	78.4	2.5	CCC+	2.7	4.0
SABHLD	Sabre GBL Inc	3.2	80.1	2.5	B-	1.1	2.6
IEP	Icahn Enterprises LP / Icahn Enterprises Finance Corp	3.1	69.3	2.2	B+	2.5	2.6
BURLN	Burford Capital Global Finance LLC	2.1	46.1	2.2	BB-	2.5	3.9
NOVHOL	Clydesdale Acquisition Holdings Inc	3.5	75.7	2.1	B+	2.9	3.6
CC	The Chemours Company	2.5	53.0	2.1	B	3.2	4.0
ACRISU	Acrisure LLC / Acrisure Finance Inc	4.3	88.6	2.1	B	3.0	2.9
ZENARC	Zenith Arc LLC	2.3	39.7	1.8	BB	0.0	4.0
WHR	Whirlpool Corp	3.9	68.4	1.8	BB-	1.8	4.3
BSPEED	Connect Holding II LLC	2.3	38.5	1.7	CCC+	1.2	3.5
ZIGGO	VZ Secured Financing BV	4.2	70.6	1.7	B+	4.4	4.2
NCLH	NCL Corp Ltd	5.0	83.2	1.7	B-	2.1	3.9
GALAXY	Galaxy Helios Data Centers II LLC	3.5	57.3	1.6	BB-	0.1	3.9
CRWV	CoreWeave Inc	7.8	120.3	1.6	B+	0.7	3.7
TRACTD	PR RNO Property Owner 1 LLC	4.6	70.6	1.5	BB	0.3	3.9
GT	The Goodyear Tire & Rubber Company	4.0	59.4	1.5	B+	3.1	4.0
NAVI	Navient Corp	3.3	47.4	1.5	BB-	3.2	2.6
AXL	American Axle & Manufacturing Inc	2.7	38.8	1.4	B+	1.8	4.7

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

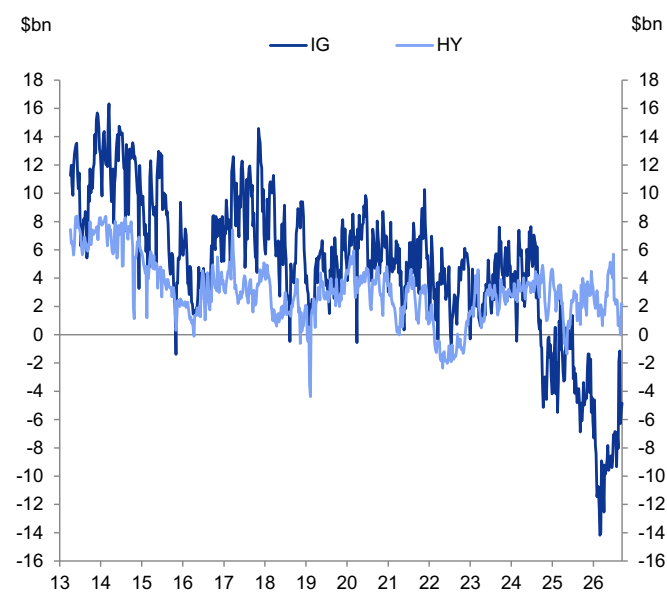
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 68: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



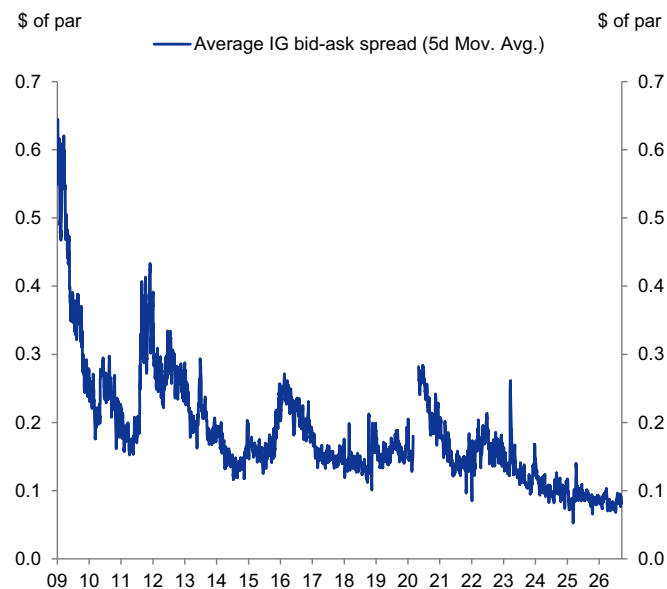
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 69: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

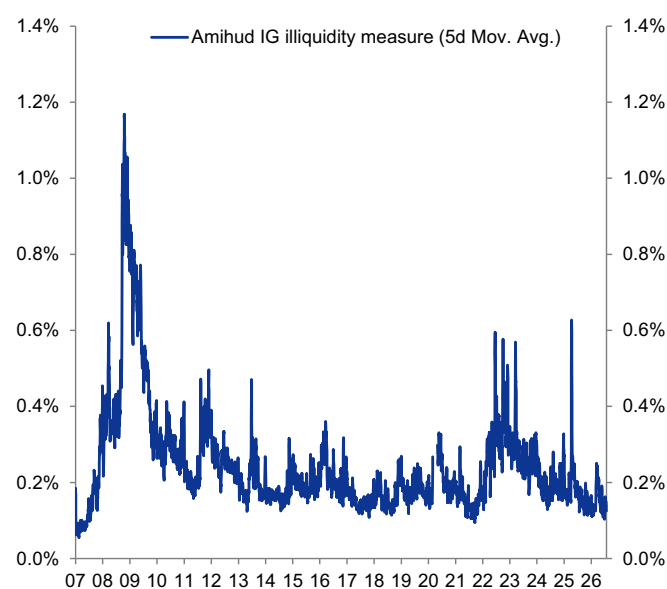
Exhibit 70: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 71: Amihud IG illiquidity measure 5-day moving average

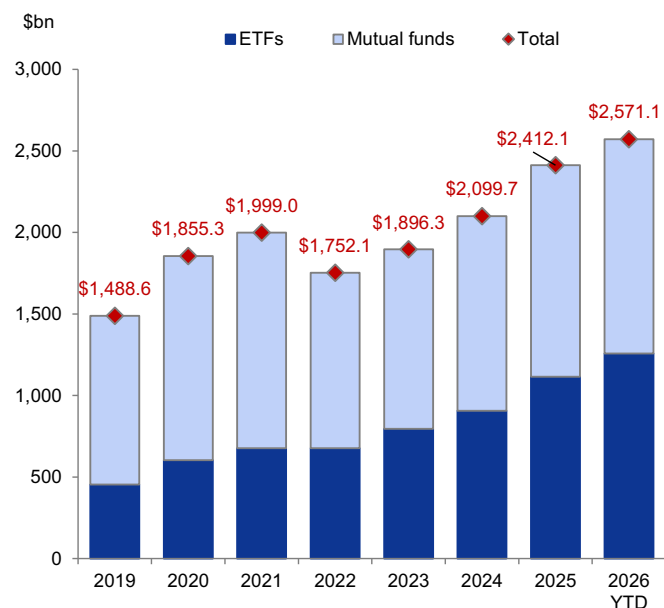


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

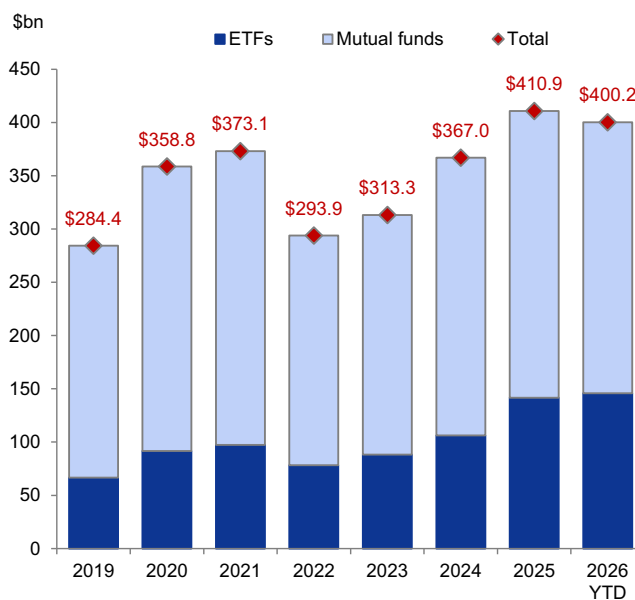
Flow dashboard: Mutual fund and ETF AUM

Exhibit 72: AUM in USD IG-focused Mutual Funds and ETFs



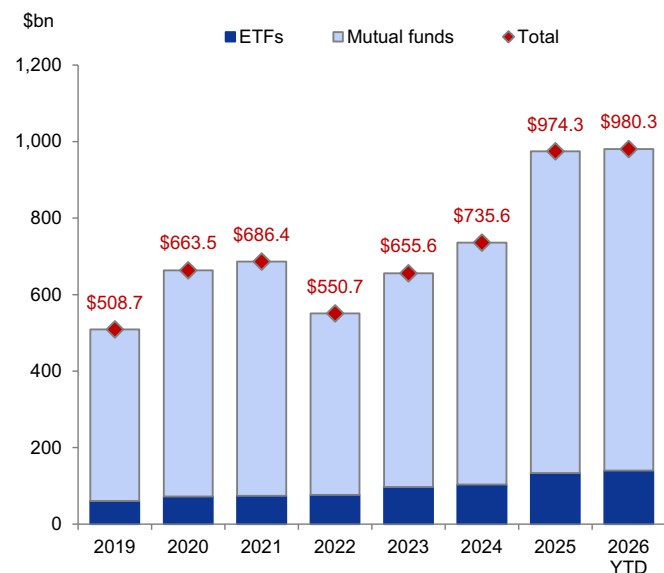
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 73: AUM in USD HY-focused Mutual Funds and ETFs



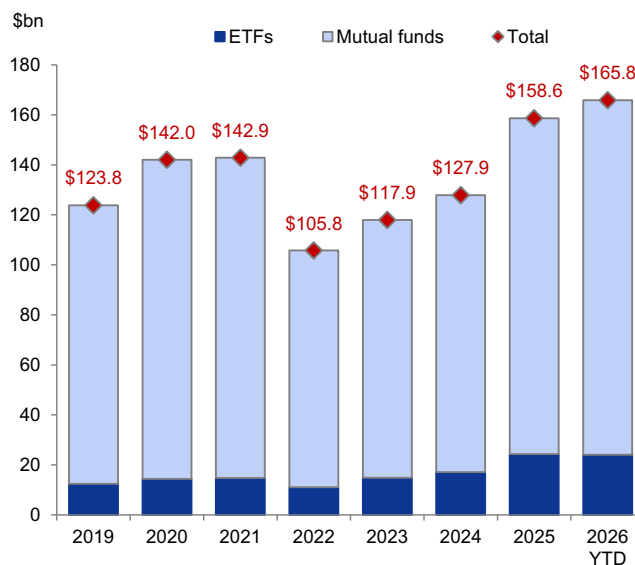
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 74: AUM in EUR IG-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

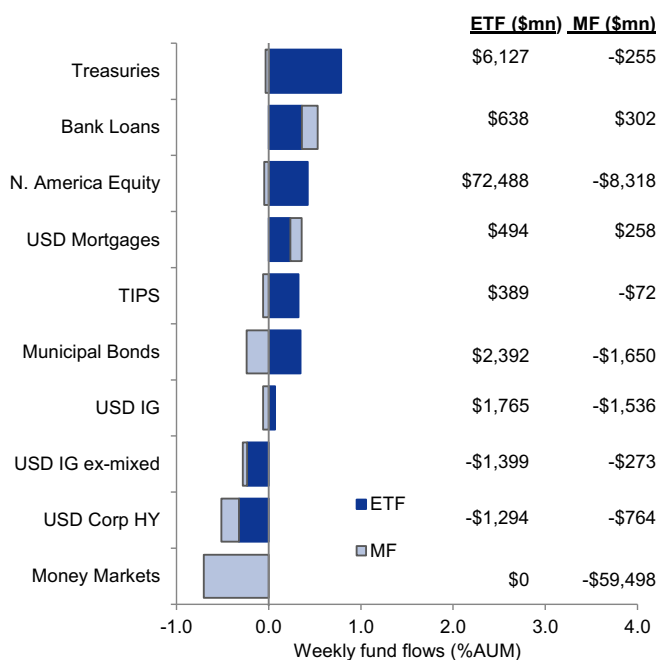
Exhibit 75: AUM in EUR HY-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

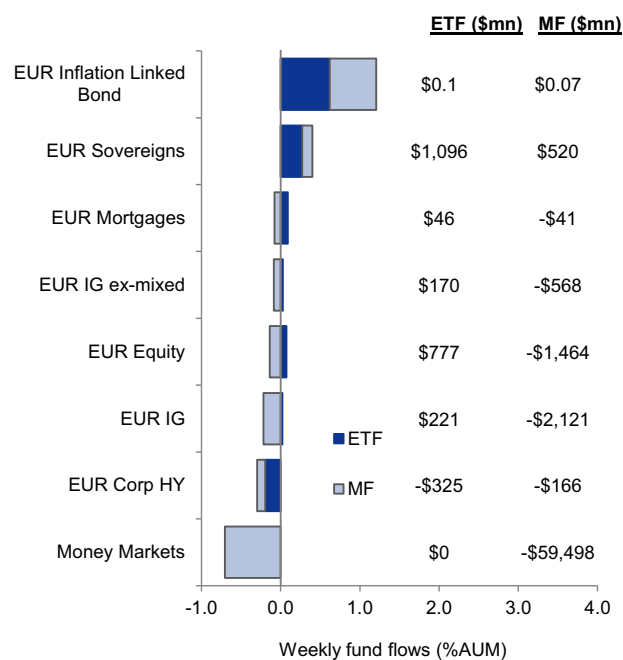
Flow dashboard: Mutual fund and ETF flows

Exhibit 76: Weekly mutual fund flows into US fixed income and equity markets



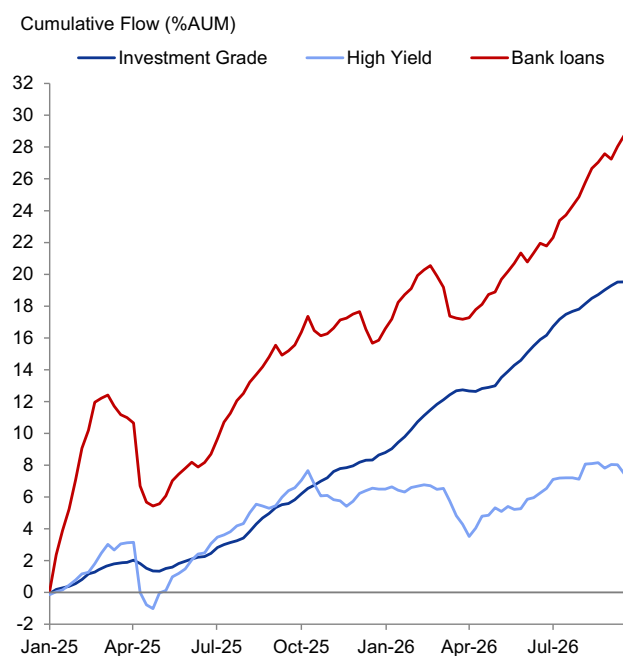
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 77: Weekly mutual fund flows into European fixed income and equity markets



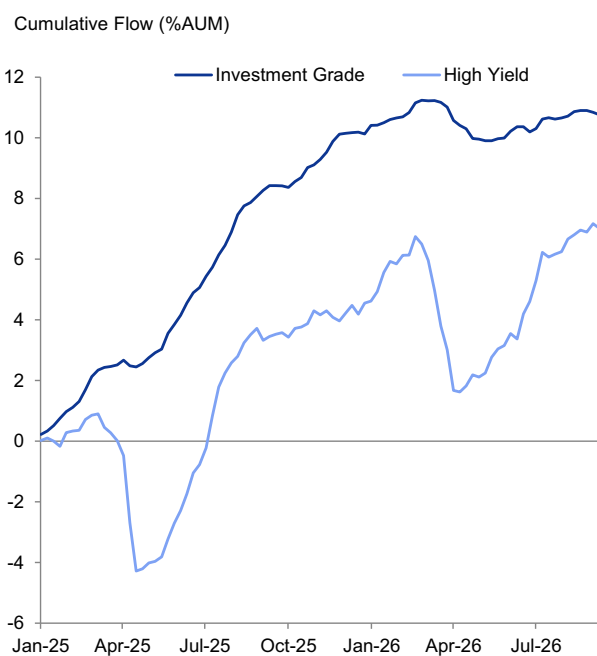
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 78: Cumulative flows as a percentage of beginning of the year AUM for US credit funds



Source: EPFR, Goldman Sachs Global Investment Research

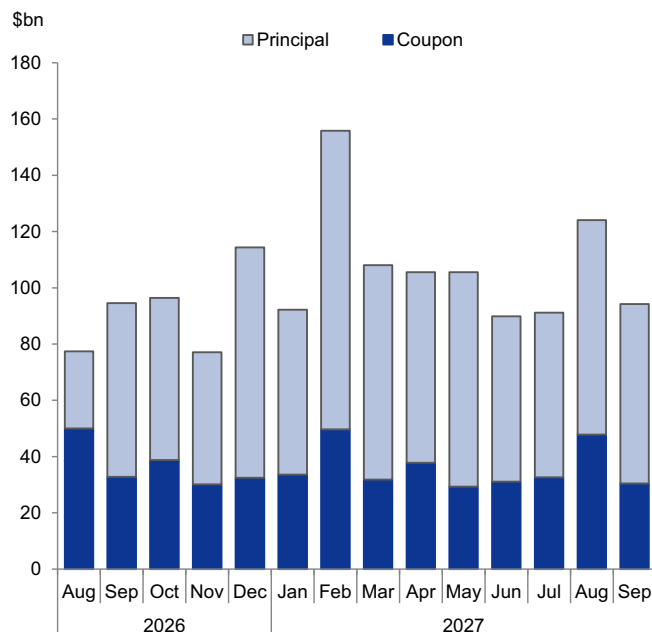
Exhibit 79: Cumulative flows as a percentage of beginning of the year AUM for European credit funds



Source: EPFR, Goldman Sachs Global Investment Research

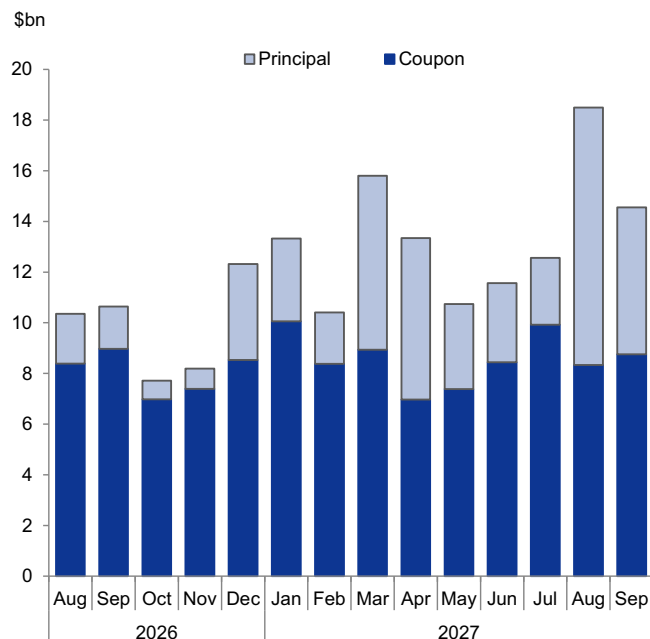
Flow dashboard: Upcoming coupon and principal payments

Exhibit 80: Our estimates of upcoming coupon and principal payments in the USD IG market



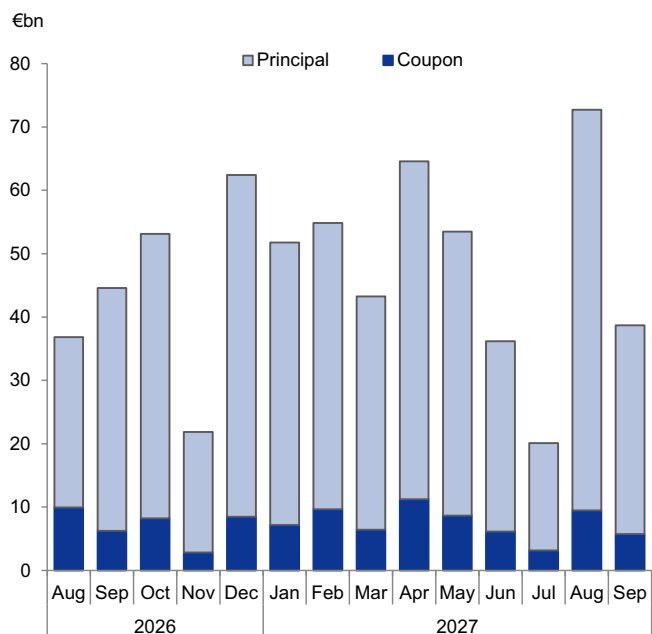
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 81: Our estimates of upcoming coupon and principal payments in the USD HY market



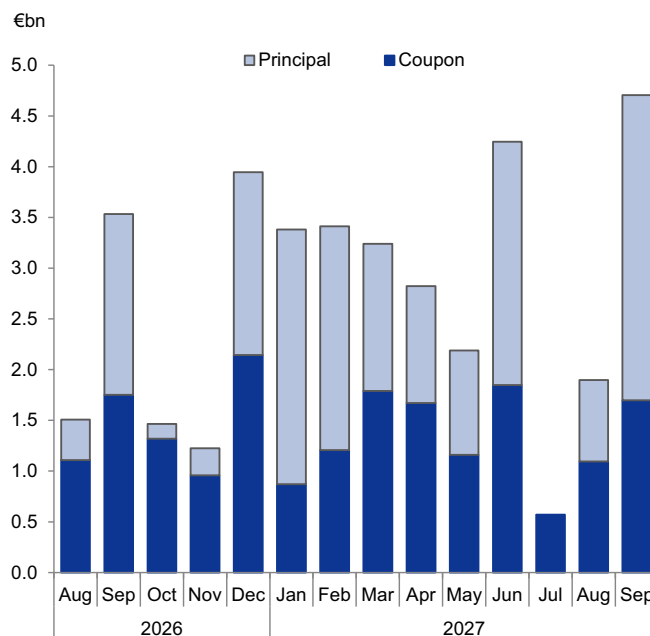
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 82: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

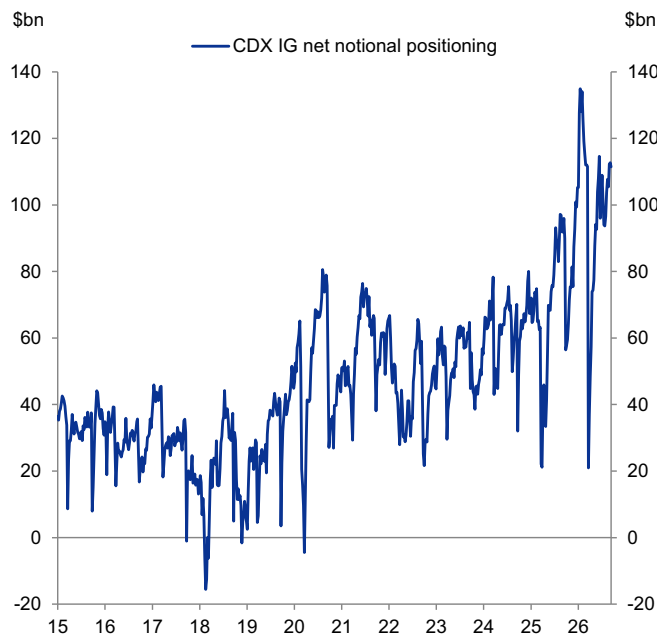
Exhibit 83: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 84: Net notional positioning in the CDX IG 5-year on-the-run contract



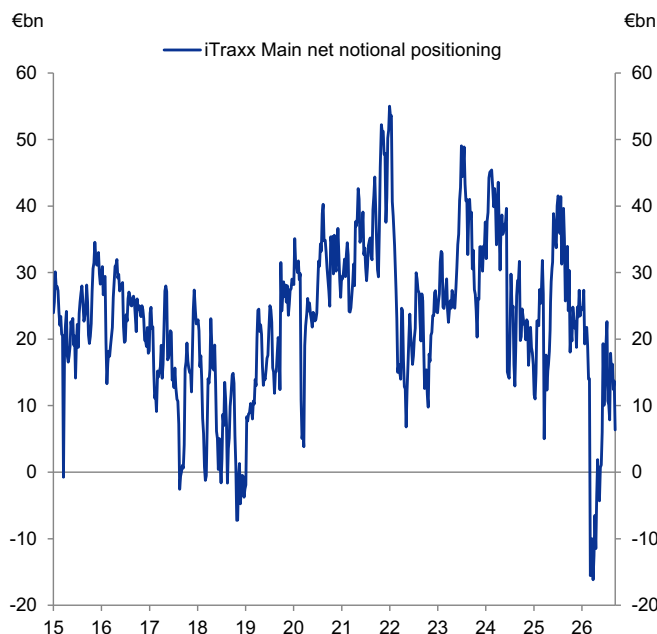
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 85: Net notional positioning in the CDX HY 5-year on-the-run contract



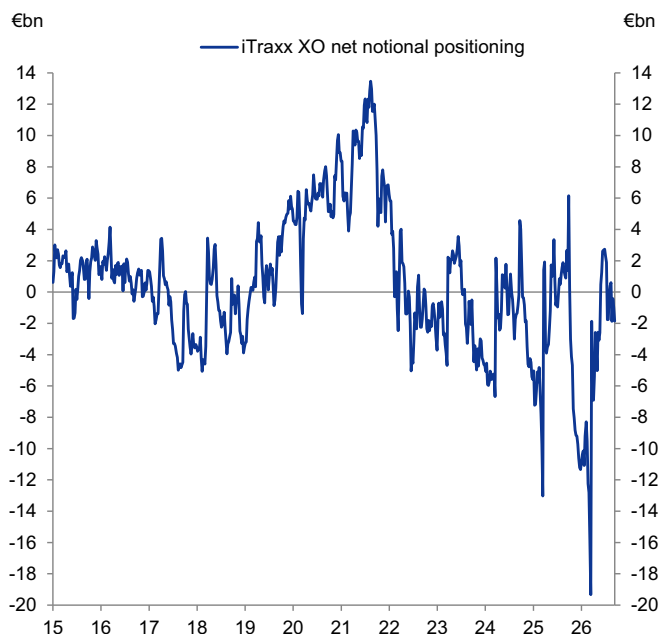
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 86: Net notional positioning in the iTraxx Main 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

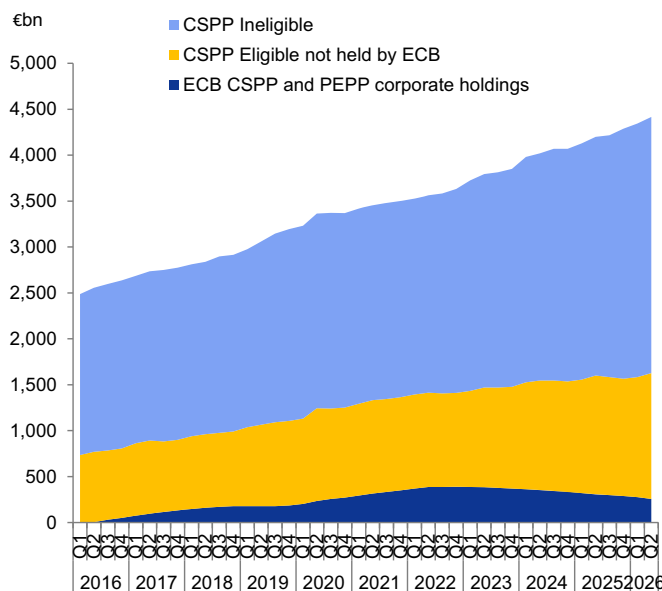
Exhibit 87: Net notional positioning in iTraxx over 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

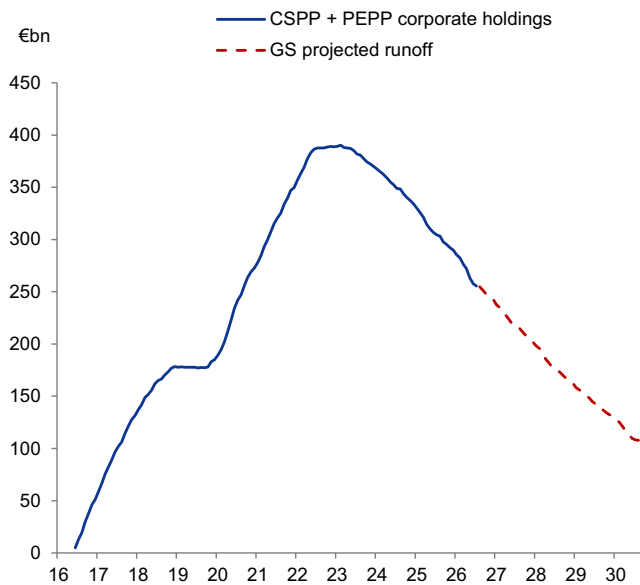
Flow dashboard: ECB QT

Exhibit 88: Ownership share of the ECB in the EUR IG market



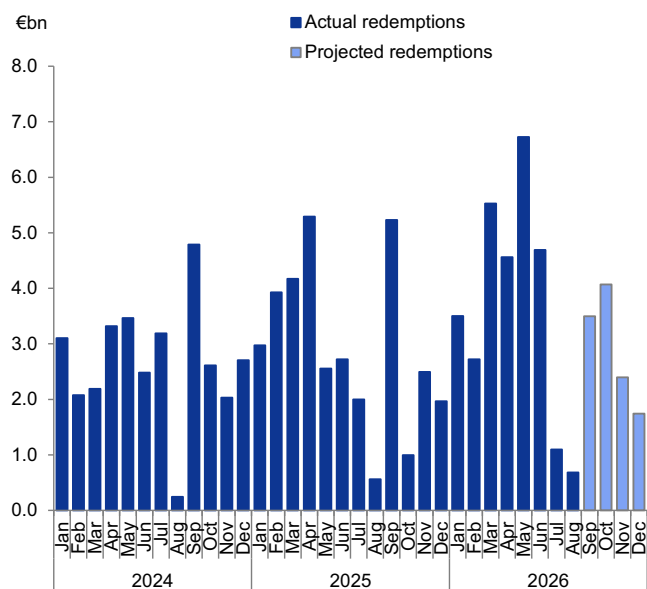
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 89: Run-off projections for the ECB corporate bond holdings



Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 90: CSPP monthly run-off pace



Source: ECB, Goldman Sachs Global Investment Research

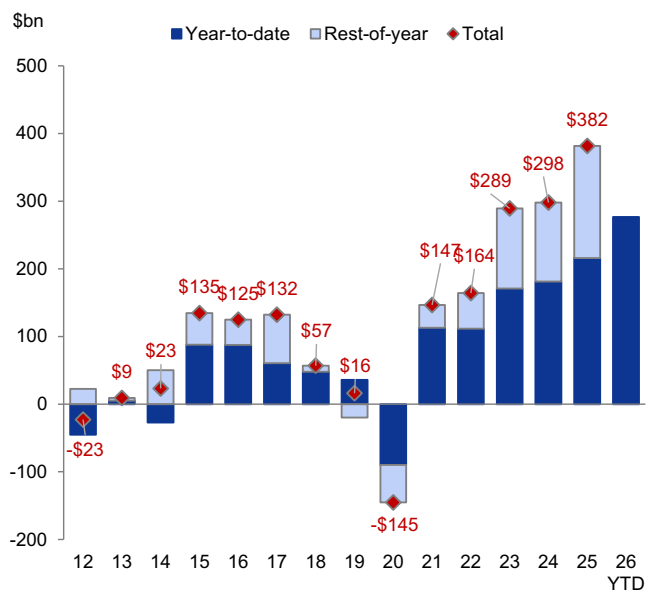
Exhibit 91: Performance of CSPP-eligible bonds vs. non-eligible bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

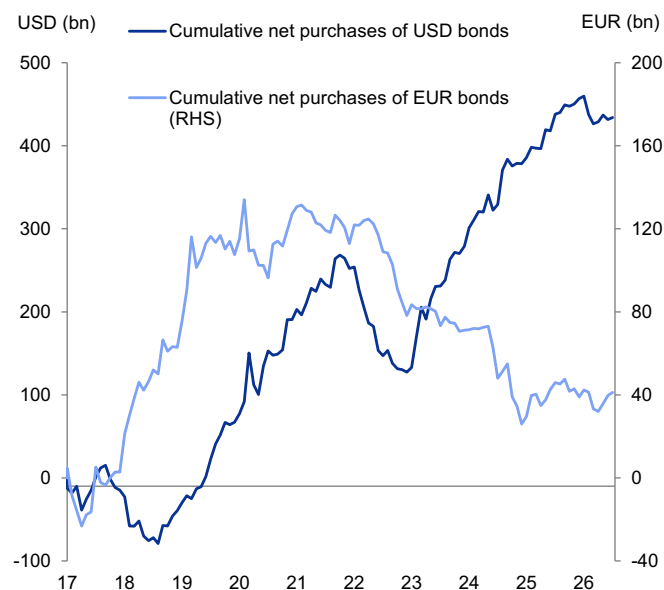
Exhibit 92: Net purchases of USD corporate bonds by foreign investors



Note: As of July 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

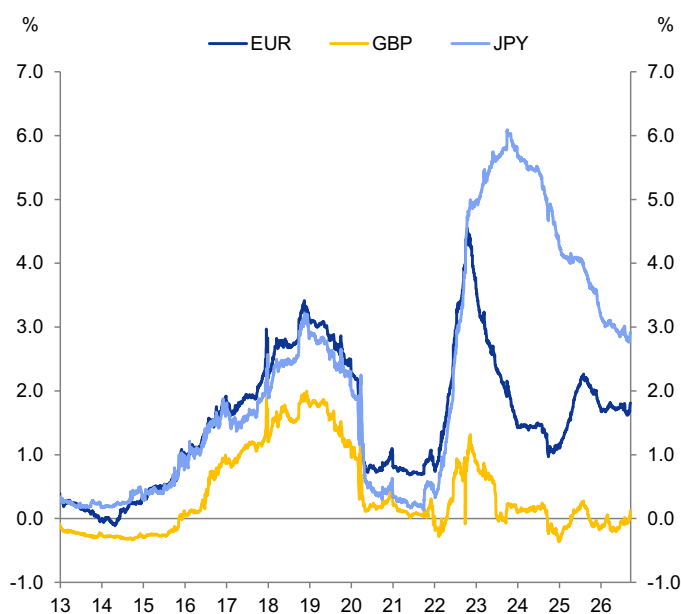
Exhibit 93: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of July 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 94: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

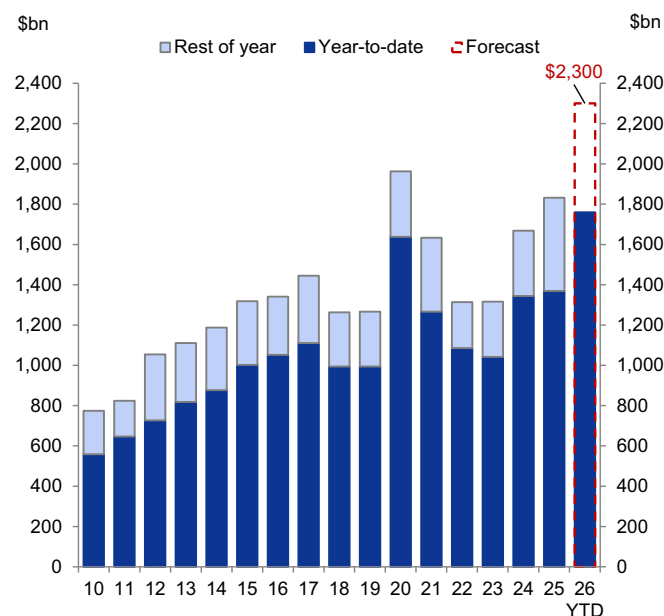


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 95: USD IG gross issuance

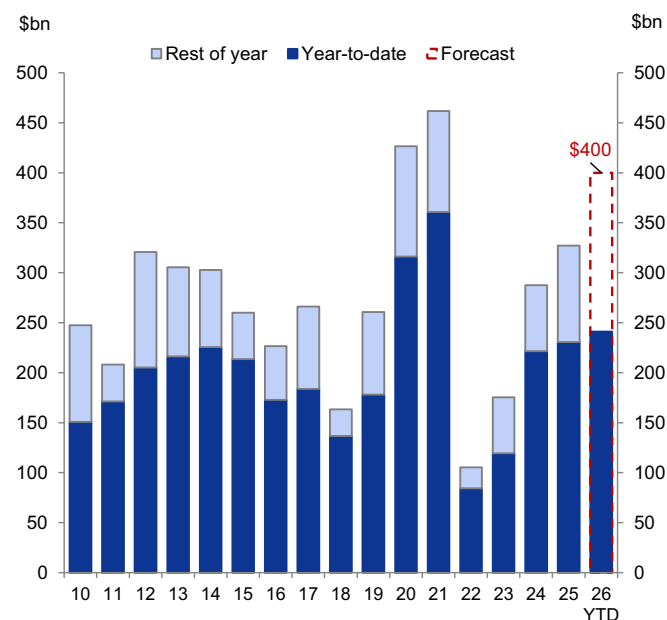
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 96: USD HY gross issuance

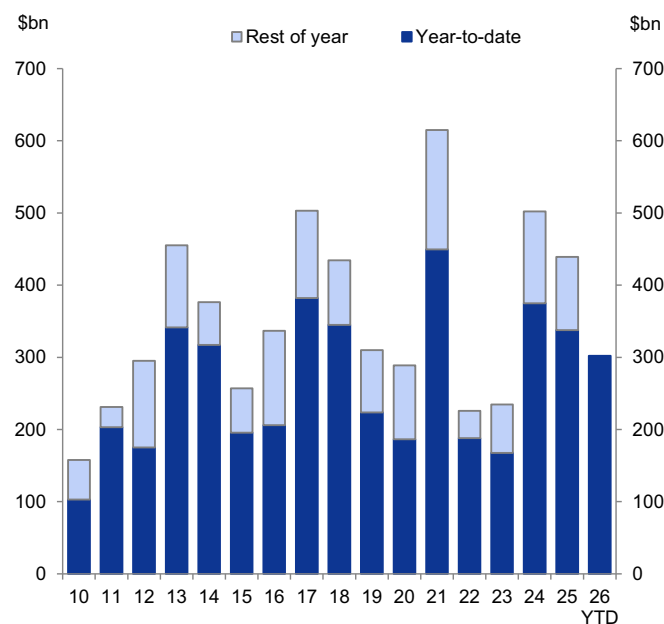
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 97: USD institutional leveraged loan gross issuance

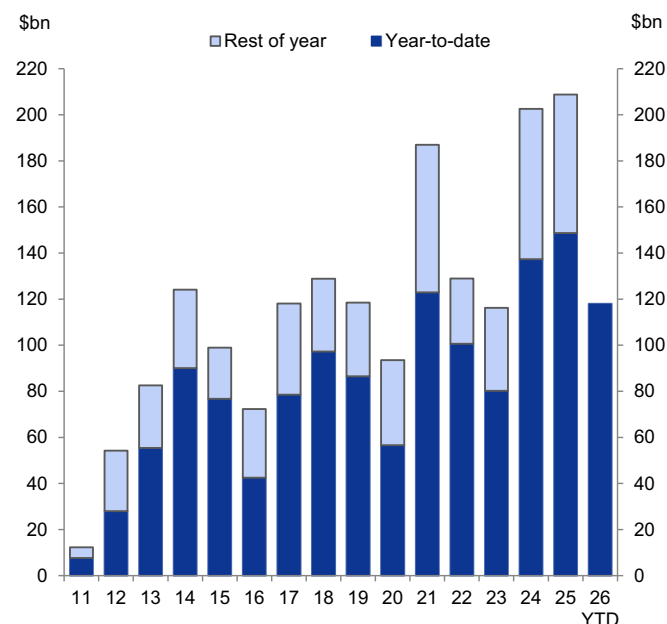
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 98: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

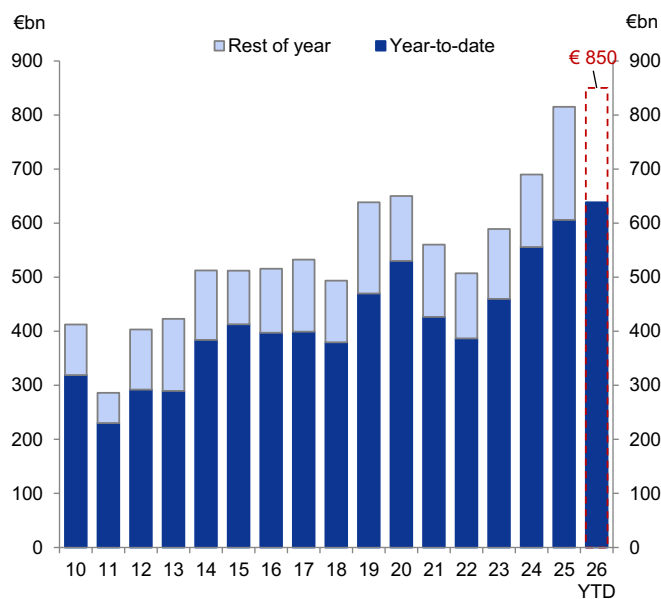


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 99: EUR IG gross issuance

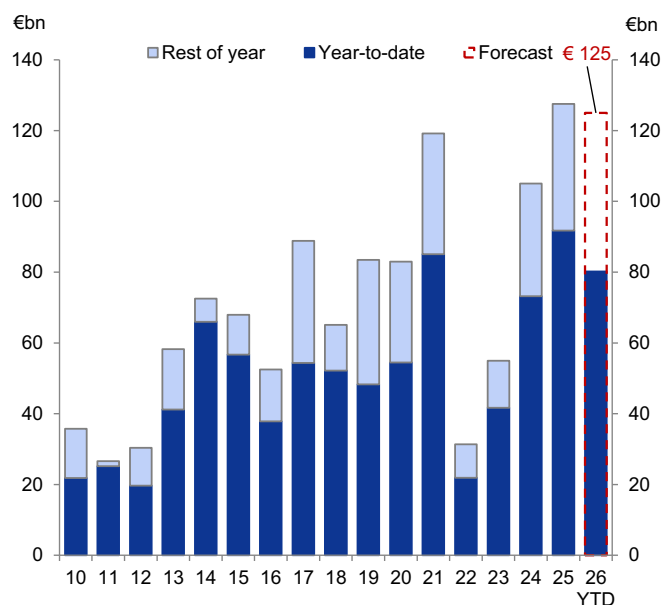
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 100: EUR HY gross issuance

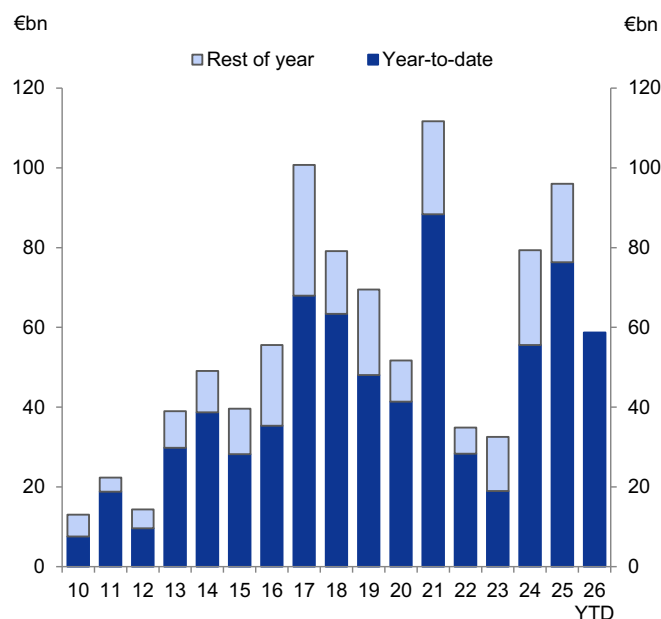
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 101: EUR Leveraged loan gross issuance

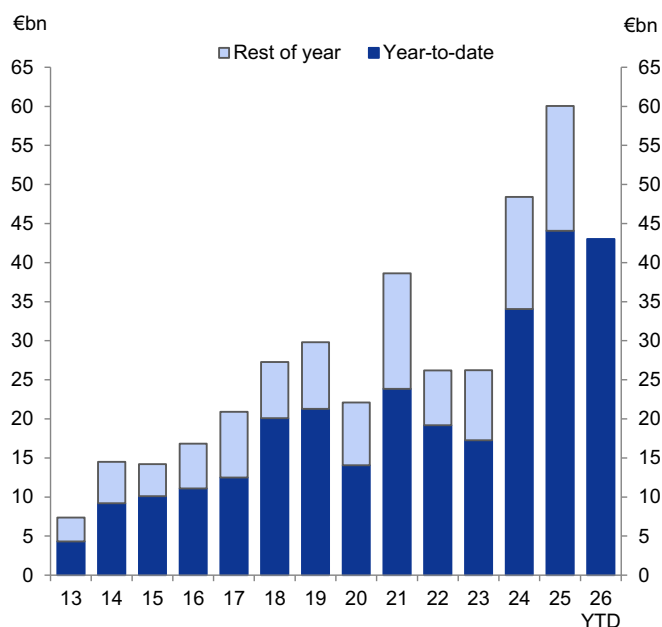
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 102: EUR CLO creation

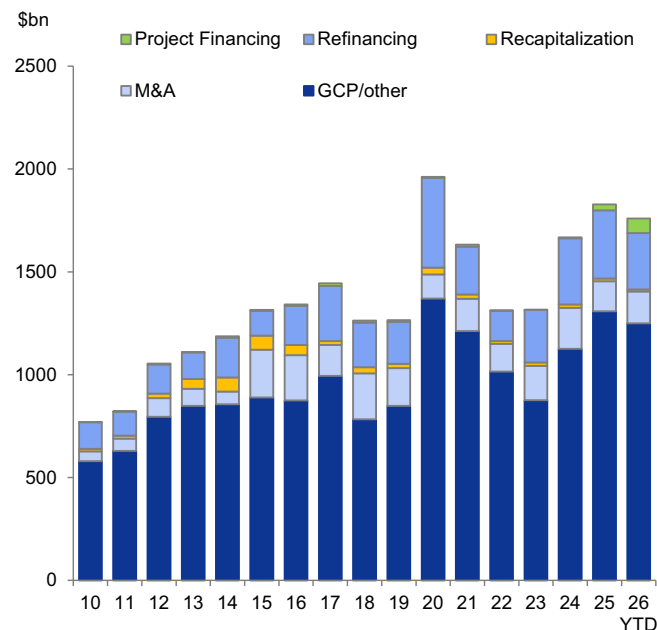
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

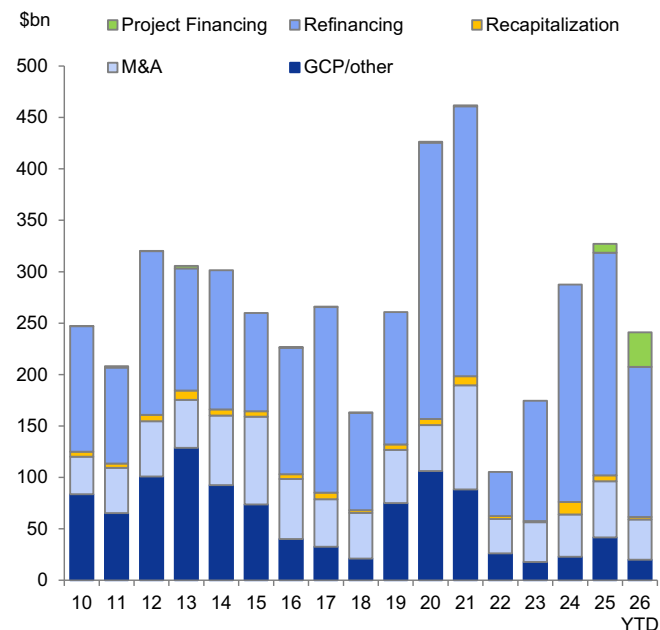
Primary market dashboard: Use of proceeds

Exhibit 103: Annual USD IG issuance volumes by use of proceeds



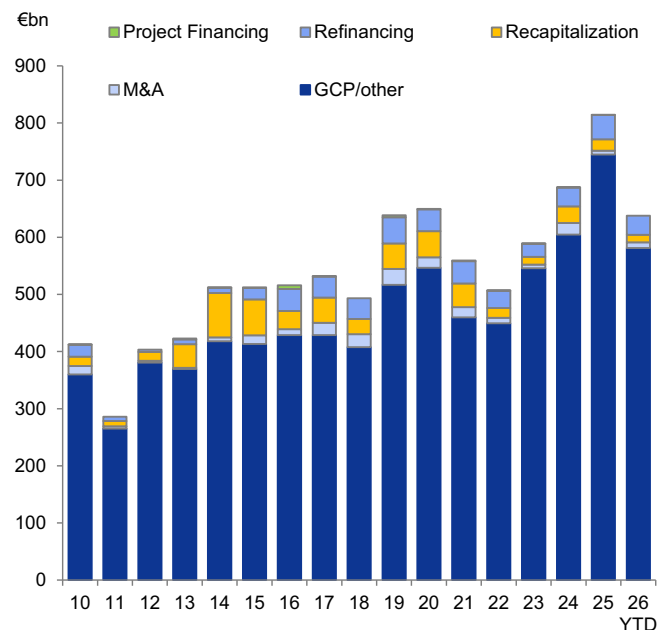
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 104: USD HY issuance volumes by use of proceeds



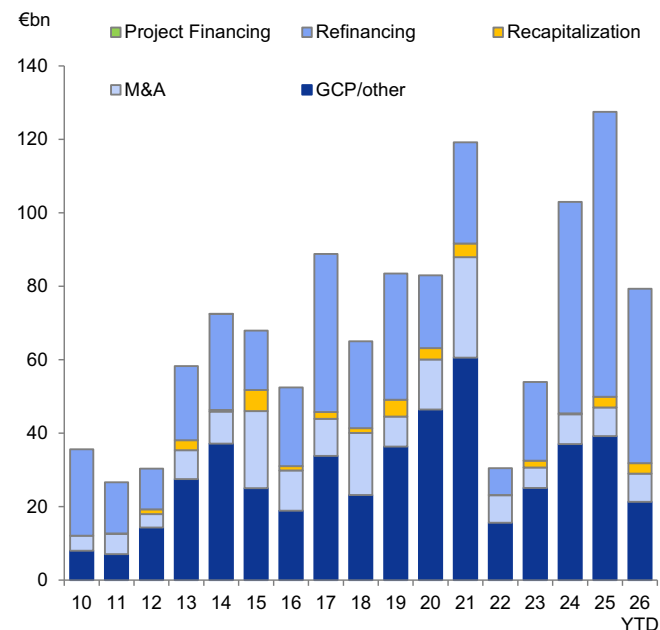
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 105: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 106: EUR HY issuance volumes by use of proceeds

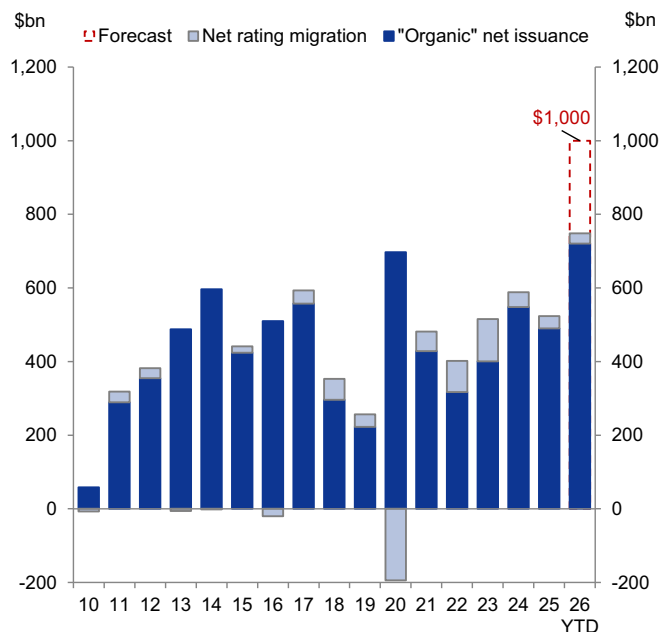


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

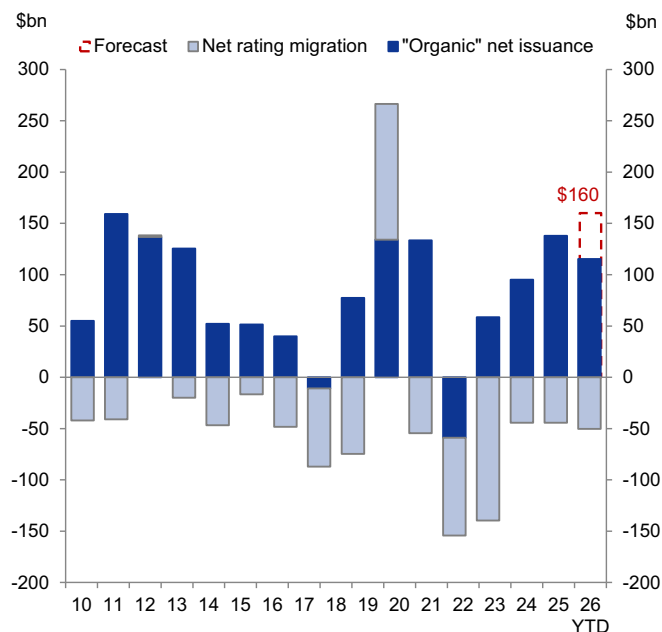
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 107: Annual USD IG net issuance



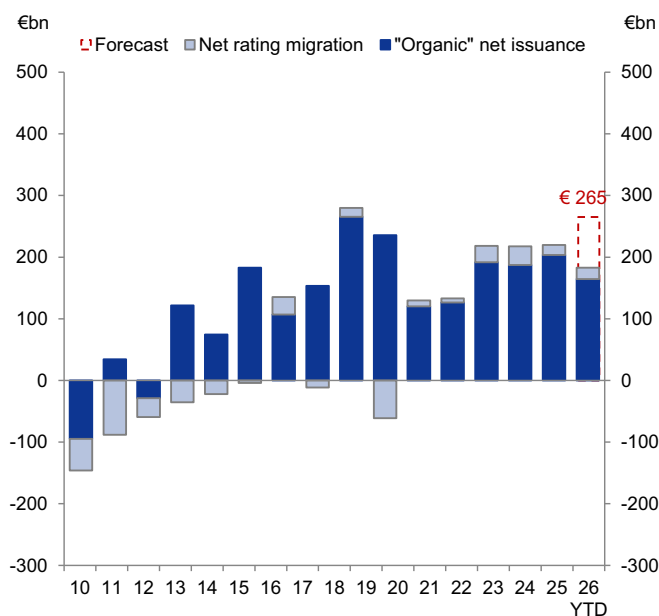
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 108: Annual USD HY net issuance



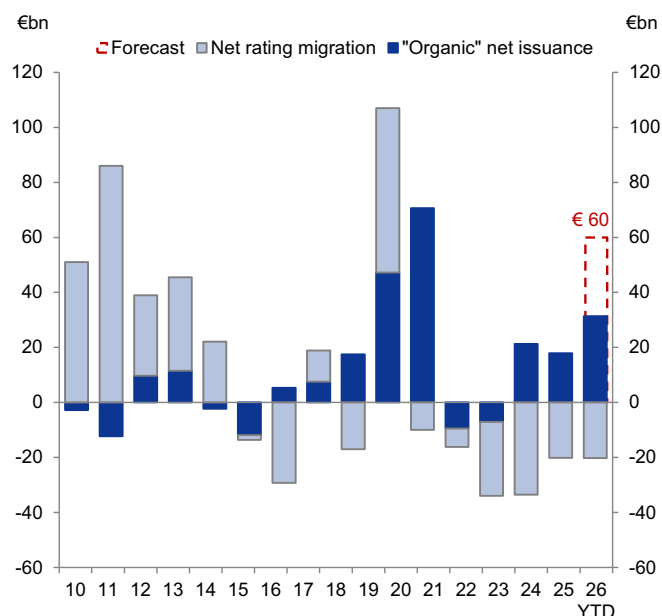
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 109: Annual EUR IG net issuance



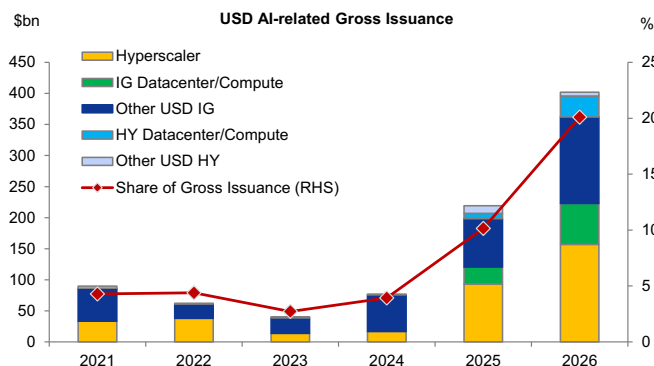
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 110: Annual EUR HY net issuance

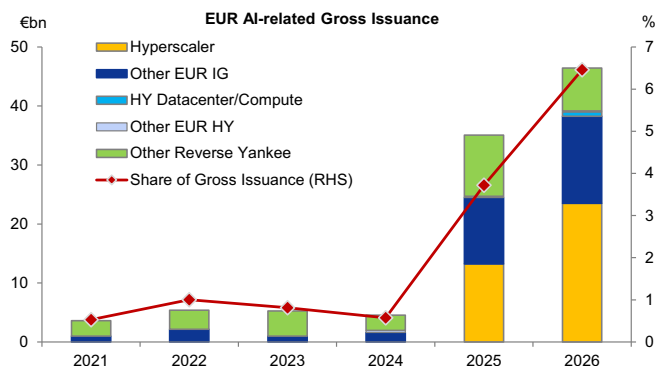


Source: Bloomberg, Goldman Sachs Global Investment Research

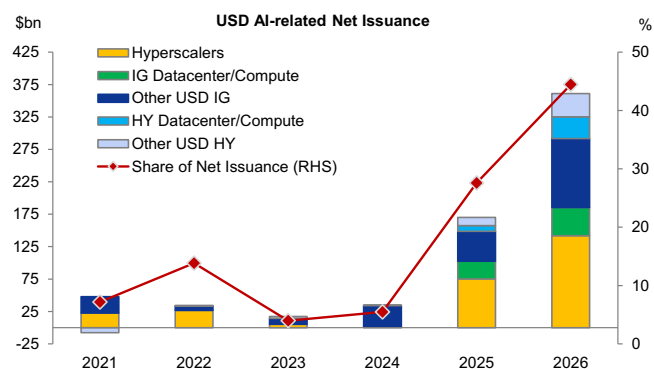
Primary market dashboard: AI-related issuance for the USD and EUR markets

Exhibit 111: USD AI-related Gross Issuance


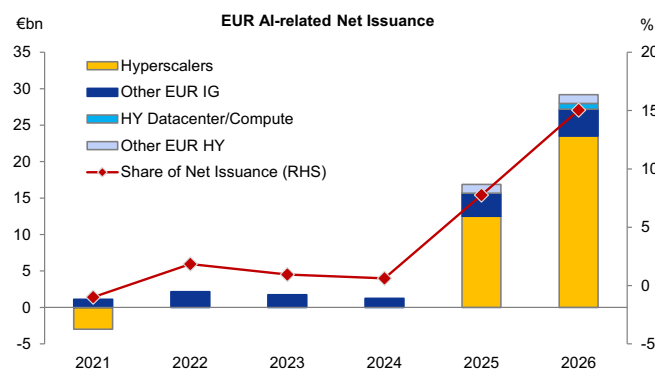
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 112: EUR AI-related Gross Issuance


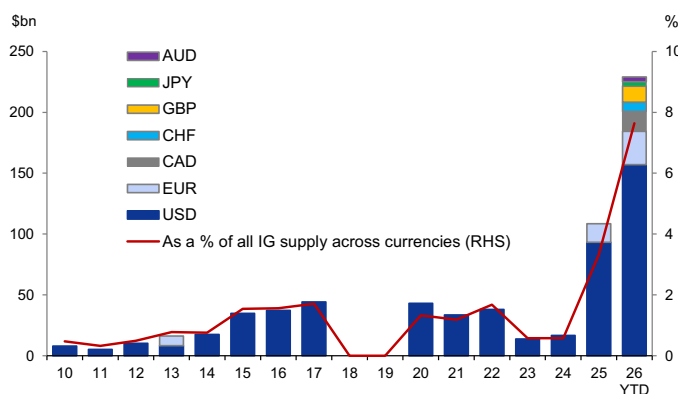
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 113: USD AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 114: EUR AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 115: Hyperscaler debt issuance across currencies (in USD-equivalent)


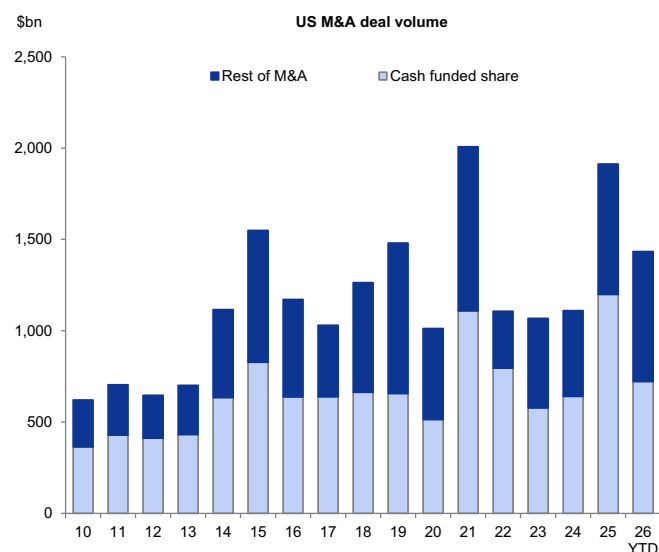
Note: Captures debt issued by the following companies: Alphabet (Google), Amazon, Microsoft, Meta, and Oracle.

Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume

Exhibit 116: Annual US M&A deal volume

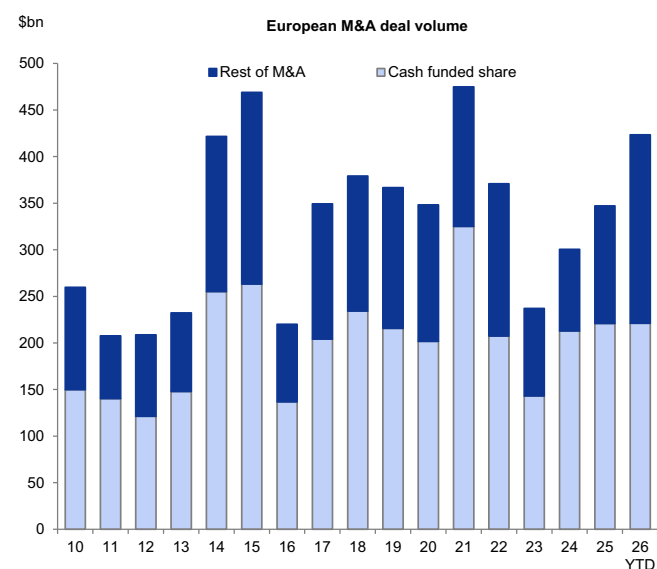
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 117: Annual European M&A deal volume

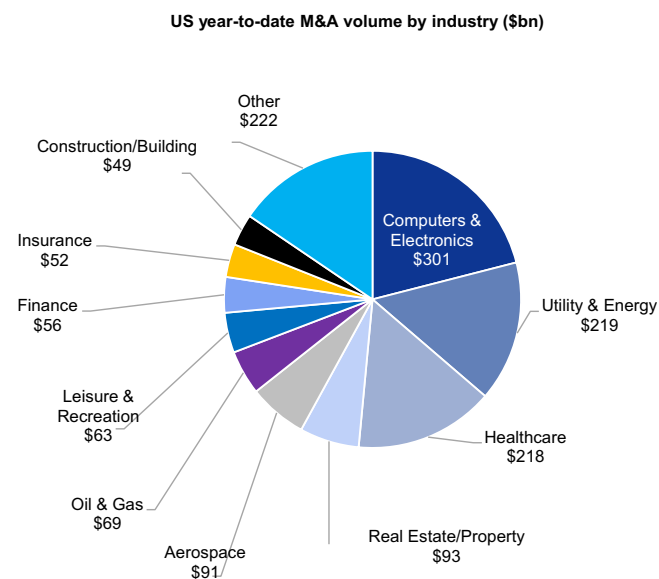
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 118: Year-to-date US M&A volume by sector

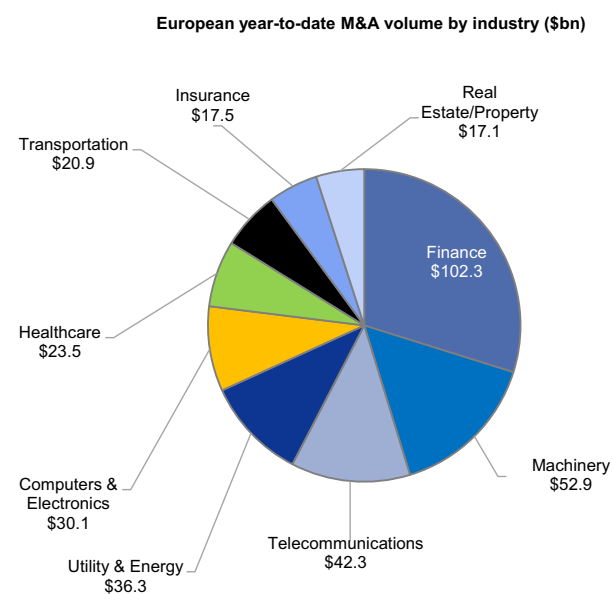
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 119: Year-to-date European M&A volume by sector

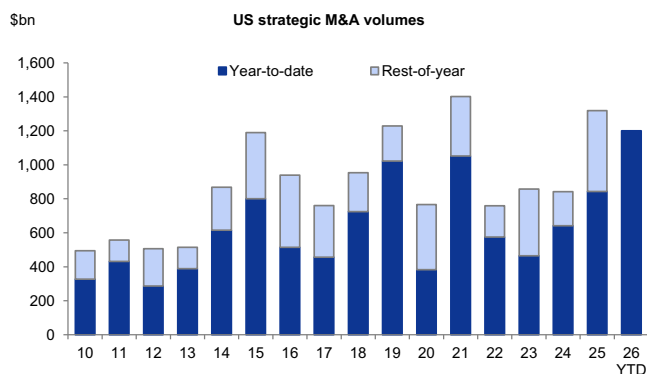
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume (Strategic deals only)

Exhibit 120: Annual US strategic M&A volumes



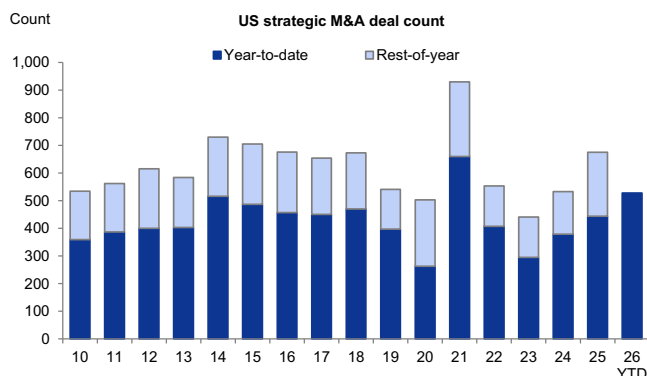
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 121: Annual European strategic M&A volumes



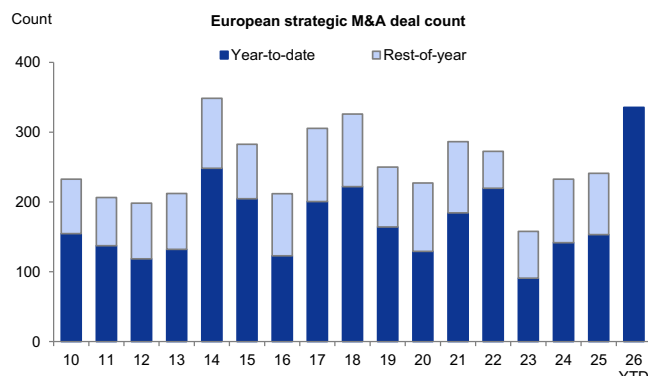
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 122: Annual US strategic M&A deal count



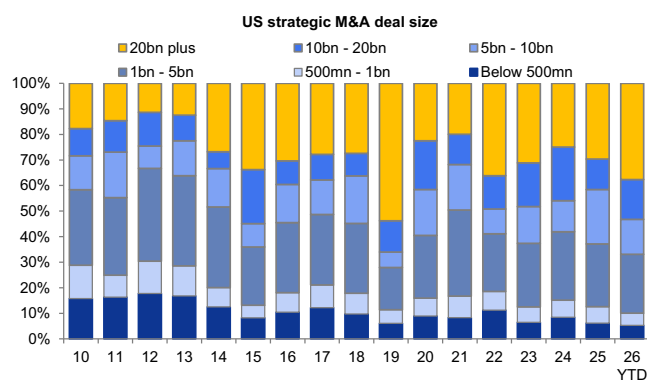
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 123: Annual European strategic M&A deal count



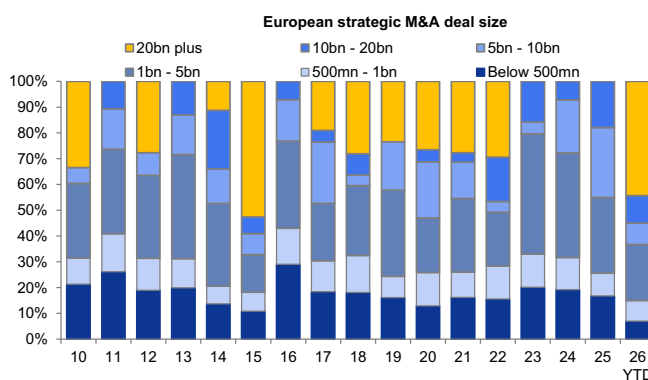
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 124: Annual US strategic M&A deal size
Percentages based on deal value, not count



Source: Dealogic, Goldman Sachs Global Investment Research

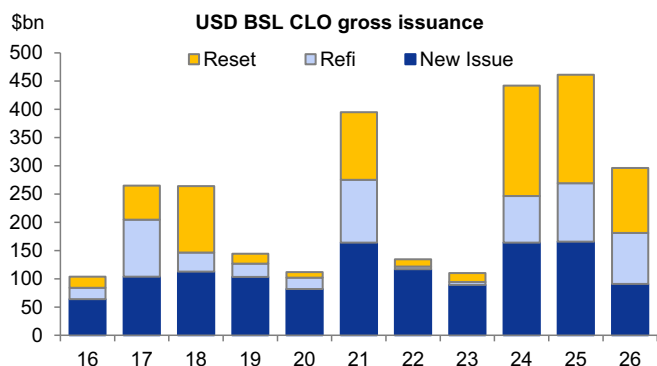
Exhibit 125: Annual European strategic M&A deal size
Percentages based on deal value, not count



Source: Dealogic, Goldman Sachs Global Investment Research

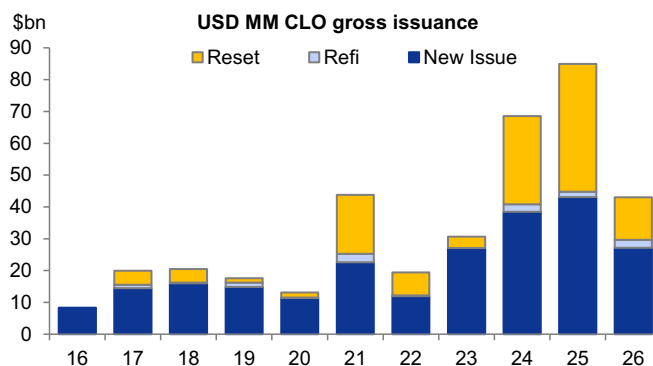
Primary market dashboard: CLO creation and refinancing

Exhibit 126: Annual USD broadly syndicated loan CLO creation



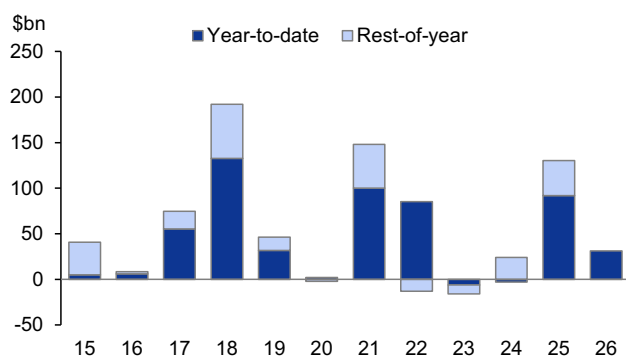
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 127: Annual USD middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

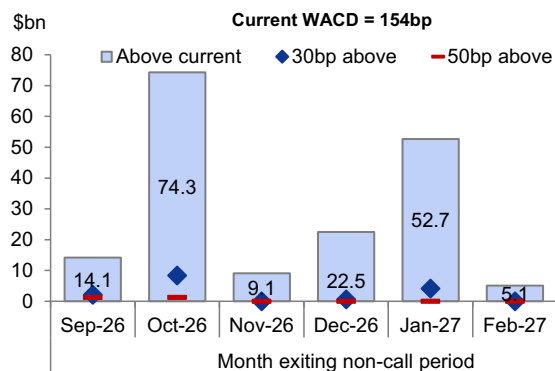
Exhibit 128: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 129: USD CLO refinance monitor

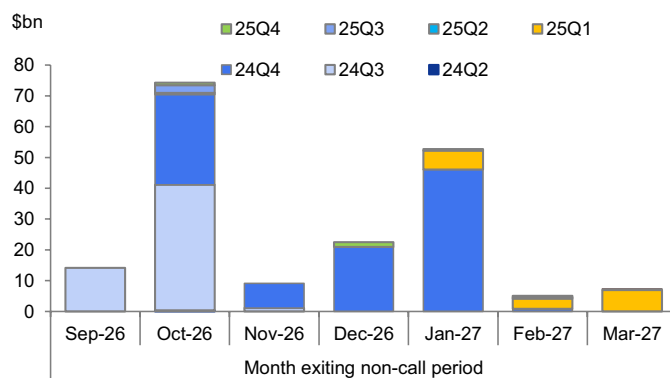
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 130: USD CLO refinance monitor

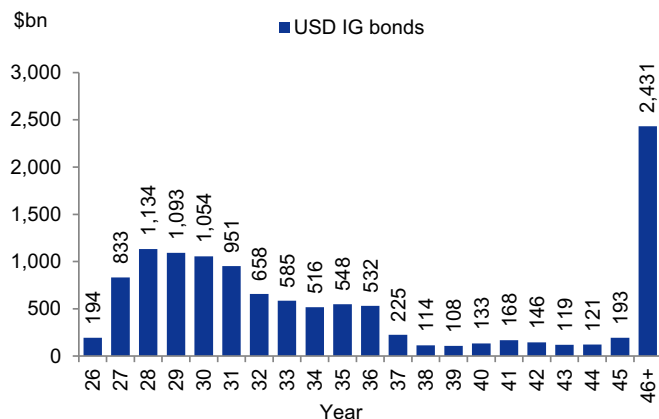
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

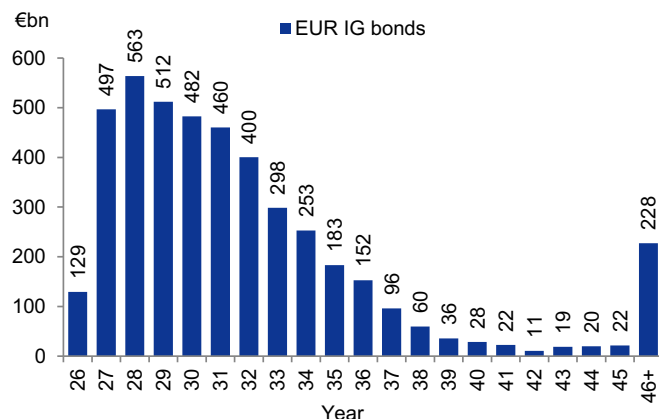
Primary market dashboard: Maturity walls

Exhibit 131: USD IG bond market maturity wall



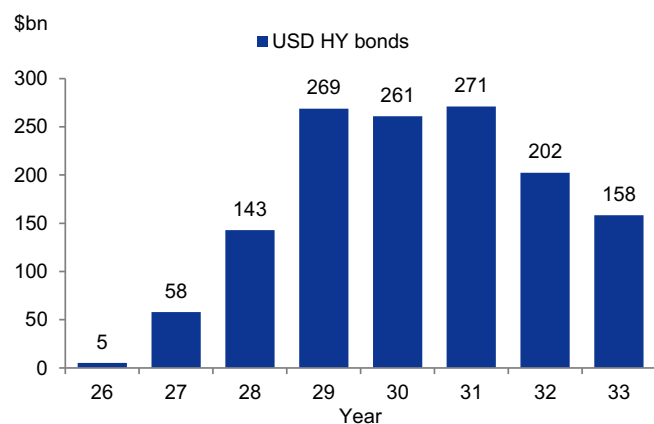
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 132: EUR IG bond market maturity wall



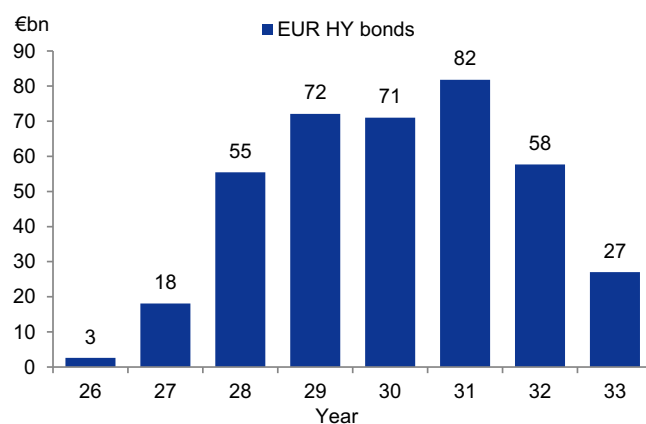
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 133: USD HY bond market maturity wall



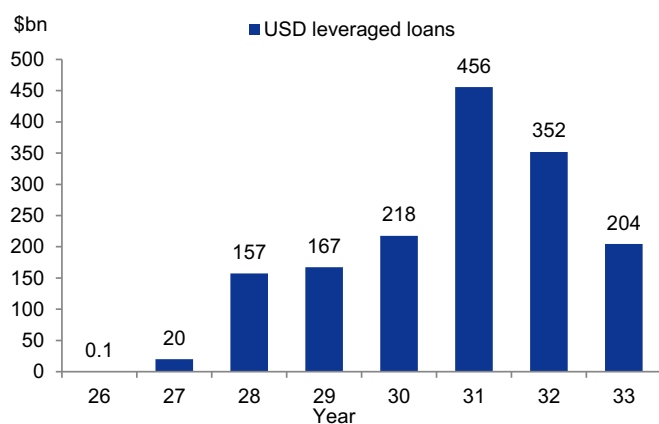
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 134: EUR HY bond market maturity wall



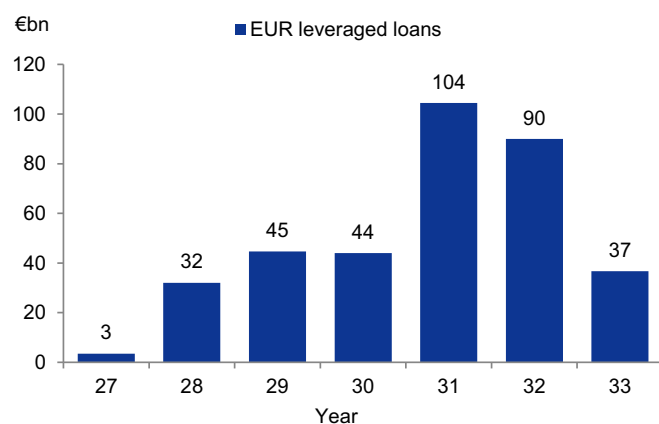
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 135: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

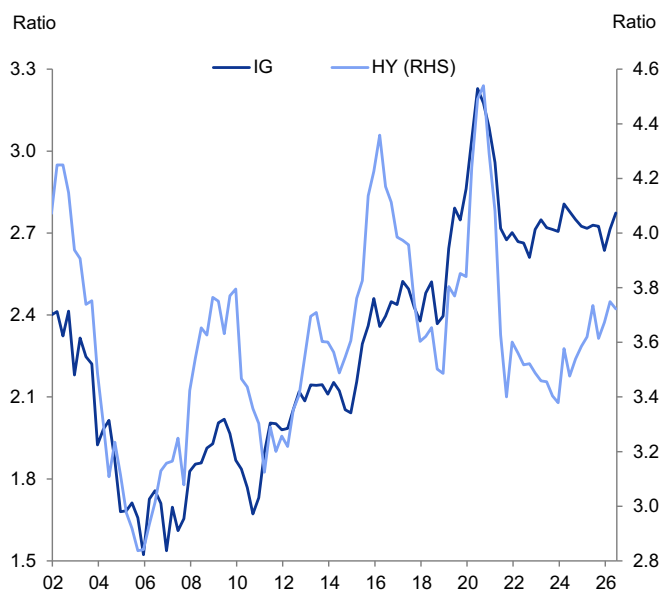
Exhibit 136: EUR leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

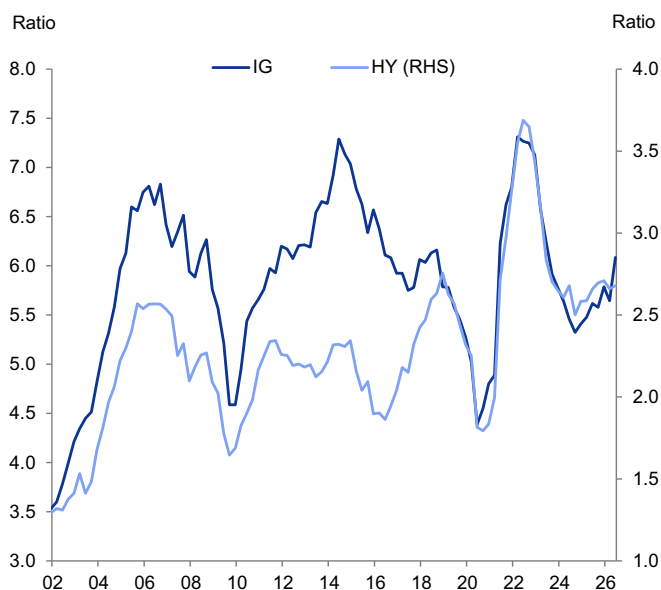
Fundamental dashboard: Credit metrics for US non-financial corporations

Exhibit 137: Net debt to EBITDA ratios for the median US IG and HY-rated non-financial corporations



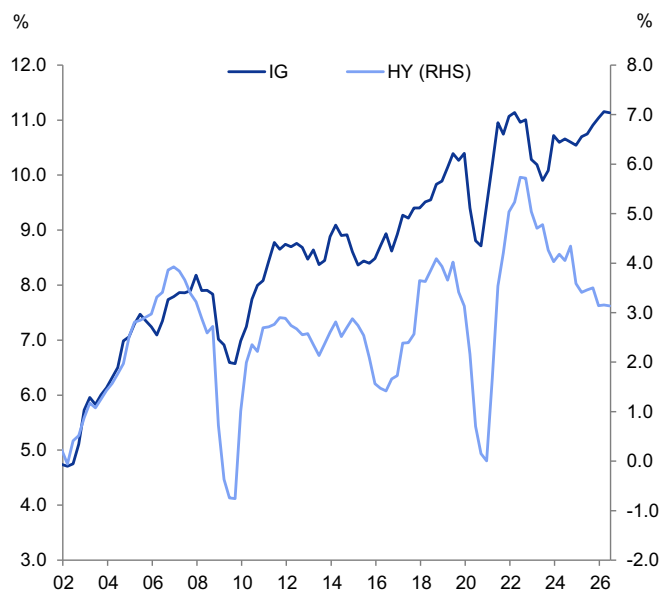
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 138: Interest coverage ratios for the median US IG and HY-rated non-financial corporations



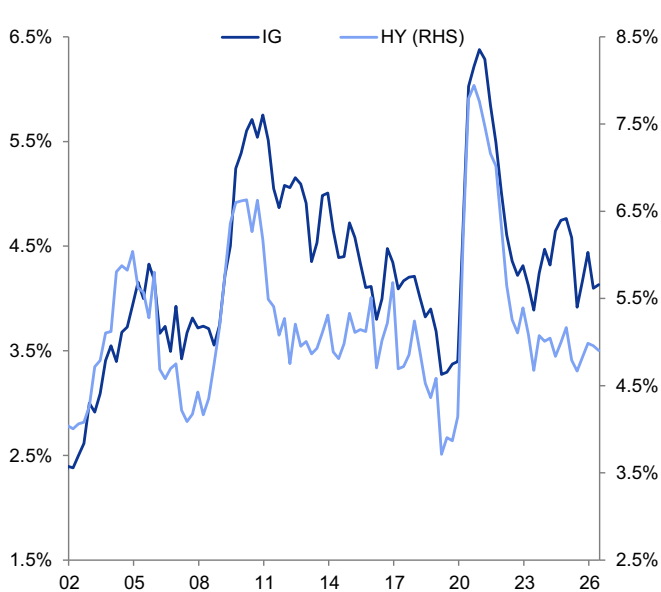
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 139: Net margins for the median US IG and HY-rated non-financial corporations



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 140: Cash to total assets ratios for the median US IG and HY-rated corporations

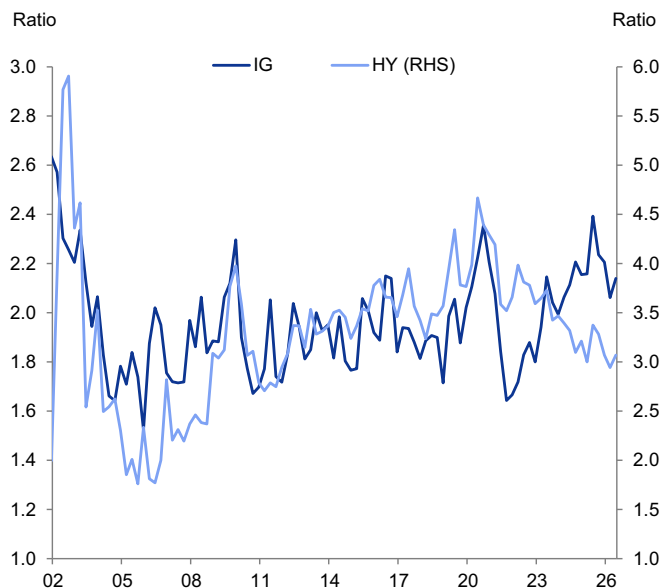


Source: FactSet, Goldman Sachs Global Investment Research

Fundamental dashboard: Credit metrics for European non-financial corporations

Exhibit 141: Net debt to EBITDA ratios for the median European IG and HY-rated non-financial corporations

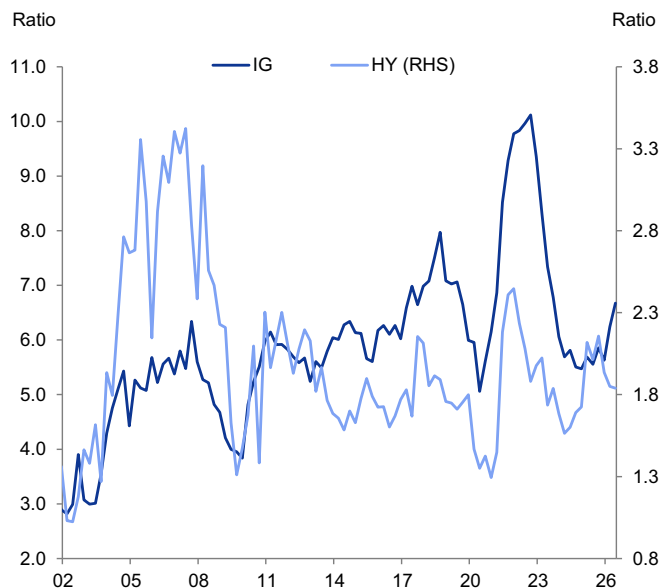
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 142: Interest coverage ratios for the median European IG and HY-rated non-financial corporations

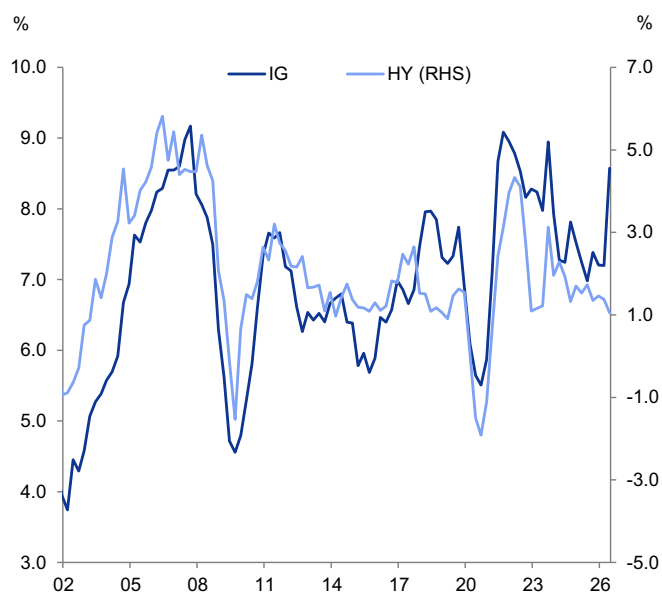
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 143: Net margins for the median European IG and HY-rated non-financial corporations

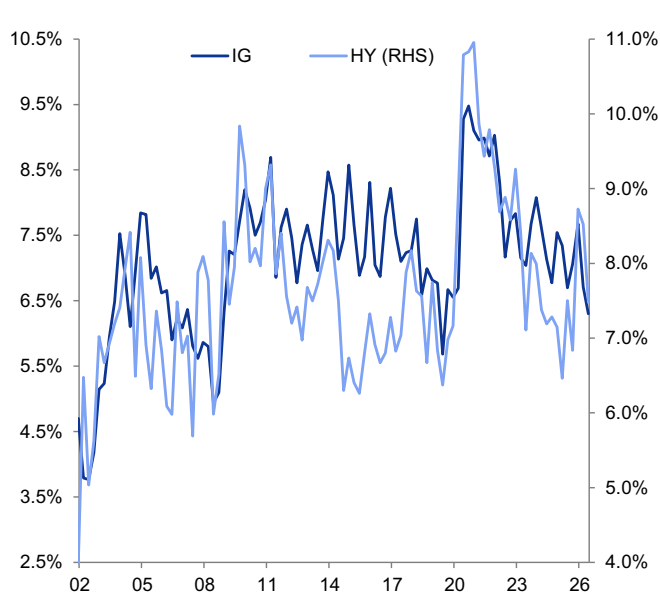
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 144: Cash to total assets ratios for the median European IG and HY-rated non-financial corporations

We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway

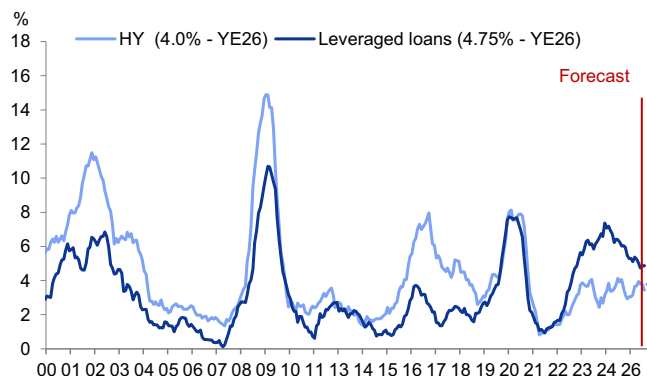


Source: FactSet, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 145: Default rates for US HY bond and leveraged loan issuers

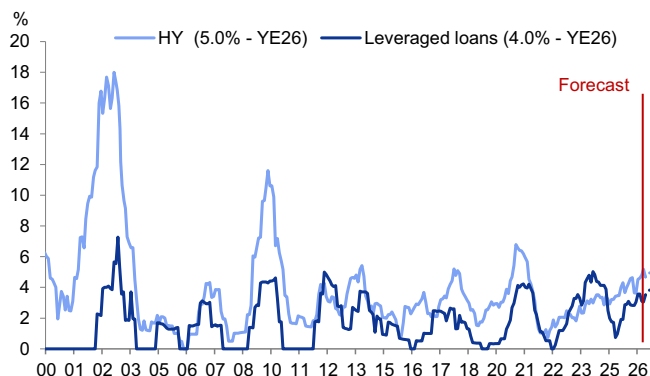
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 146: Default rates for European bond and leveraged loan issuers

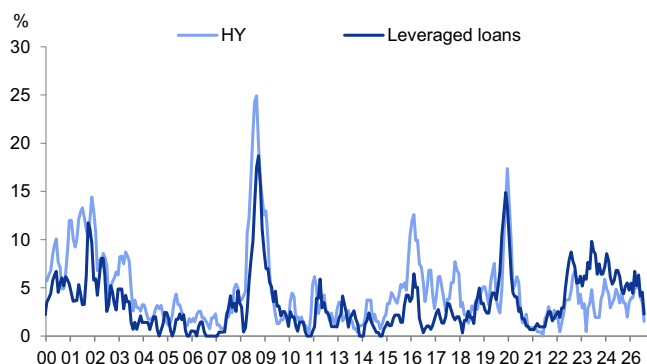
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 147: High frequency default rates for US HY bond and leveraged loan issuers

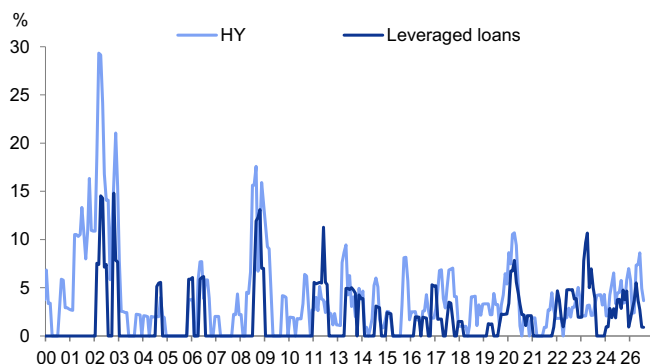
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

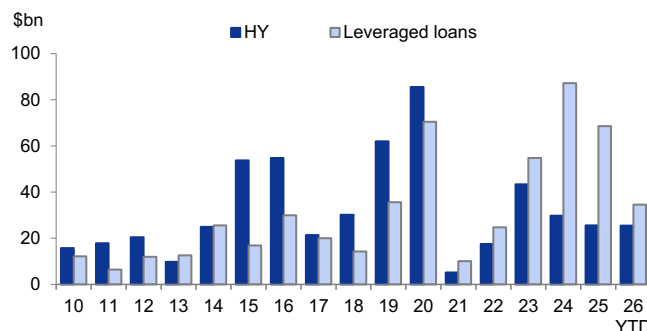
Exhibit 148: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



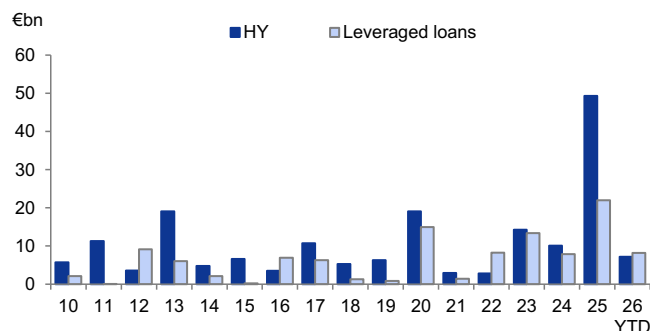
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 149: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 150: Annual par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 151: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	6/30/2026	DISH DBS Corporation	Media: Broadcasting & Subscription	Prepackaged Chapter 11	9.8
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.1
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.9
	2/17/2026	Vibrantz Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6
	12/8/2025	Newfold Digital Holdings Group, Inc	High Tech Industries	Distressed exchange	2.5
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.3

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

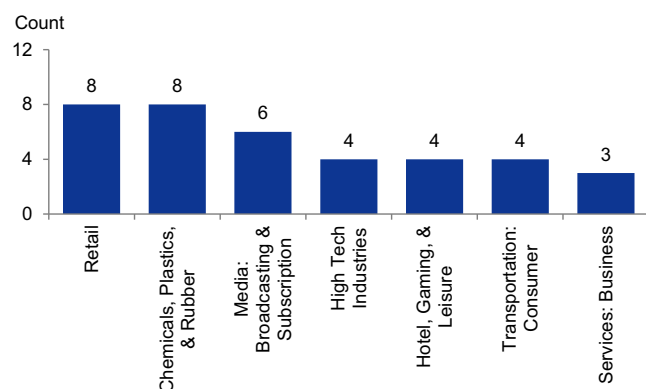
Exhibit 152: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
	4/29/2026	La Financiere ATALIAN S.A.S.	Services: Business	Missed interest payment	1.0
	11/12/2025	ARD Finance S.A.	Containers, Packaging, & Glass	Distressed exchange	0.9
	11/4/2025	Klockner Pentaplast of America, Inc	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8
Loans	4/30/2026	Poseidon BidCo S.A.S	High Tech Industries	Missed interest payment	1.6
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	7/10/2026	Sirona BidCo France	Chemicals, Plastics, & Rubber	Missed interest payment	1.2
	8/17/2026	Foundever Group	Services: Business	Distressed exchange	1.2
	4/7/2026	Biscuit Holding S.A.S.	Beverage, Food, & Tobacco	Missed interest payment	1.1

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 153: Industry default count in the US

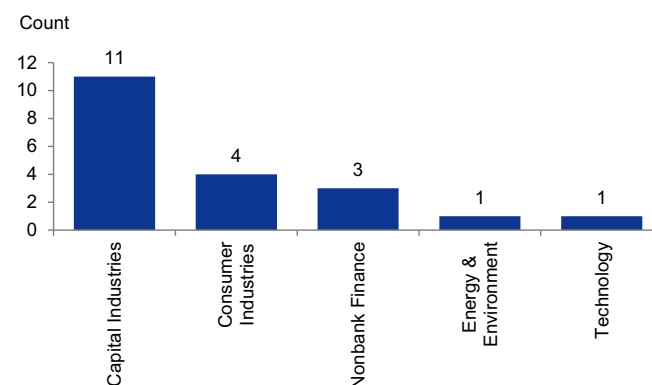
Trailing 12-month counts



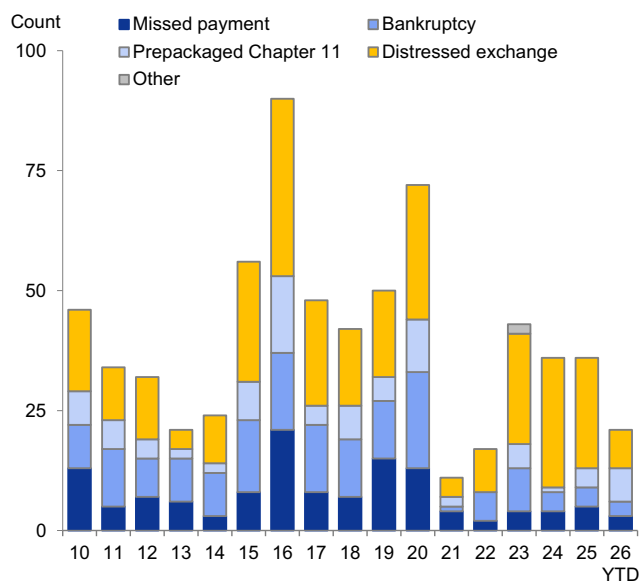
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 154: Industry default count in Europe

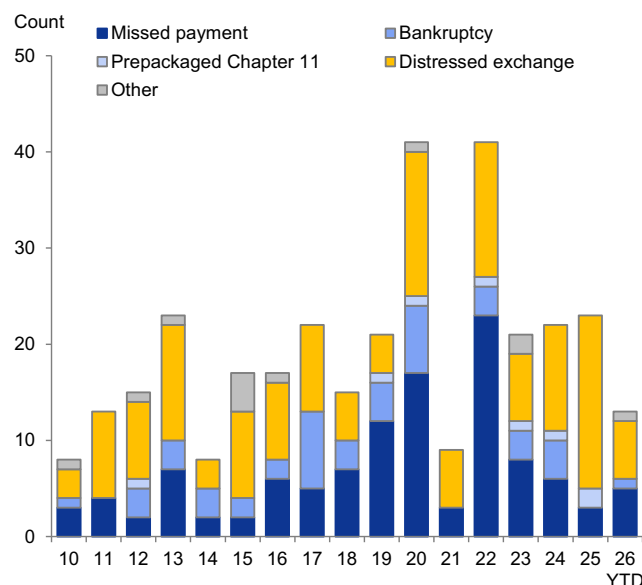
Trailing 12-month counts



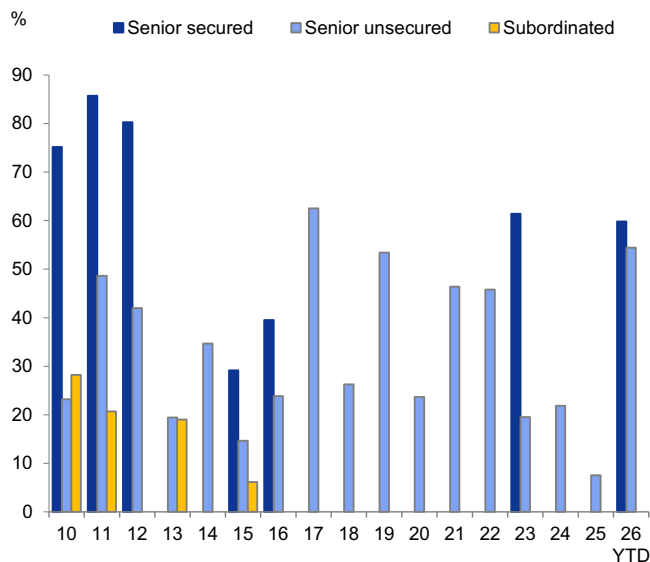
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 155: Composition of defaults in the USD HY bond market by type

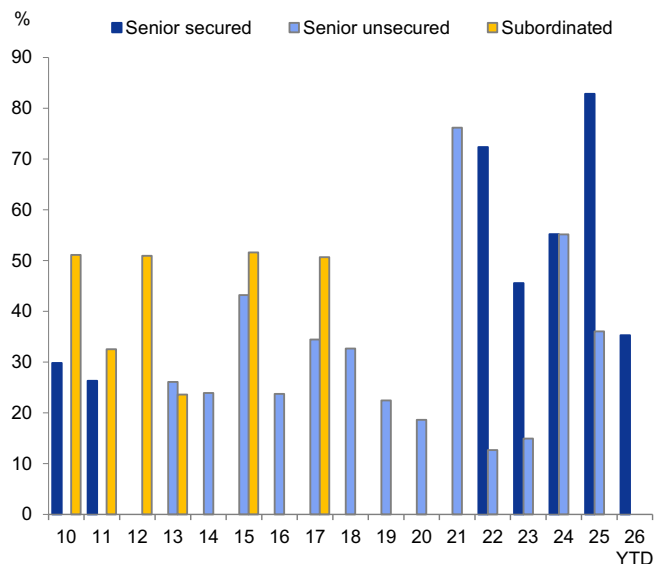
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 156: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

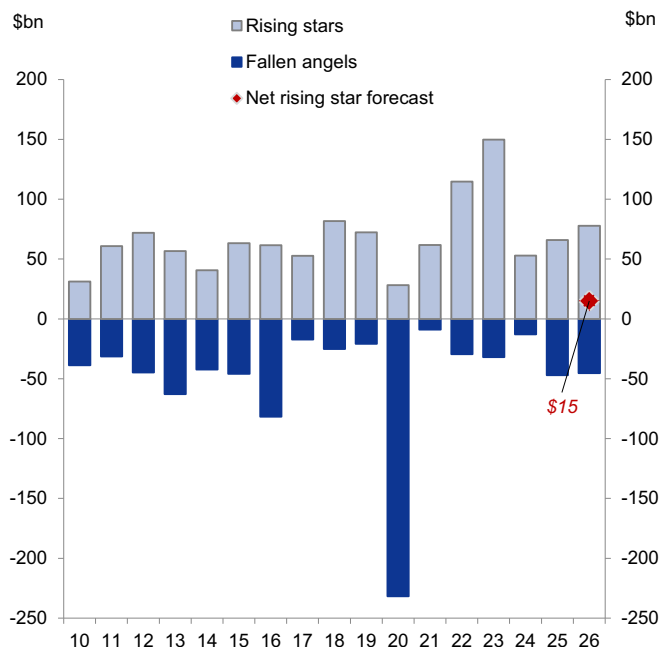
Exhibit 157: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

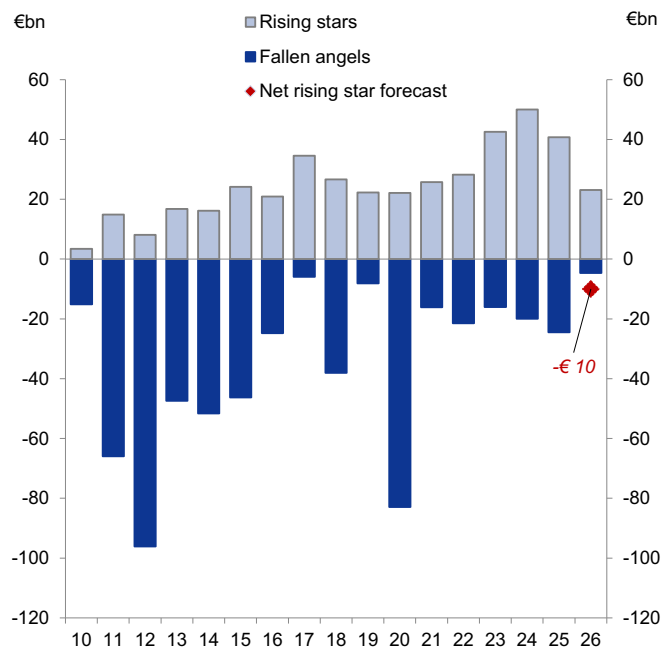
Exhibit 158: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

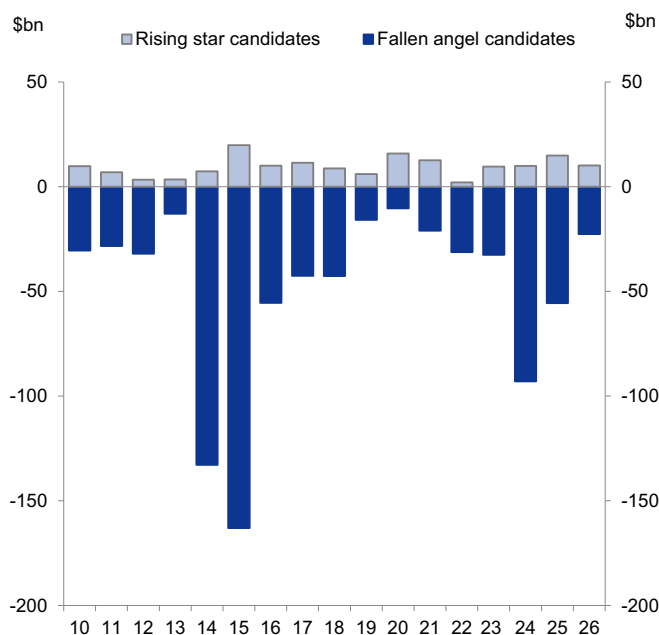
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 159: Fallen angels vs. rising stars in the USD market


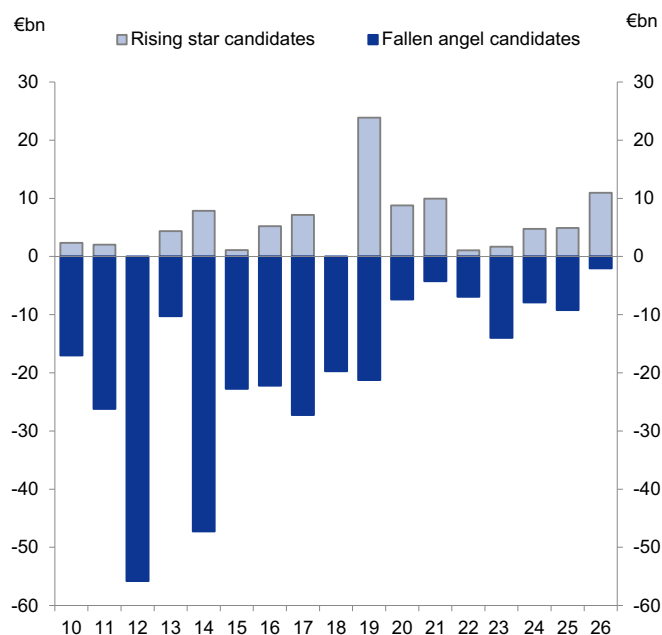
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 160: Fallen angels vs. rising stars in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

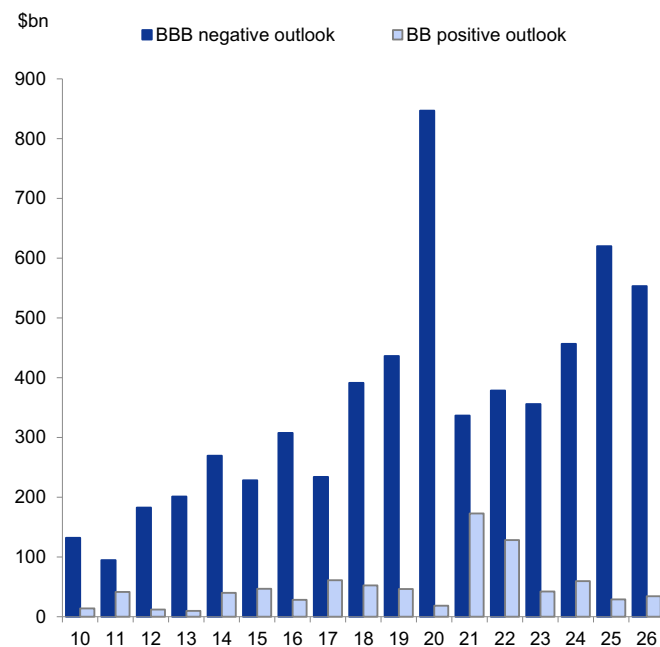
Exhibit 161: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 162: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


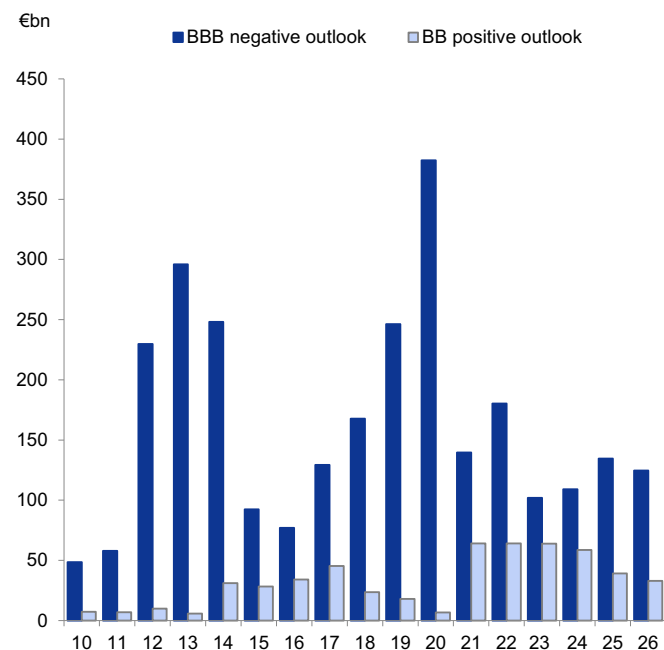
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 163: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 164: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 165: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Cox Communications Inc	7.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Flowers Foods Inc	1.7
Coty/Hfc Prestige/Int Us	1.7
Leggett & Platt Inc	1.5
Electronic Arts Inc	1.4
Banco Santander Mexico	1.3

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 166: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Carnival Corp Ltd	8.5
Teva Pharmaceuticals Ne	7.4
Vistra Operations Co Llc	6.3
Medline Borrower Lp	4.5
Office Cherifien Des Pho	4.1
1011778 Bc / New Red Fin	4.0
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Sba Communications Corp	3.0
Frontier Communications	2.8

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 167: Top EUR Fallen angel issues in the last 12 months

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Adecco Int Financial Svs	0.5
Coty Inc	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 168: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Teva Pharm Fnc Nl Ii	6.2
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Carnival Uk Ltd	1.6
Unibail-Rodamco-Westfld	1.5
Unipol Assicurazioni Spa	1.3
Metro Ag	1.1
Bper Banca	1.0
Piraeus Bank Sa	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

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